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**PROJECT TITLE**

NU Racing Powertrain Cooling System Design

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FINAL YEAR PROJECT

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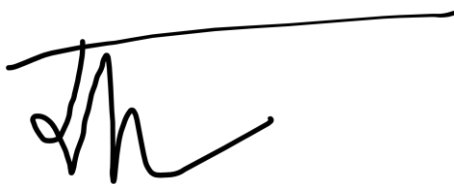
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# List of Terms and Abbreviations

**FSAE:** Formula SAE

**Formula SAE:** A competition organised by the Society of Automotive Engineers (SAE) where university students design and manufacture a Formula-style race car

**Endurance Event:** Dynamic competition event where the car is required to complete 22km

**Autocross Event:** Dynamic competition event where the car is required to complete approximately 1 km track as fast as possible

**NU Racing:** The University of Newcastle's Formula SAE team

**NU23:** The 2023 NU Racing FSAE competition car

**NU24:** The 2024 NU Racing FSAE competition car

**MoTeC:** Manufacturer of automotive data collection systems

**Bleeding:** The process of removing air from the system

**LPM:** Litres Per Minute

**Powertrain:** The system responsible for delivering power from the accumulator/batteries to the wheels

**Accumulator:** The energy storage device used to deliver power to the car

**Track Day:** A day or session dedicated to testing the car out on a track

**CEN:** Central Electronic Node low voltage electrical circuit board

**FPI:** Fins per inch

**PETG:** Polyethylene terephthalate glycol, a thermoplastic polyester

**CFD:** Computational fluid dynamics

**FEA:** Finite Element Analysis

**AUD:** Australian Dollars

**CAD:** Computer-Aided Design

# Dot Point Summary

Throughout the Final Year Project of 2024, the contributions to NU Racing consisted of:

- Initial Testing and Research
  - Theoretical flowrate calculations and real flowrate measurements for NU23 cooling system with addition of NU24s motor, EMRAX 228.
  - Calculations for possible cooling system configurations and new pump and radiator research.
- New Radiator Decisions and Design
  - Back and forth theoretical temperature measurements and design work for different configurations of a new radiator to be fit onto NU24 or if not delivered in time for NU25.
- Radiator Mounting Brackets Design
  - Radiator bracket designs with and without jacking point with decisions based on rules and FEA to determine the best option for mounting the radiator and other parts of the cooling system.
- Pump and Overflow Mounting Design
  - Designing a new pump and overflow mounting/enclosure to be fit onto the radiator bracket and hold said part
  - Decisions on pump mounting due to small clearances with the rear chassis and brackets
- Radiator Fan Sizing and Comparison
  - Investigating the possibility of a different radiator fan with different sizing than the current used fan, comparisons with the usage of flow-pressure curves and mass
- Shroud Study
  - Exploring the usage of a shroud and exactly how effective they are for our cooling

system using CFD analyses and real-world data during track days using a 3D printed model.

- Enclosures and Manifolds Designing
  - Designed the initial CEN enclosure which was printed off and used for initial testing on NU23.
  - Designed the fan manifold to fit the new accumulator enclosure to allow for ease of removal and optimal airflow through the batteries.
- Cost Report Contributions
  - Contributed to the cost report by completing cooling system costing and other systems as well.
  - Created models and drawings for wiring, brake lines and hoses to be used in demonstration of cost report.
- 2024 FSAE-A Competition and Cooling System Performance
  - Attended the 2024 FSAE-A competition where the cooling system was tested in multiple events where the performance was shown to be above what is required.
- NU25 Cooling System Design
  - Designed a new cooling system with the possibility of new and better components to optimize the system itself.
  - Designed a new mounting for the radiator, jacking point and pumps to allow for better bleeding to the system.
  - Proposed the idea of a single loop cooling system with careful consideration of motor controller temperatures.

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# **1 Introduction**

This report focuses on the development of the powertrain cooling system for NU Racings 2024 FSAE electric vehicle, NU24 by expanding on existing knowledge and designs and enhancing the system's performance. This report outlines the calculations, design choices and decisions regarding the components to be included in NU24s cooling system. Also, included is the design and manufacturing process enclosures for components to be tested on NU23 and to be used on NU24.

The primary aim for improving the cooling system for 2024 consists of increasing flow rates to keep the components which have a minimum flow rate safer and increasing air flow through radiator to provide overall better cooling. Several variables need to be examined in the optimization process of the cooling system including but not limited to, flowrates, pressures, temperatures, current and voltages. Keeping the number of components and the total weight at a minimum is also a major consideration in the design process.

## **1.1 NU23 Cooling Setup**

The NU23 cooling system consists of two separate cooling loops, one cooling the motor and the other cooling the motor controller with each loop having one pump but both going through the same radiator as shown in Figure 1.



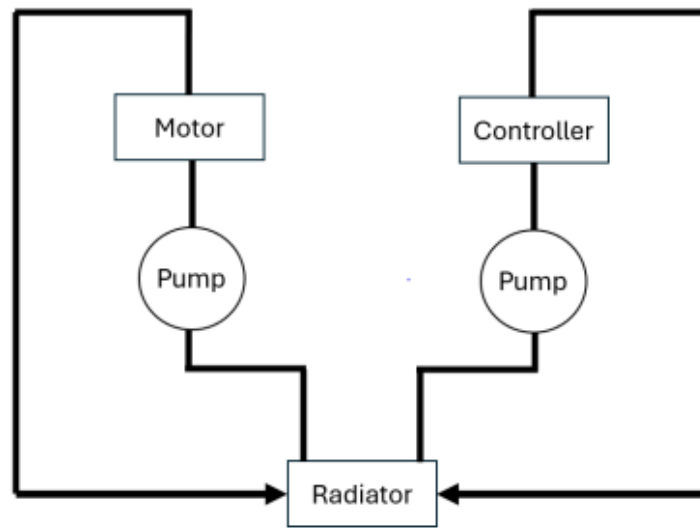


Figure 1: NU23 Cooling Setup

The motor which was used was the Emrax-188 which will be replaced by an Emrax-228 for NU24, the motor controller was the Cascadia CM200 (Cascadia Motion, 2024) which will be used in NU24, the radiator was a dual pass single loop radiator converted into a single pass dual loop radiator from PWR, a 7.5 inch SPAL pulling fan, and the pumps were Davies-Craig 12V EBP40s. The relevant flow, pressure, temperature and dimension details of these components from the manufacturer or last year's calculations can be seen below in Table 1, Figure 2 and Figure 3.

Table 1: Cooling System Components Information (Emrax, 2024)

| Component                                        | Rated Flow (LPM) | Pressure loss (bar)                      | Resistance                        | Maximum Inlet Coolant Temp (°C) | Max Pressure (bar) |
|--------------------------------------------------|------------------|------------------------------------------|-----------------------------------|---------------------------------|--------------------|
| Emrax-188                                        | 6-8              | 0.5 @ 7 LPM                              | 0.0102                            | 50                              | 2                  |
| Emrax-228                                        | 6-8              | 0.9 @ 7 LPM                              | 0.0184                            | 50                              | 2                  |
| CM200                                            | >12              | 0.3 @ 12 LPM                             | 0.00208                           | 80                              | 3                  |
| Radiator                                         | -                | 0.0125 @ 8 LPM (dual pass, single loop)  | 0.000195 (dual pass, single loop) | -                               | -                  |
| Radiator and other components (hose, barbs, etc) | -                | 0.61 @ 14.8 LPM (dual pass, single loop) | 0.00195 (single pass, dual loop)  | -                               | -                  |

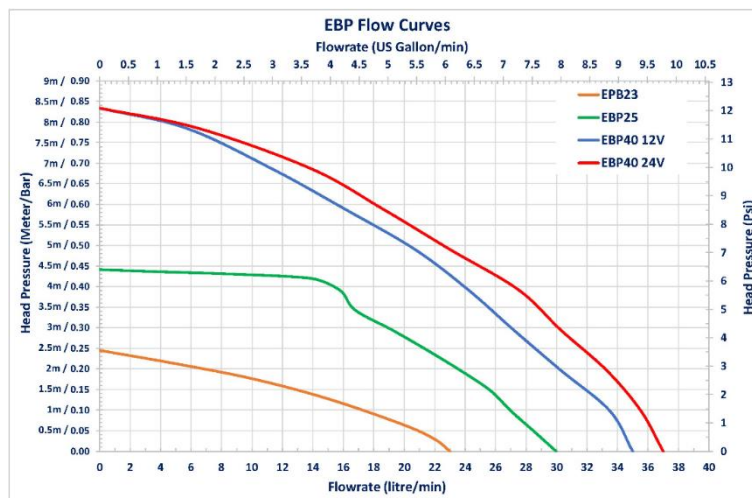


Figure 2: EBP40 Flow Curve (Davies Craig, 2024)

| suction / aspirante                                                |                                                |                                                 |                                  |
|--------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------|----------------------------------|
| Static pressure<br><i>Pressione statica</i><br>mm H <sub>2</sub> O | Airflow<br><i>Portata</i><br>m <sup>3</sup> /h | Current input<br><i>Corrente assorbita</i><br>A | Airflow<br><i>Portata</i><br>CFM |
| 0                                                                  | 730                                            | 6,9                                             | 431                              |
| 5                                                                  | 650                                            | 7,1                                             | 384                              |
| 7,5                                                                | 610                                            | 7,2                                             | 360                              |
| 10                                                                 | 570                                            | 7,3                                             | 336                              |
| 12,5                                                               | 510                                            | 7,2                                             | 301                              |
| 15                                                                 | 450                                            | 7,3                                             | 266                              |
| 20                                                                 | 330                                            | 7,4                                             | 195                              |
| 25                                                                 | 240                                            | 7,8                                             | 142                              |
| 35                                                                 | 0                                              | 8,4                                             | 0                                |

Figure 3: SPAL Fan Specs (SPAL Automotive, 2023)

Table 2: NU23 Radiator Dimensions

| Radiator properties | Value         |
|---------------------|---------------|
| Stack Height        | 328 mm        |
| Mean Channel Length | 175 mm        |
| Core Thickness      | 26 mm         |
| Fin type            | 4.85 mm R-fin |
| Fin density         | 16 FPI        |

## 2 Initial Testing and Research

For the scope of this project, the flow rates were to be validated for NU23, research on pumps to see if there are better options for NU24, investigate a new radiator to be used and determine if there are any other ways to set up the cooling system.

### 2.1 Flow Rates and Pressures

Due to the last-minute inverter change last year, it was pivotal to verify the previous year's results concerning flow rates, pressures, resistances, etc. to assist with future improvements and calculations for NU24. This was accomplished by doing calculations using the current pumps' flow curve versus system curves (Michael Smith Engineers, 2024) and measuring the flow through each cooling loop on NU23 using a flow rate meter shown in figure 4. To read the results from the flow meter a teensy with Arduino code (Willtronics, 2024) was to be used, the code is provided in Appendix A.



Figure 4: Flow Meter

The calculations were done by finding the flow rates operating point of each loop, this was done by using the formula shown.

$$\Delta p = Q^2 R$$

Where  $\Delta p$  is the pressure loss,  $Q$  is the flow rate and  $R$  is the resistance.

Using this formula pressure losses were found at each flow rate interval to plot a line on the EBP40 12V flow curve. The point at which these two curves intersect would be the flow rate which we should see in each loop.

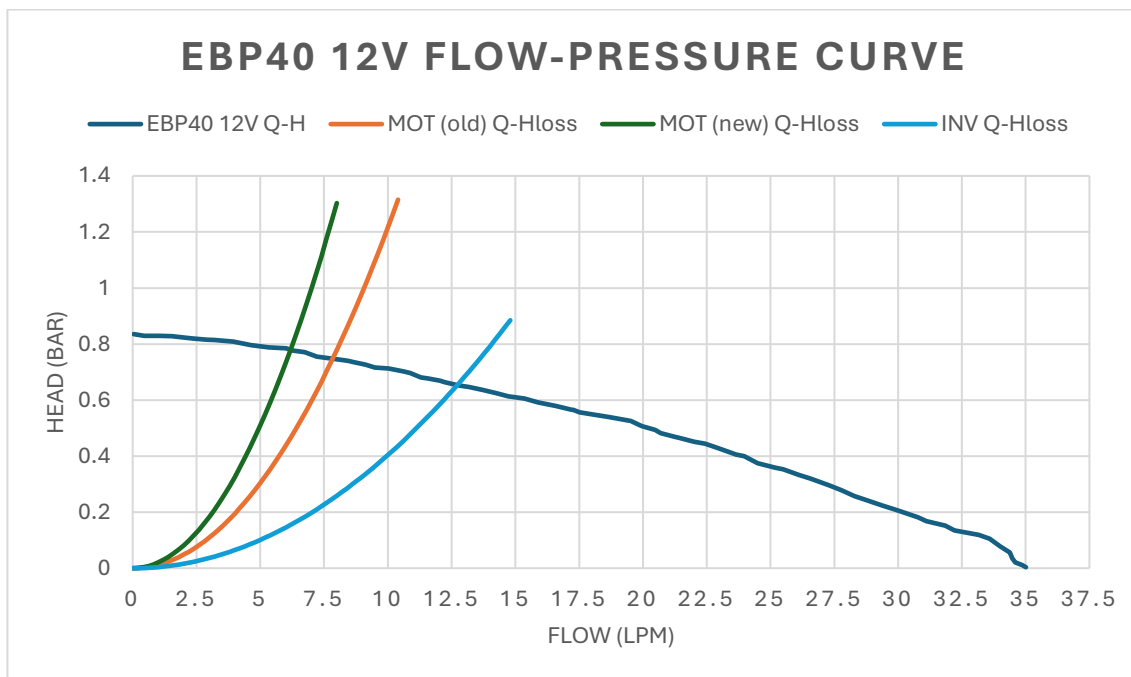


Figure 5: EBP40 Flow Curve vs System Curves

From Figure 5, the flowrate for the motor controller loop would be roughly 12.6 LPM, the NU23 motor loop was 7.8 LPM and the new motor for NU24 will expect 6.2 LPM.

To use the flow meter on NU23, the flow meter had to be validated. This was done by using a stopwatch to time the flow from a hose through the flow meter and into a bucket which was being weighed with a scale to measure the liters. The first tests results showed that the flow meter was measuring half of the flow, this was due to the code using 'RISING' which only records data when the pin goes from low to high instead of 'CHANGE' which records data whenever the pin changes value. A second test using 'CHANGE' showed accurate results thus making it able to be used on NU23.



Figure 6: Flow Meter Validation

The flow meter was then connected to the motor controller loop and read 12.6 LPM of flow. Therefore, previous calculations were correct, meaning the NU23 motor loop did not need to be measured. The NU24 motor loop was tested due to the closeness of expected flow to minimal required flow for the motor except the flowmeter was shown to be measuring around sixty percent of the actual flow.



Figure 7: Flow Meter on Motor Controller Loop

## 2.2 Cooling Loop Setups

Cooling setups other than the current setup which was used on NU23 were explored to try and find a more efficient setup with less components. Knowing that we need a minimum of 6 LPM for the motor and 12 LPM for the motor controller for these systems to be considered safe according to their data sheets, therefore if we were to use a single loop system the minimum would need to be 12 LPM total therefore we can calculate the pressure loss for the system.

$$\Delta p = 12^2 \cdot 0.0194$$

$$\Delta p = 2.794 \text{ bar}$$

Thus, the head pressure for this system would have to equal that of this calculated pressure loss or be larger but the max inlet pressure for the motor is 2 bar meaning a single loop setup would be unsafe for NU24s powertrain risking harm to some of the most expensive components on the car.

This concluded that the only setup option for the new car was to use the same dual-loop configuration as what was put on NU23 due to the high resistance of the motor mixed with the high flow rate required for the motor controller.

## 2.3 Pumps

The current pumps which are on NU23 were shown to perform well and produce flow rates within range for the new motor and motor controller except they are on the lower end of the limits, therefore it was feasible to scope out more powerful pumps to keep both these systems safer.

A more powerful pump to be used would have to deliver at least 7 LPM at 0.9 bar of head to allow the motor loop to be at least in the middle of the desired flowrate, the pump would also have to have a current draw of around 6 Amps and a voltage of 12 V. There was only a limited selection of pumps which were readily available and able to produce this amount of flow at the large head pressure by themselves; Bosch PCE-XL, Pierburg CWA100-2, CWA100-3 and CWA150 (Tecomotive, 2024). These pumps were quite powerful but at this power the price was quite high. A comparison of the data for these pumps and the current pump can be shown below.

Table 3: Pump Comparison

| Pumps                  | Price (AUD) + Shipping | Head @ 7LPM (bar) | Max Current (A) | Voltage (V) | Mass (kg) |
|------------------------|------------------------|-------------------|-----------------|-------------|-----------|
| Davies-Craig 12V EBP40 | ~250                   | 0.75              | 4.6             | 12          | 0.57      |
| Bosch PCE-XL           | ~458                   | 0.96              | 8.5             | 12          | 0.66      |
| Pierburg CWA100-2      | ~513                   | 1                 | 8.5             | 8-16        | 1         |
| Pierburg CWA100-3      | ~472                   | 1.11              | 8.5             | 8-16        | 1         |
| Pierburg CWA150        | ~404                   | 1.7               | 15              | 9-16        | 1         |

Each of these pumps should be able to be used at around 6 Amps. All pumps with a maximum of 8.5 Amps were assumed to perform the same as the Bosch PCE-XLs current curve shown in figure 8 due to Pierburg pumps not having a current curve. The CWA150 was unknown how it would perform at 6 Amps. Table 3 showed that the prices for a more powerful pump were at a minimum of a \$154 AUD increase compared to that of the EBP40s. Therefore, a new pump was only to be ordered if it was truly necessary.



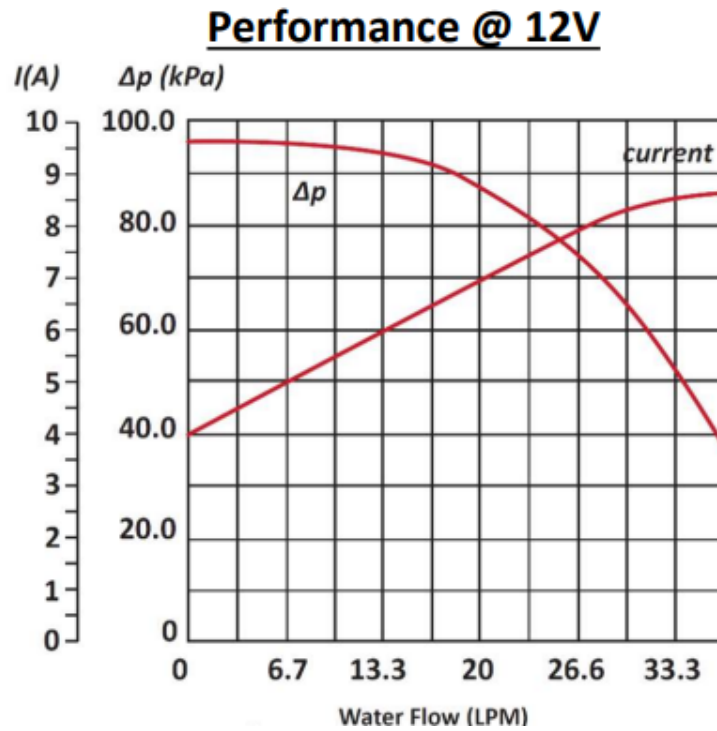


Figure 8: PCE-XL Flow Curve (Cascadia Motion, 2024)

## 2.4 Initial Radiator Sizing

The initial plan for the radiator was to purchase a new radiator with the face area of its core to be roughly the size of the fan which has a diameter of 190 mm as this will increase the area of air flow and in turn improve cooling.

A radiator with the core sizing of 200x200 mm was to be explored if it would be efficient, the air flow area will be close to maximum but as the core size reduces for a radiator the cooling capacity does go down. The current radiator also only has a split at the bottom tank which means mixing at the top tank is possible, a complete split down the middle or separate radiators was explored as well.

The split position was also explored whether this should change using the effectiveness-NTU method (see Appendix B) using the following values from the current radiator in Table 4.

Table 4: Radiator Properties (Symons, 2023)

| Property                                                                    | Symbol           | Value                                                     |
|-----------------------------------------------------------------------------|------------------|-----------------------------------------------------------|
| Radiator Inlet Water Temperature                                            | $T_{Hot\ Water}$ | Variable Input °C                                         |
| Ambient Air Temperature                                                     | $T_{Cold\ Air}$  | Variable Input °C                                         |
| Specific Heat Capacity of Water                                             | $C_{p\ Water}$   | Changes due to temperature $\frac{J}{kg \cdot K}$         |
| Specific Heat Capacity of Air                                               | $C_{p\ Air}$     | Changes due to temperature $\frac{J}{kg \cdot K}$         |
| Air Velocity Entering Radiator                                              | $V_{Air}$        | 4.5 m/s                                                   |
| Air Density                                                                 | $\rho_{Air}$     | Changes due to temperature $\frac{kg}{m^3}$               |
| Water Density                                                               | $\rho_{Water}$   | Changes due to temperature $\frac{kg}{m^3}$               |
| Volumetric Flowrate of Water                                                | $Q_{Water}$      | 6 LPM for Motor Side and 12 LPM for Motor Controller side |
| Nusselt Number for Convection with a Uniform Wall Temperature Approximation | $Nu_{Water}$     | 3.66                                                      |
| Radiator Tube Hydraulic Diameter                                            | $D_{Hydraulic}$  | 1.4 mm                                                    |
| Number of Radiator Fins                                                     | N                | 5610 (110x51)                                             |
| Radiator Fin Surface Area                                                   | $A_f$            | $1.75\ m^2$                                               |
| Total Radiator Surface Area                                                 | $A_t$            | $2.22\ m^2$                                               |
| Fin Length                                                                  | $F_L$            | 0.003 m                                                   |
| Fin Thickness                                                               | t                | 0.0002 m                                                  |
| Thermal Conductivity of Aluminium                                           | $k_f$            | $237\ \frac{W}{m \cdot K}$                                |
| Thermal Conductivity of Air                                                 | $k_{Air}$        | Changes due to temperature $\frac{W}{m \cdot K}$          |
| Thermal Conductivity of Water                                               | $k_{Water}$      | Changes due to temperature $\frac{W}{m \cdot K}$          |

This analysis concluded that a top split is recommended rather than at the bottom due to the outlet temperatures being slightly better with the top split having an average outlet temperature roughly 0.3 °C below that of the bottom split.

Therefore, communications over email with PWR about custom manufacturing began, to get their input and capabilities of a radiator core at the scoped size to eventually come to a solution and thus a new radiator.

### 3 New Radiator

The proposed new radiator will be set up for two separate cooling loops much like the final setup of the radiator on NU23. Following the initial sizing of a new radiator communications with PWR were conducted to go through all edits to the radiator core to maximize the cooling capabilities of the system accounting for appropriate pressure drops, flowrates and resistances. For reference, the data for a 328x175x26 mm 16FPI NU23 radiator is shown below in figure 9 and 10, this data is assuming a complete split therefore we can assume the cooling capacity is slightly less.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 0.76                               | 76                             | 3.00                      | 0.102                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 49.4                          | 6                               | 1.59                           | 2.081                                          | 0.060              | 12.4           |
| 26.0                          | 0.77                               | 77                             | 3.00                      | 0.102                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.4                          | 5                               | 2.21                           | 2.078                                          | 0.060              | 12.5           |
| 35.0                          | 0.74                               | 74                             | 3.00                      | 0.099                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 50.5                          | 6                               | 1.04                           | 2.041                                          | 0.059              | 12.2           |
| 35.0                          | 0.75                               | 75                             | 3.00                      | 0.099                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.6                          | 5                               | 1.64                           | 2.038                                          | 0.058              | 12.2           |
| 26.0                          | 1.49                               | 149                            | 4.50                      | 0.153                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 48.4                          | 6                               | 2.07                           | 2.696                                          | 0.077              | 16.1           |
| 26.0                          | 1.50                               | 150                            | 4.50                      | 0.153                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 57.0                          | 5                               | 2.86                           | 2.691                                          | 0.077              | 16.1           |
| 35.00                         | 1.45                               | 145                            | 4.50                      | 0.148                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 49.9                          | 6                               | 1.34                           | 2.646                                          | 0.076              | 15.8           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.148                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 58.6                          | 5                               | 2.12                           | 2.642                                          | 0.076              | 15.8           |
| 26.0                          | 2.45                               | 245                            | 6.00                      | 0.204                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 47.6                          | 6                               | 2.47                           | 3.223                                          | 0.092              | 19.2           |
| 26.0                          | 2.47                               | 247                            | 6.00                      | 0.204                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 55.9                          | 5                               | 3.42                           | 3.217                                          | 0.092              | 19.3           |
| 35.0                          | 2.39                               | 239                            | 6.00                      | 0.198                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 49.4                          | 6                               | 1.61                           | 3.164                                          | 0.091              | 18.9           |
| 35.0                          | 2.41                               | 241                            | 6.00                      | 0.198                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 57.7                          | 5                               | 2.54                           | 3.158                                          | 0.091              | 18.9           |

Figure 9: 164x175x26 mm 16 FPI Single Pass Core Data (Motor Side)

|      |      |     |      |       |      |      |      |      |      |    |      |        |        |      |
|------|------|-----|------|-------|------|------|------|------|------|----|------|--------|--------|------|
| 26.0 | 0.74 | 74  | 3.00 | 0.102 | 1.18 | 13.0 | 0.22 | 34.0 | 33.4 | 17 | 0.54 | 2.366  | 0.068  | 7.5  |
| 26.0 | 0.76 | 76  | 3.00 | 0.102 | 1.18 | 13.0 | 0.22 | 46.0 | 44.5 | 14 | 1.36 | 2.364  | 0.068  | 7.6  |
| 35.0 | 0.73 | 73  | 3.00 | 0.099 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 17 | 0.07 | -2.315 | -0.066 | 7.4  |
| 35.0 | 0.74 | 74  | 3.00 | 0.099 | 1.15 | 13.0 | 0.22 | 46.0 | 45.2 | 14 | 0.73 | 2.314  | 0.066  | 7.4  |
| 26.0 | 1.46 | 146 | 4.50 | 0.153 | 1.18 | 13.0 | 0.22 | 34.0 | 33.2 | 17 | 0.72 | 3.140  | 0.090  | 10.0 |
| 26.0 | 1.48 | 148 | 4.50 | 0.153 | 1.18 | 13.0 | 0.22 | 46.0 | 44.0 | 14 | 1.80 | 3.138  | 0.090  | 10.0 |
| 35.0 | 1.43 | 143 | 4.50 | 0.148 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 17 | 0.09 | -3.078 | -0.088 | 9.8  |
| 35.0 | 1.44 | 144 | 4.50 | 0.148 | 1.15 | 13.0 | 0.22 | 46.0 | 44.9 | 14 | 0.97 | 3.075  | 0.088  | 9.8  |
| 26.0 | 2.42 | 242 | 6.00 | 0.204 | 1.18 | 13.0 | 0.22 | 34.0 | 33.0 | 17 | 0.87 | 3.800  | 0.109  | 12.1 |
| 26.0 | 2.45 | 245 | 6.00 | 0.204 | 1.18 | 13.0 | 0.22 | 46.0 | 43.6 | 14 | 2.18 | 3.797  | 0.109  | 12.2 |
| 35.0 | 2.35 | 235 | 6.00 | 0.198 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 17 | 0.11 | -3.727 | -0.107 | 11.9 |

Figure 10: 164x175x26 mm 16 FPI Single Pass Core Data (Inverter Side)

#### 3.1 Radiator Configurations

Initially it was thought that we could use a core at 200x175x26 mm, but PWR strongly suggested that at this size the cooling would be decreased too much to the point of extra air flow wouldn't be beneficial, shown in figure 11 and 12, therefore, sizes up to 300x300 mm were shown by PWR, although anything larger in size then the NU23 radiator size would not be considered.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.51                               | 151                            | 4.50                      | 0.093                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.9                          | 9.8                             | 1.98                           | 3.056                                          | 0.053              | 11.2           |
| 30.0                          | 1.49                               | 149                            | 4.50                      | 0.092                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 59.3                          | 9.8                             | 1.75                           | 3.029                                          | 0.053              | 11.1           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.090                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.9                          | 9.8                             | 1.47                           | 2.996                                          | 0.052              | 11.0           |

Figure 11: 100x175x26 mm 16 FPI Single Pass Core Data (Motor Side)

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.47                               | 147                            | 4.50                      | 0.093                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.5                          | 31.3                            | 0.47                           | 3.382                                          | 0.059              | 6.6            |
| 30.0                          | 1.45                               | 145                            | 4.50                      | 0.092                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.7                          | 31.3                            | 0.23                           | 3.350                                          | 0.059              | 6.5            |
| 35.0                          | 1.43                               | 143                            | 4.50                      | 0.090                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 31.2                            | 0.06                           | -3.311                                         | -0.058             | 6.4            |

Figure 12: 100x175x26 16FPI Single Pass Core Data (Inverter Side)

Increasing the height was explored to allow for a better fitting for the 7.5 inch fan and increase in filler point height to allow for a better bleeding position, with sizes of 200x200 mm (figures 13 and 14) and 250x200 mm, (figures 15 and 16) were investigated, the 200x200 mm showed better results than the 200x175 but still was lacking in cooling capacity, the 250x200 mm showed much better results, therefore this size was chosen and edits using this size were made. These edits include asymmetrical cores, core thickness, fin density and dual pass configuration.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.51                               | 151                            | 4.50                      | 0.107                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.3                          | 11.2                            | 2.24                           | 3.032                                          | 0.061              | 12.7           |
| 30.0                          | 1.49                               | 149                            | 4.50                      | 0.105                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 58.9                          | 11.2                            | 1.98                           | 3.006                                          | 0.060              | 12.6           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.103                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.5                          | 11.2                            | 1.67                           | 2.973                                          | 0.059              | 12.4           |

Figure 13: 100x200x26 mm 16 FPI Single Pass Core Data (Motor Side)

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.47                               | 147                            | 4.50                      | 0.107                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.4                          | 35.8                            | 0.54                           | 3.367                                          | 0.067              | 7.5            |
| 30.0                          | 1.45                               | 145                            | 4.50                      | 0.105                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.7                          | 35.7                            | 0.27                           | 3.336                                          | 0.067              | 7.4            |
| 35.0                          | 1.43                               | 143                            | 4.50                      | 0.103                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 35.6                            | 0.07                           | -3.297                                         | -0.066             | 7.3            |

Figure 14: 100x200x26 mm 16FPI Single Pass Core Data (Inverter Side)

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.51                               | 151                            | 4.50                      | 0.133                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 57.4                          | 8.5                             | 2.66                           | 2.875                                          | 0.072              | 15.0           |
| 30.0                          | 1.49                               | 149                            | 4.50                      | 0.131                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 58.1                          | 8.5                             | 2.35                           | 2.851                                          | 0.071              | 14.9           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.129                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 58.9                          | 8.4                             | 1.97                           | 2.821                                          | 0.071              | 14.7           |

Figure 15: 125x200x26 mm 16 FPI Single Pass Core Data (Motor Side)

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 1.46                               | 146                            | 4.50                      | 0.133                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.3                          | 26.8                            | 0.65                           | 3.267                                          | 0.082              | 9.1            |
| 30.0                          | 1.45                               | 145                            | 4.50                      | 0.131                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.6                          | 26.7                            | 0.32                           | 3.237                                          | 0.081              | 9.0            |
| 35.0                          | 1.43                               | 143                            | 4.50                      | 0.129                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 26.6                            | 0.08                           | -3.201                                         | -0.080             | 8.9            |

Figure 16: 125x200x26 16 FPI Single Pass Core Data (Inverter Side)

Having asymmetrical lengths for the radiator core was explored to see how this would affect the cooling of either side. This was a valid configuration which allowed more balanced cooling between both sides of the core but didn't change all that much and for simplicity the core was decided to be symmetrical. Data is shown in figure 17 and 18 below.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26                            | 0.99                               | 99                             | 3.50                      | 0.083                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 59.0                          | 11                              | 1.90                           | 2.561                                          | 0.051              | 10.7           |
| 30.0                          | 0.98                               | 98                             | 3.50                      | 0.082                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 59.5                          | 11                              | 1.67                           | 2.538                                          | 0.051              | 10.6           |
| 35.0                          | 0.97                               | 97                             | 3.50                      | 0.080                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 60.1                          | 11                              | 1.41                           | 2.509                                          | 0.050              | 10.5           |
| 26.0                          | 1.51                               | 151                            | 4.50                      | 0.107                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.3                          | 11                              | 2.24                           | 3.032                                          | 0.061              | 12.7           |
| 30.0                          | 1.49                               | 149                            | 4.50                      | 0.105                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 58.9                          | 11                              | 1.98                           | 3.006                                          | 0.060              | 12.6           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.103                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.5                          | 11                              | 1.67                           | 2.973                                          | 0.059              | 12.4           |
| 26.0                          | 2.13                               | 213                            | 5.50                      | 0.130                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 57.7                          | 11                              | 2.56                           | 3.455                                          | 0.069              | 14.4           |
| 30.0                          | 2.11                               | 211                            | 5.50                      | 0.128                         | 1.17                  | 7.0                          | 0.11                    | 63.0                         | 58.3                          | 11                              | 2.26                           | 3.426                                          | 0.069              | 14.3           |
| 35.0                          | 2.08                               | 208                            | 5.50                      | 0.126                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.0                          | 11                              | 1.90                           | 3.390                                          | 0.068              | 14.2           |

Figure 17: 100x200x26 mm 16 FPI Single Pass Core Data (Motor Side)

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26                            | 0.96                               | 96                             | 3.50                      | 0.124                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.3                          | 21                              | 0.64                           | 2.665                                          | 0.080              | 8.9            |
| 30.0                          | 0.95                               | 95                             | 3.50                      | 0.123                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.6                          | 21                              | 0.32                           | 2.640                                          | 0.079              | 8.8            |
| 35.0                          | 0.93                               | 93                             | 3.50                      | 0.121                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 21                              | 0.08                           | -2.610                                         | -0.078             | 8.7            |
| 26.0                          | 1.46                               | 146                            | 4.50                      | 0.160                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.2                          | 21                              | 0.76                           | 3.171                                          | 0.095              | 10.6           |
| 30.0                          | 1.45                               | 145                            | 4.50                      | 0.158                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.6                          | 21                              | 0.38                           | 3.143                                          | 0.094              | 10.5           |
| 35.0                          | 1.43                               | 143                            | 4.50                      | 0.155                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 21                              | 0.09                           | -3.108                                         | -0.093             | 10.3           |
| 26.0                          | 2.07                               | 207                            | 5.50                      | 0.195                         | 1.18                  | 13.0                         | 0.22                    | 34.0                         | 33.0                          | 21                              | 0.87                           | 3.625                                          | 0.109              | 12.1           |
| 30.0                          | 2.05                               | 205                            | 5.50                      | 0.193                         | 1.17                  | 13.0                         | 0.22                    | 34.0                         | 33.5                          | 21                              | 0.43                           | 3.593                                          | 0.108              | 12.0           |
| 35.0                          | 2.02                               | 202                            | 5.50                      | 0.190                         | 1.15                  | 13.0                         | 0.22                    | 34.0                         | 34.1                          | 21                              | 0.11                           | -3.554                                         | -0.107             | 11.8           |

Figure 18: 150x200x26 mm 16FPI Single Pass Core Data (Inverter Side)

On closer inspection to the chassis, mounts to be used and where the jacking point was supposed to be for NU24 the minimum length of the core was changed to 260 mm to allow enough space for the jacking point to be added to the radiator brackets, accounting for the rules associated, indicated in Appendix C. This allowed for the rest of the changes to the radiator data to be in accordance with this length.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 0.76                               | 76                             | 3.00                      | 0.092                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 49.5                          | 9                               | 1.51                           | 2.184                                          | 0.057              | 11.8           |
| 26.0                          | 0.77                               | 77                             | 3.00                      | 0.092                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.6                          | 8                               | 2.10                           | 2.182                                          | 0.057              | 11.8           |
| 35.0                          | 0.74                               | 74                             | 3.00                      | 0.090                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 50.6                          | 9                               | 0.98                           | 2.141                                          | 0.056              | 11.6           |
| 35.0                          | 0.75                               | 75                             | 3.00                      | 0.090                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.7                          | 8                               | 1.56                           | 2.138                                          | 0.056              | 11.6           |
| 26.0                          | 1.49                               | 149                            | 4.50                      | 0.138                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 48.5                          | 9                               | 1.97                           | 2.850                                          | 0.074              | 15.4           |
| 26.0                          | 1.51                               | 151                            | 4.50                      | 0.138                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 57.3                          | 8                               | 2.74                           | 2.846                                          | 0.074              | 15.5           |
| 35.00                         | 1.45                               | 145                            | 4.50                      | 0.134                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 50.0                          | 9                               | 1.28                           | 2.797                                          | 0.073              | 15.1           |
| 35.0                          | 1.47                               | 147                            | 4.50                      | 0.134                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 58.8                          | 8                               | 2.03                           | 2.793                                          | 0.073              | 15.2           |

Figure 19: 130x100x26 mm 16 FPI Single Pass Core Data (Motor Side)

|      |      |     |      |       |      |      |      |      |      |    |      |        |        |     |
|------|------|-----|------|-------|------|------|------|------|------|----|------|--------|--------|-----|
| 26.0 | 0.74 | 74  | 3.00 | 0.092 | 1.18 | 13.0 | 0.22 | 34.0 | 33.4 | 25 | 0.51 | 2.433  | 0.063  | 7.0 |
| 26.0 | 0.76 | 76  | 3.00 | 0.092 | 1.18 | 13.0 | 0.22 | 46.0 | 44.6 | 22 | 1.26 | 2.431  | 0.063  | 7.0 |
| 35.0 | 0.73 | 73  | 3.00 | 0.090 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 25 | 0.06 | -2.380 | -0.062 | 6.9 |
| 35.0 | 0.74 | 74  | 3.00 | 0.090 | 1.15 | 13.0 | 0.22 | 46.0 | 45.2 | 22 | 0.68 | 2.378  | 0.062  | 6.9 |
| 26.0 | 1.46 | 146 | 4.50 | 0.138 | 1.18 | 13.0 | 0.22 | 34.0 | 33.3 | 26 | 0.68 | 3.248  | 0.084  | 9.4 |
| 26.0 | 1.49 | 149 | 4.50 | 0.138 | 1.18 | 13.0 | 0.22 | 46.0 | 44.1 | 22 | 1.69 | 3.245  | 0.084  | 9.4 |
| 35.0 | 1.43 | 143 | 4.50 | 0.134 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 25 | 0.08 | -3.182 | -0.083 | 9.2 |
| 35.0 | 1.45 | 145 | 4.50 | 0.134 | 1.15 | 13.0 | 0.22 | 46.0 | 45.0 | 22 | 0.91 | 3.180  | 0.083  | 9.2 |

Figure 20: 130x100x26 mm 16 FPI Single Pass Core Data (Inverter Side)

The core thickness was considered as well. A thicker core of 32 mm compared to 26 mm shown in figure 21 and 22 showed lower water side pressure drops and increase in air side pressure drop therefore, for the increase in size and weight and worse air flow it was not worth the slightly better water flow. A lower core thickness wasn't explored as the water side pressure drops were a problem as we needed as little as possible to allow for safe operating flow for both sides of the core.

| Measured Cold Side Parameters |                                    |                                |                           |                               |                       | Measured Hot Side Parameters |                         |                              |                               |                                 | Calculated Hot Side Parameters |                                                |                    |                |
|-------------------------------|------------------------------------|--------------------------------|---------------------------|-------------------------------|-----------------------|------------------------------|-------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|------------------------------------------------|--------------------|----------------|
| Inlet Temperature (°C)        | Inlet Pressure At Core Face (mbar) | Pressure Drop Across Core (Pa) | Inlet Face Velocity (m/s) | Inlet Mass Flow Rate (kg/sec) | Inlet Density (kg/m³) | Flow Rate (L/min)            | Mass Flow Rate (kg/sec) | Water Inlet Temperature (°C) | Water Outlet Temperature (°C) | Water Side Pressure Drop (mbar) | Hot Side Heat Rejection (kW)   | Av. Normalised Heat Dissipation (kW/m²/°C ETD) | Heat Duty (kW/ETD) | Efficiency (%) |
| 26.0                          | 0.94                               | 94                             | 3.00                      | 0.092                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 49.3                          | 7                               | 1.63                           | 2.348                                          | 0.061              | 12.6           |
| 26.0                          | 0.95                               | 95                             | 3.00                      | 0.092                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 58.3                          | 6                               | 2.26                           | 2.345                                          | 0.061              | 12.7           |
| 35.0                          | 0.91                               | 91                             | 3.00                      | 0.090                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 50.5                          | 7                               | 1.06                           | 2.300                                          | 0.060              | 12.4           |
| 35.0                          | 0.93                               | 93                             | 3.00                      | 0.090                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 59.5                          | 6                               | 1.67                           | 2.297                                          | 0.060              | 12.4           |
| 26.0                          | 1.84                               | 184                            | 4.50                      | 0.138                         | 1.18                  | 7.0                          | 0.12                    | 52.7                         | 48.2                          | 7                               | 2.13                           | 3.079                                          | 0.080              | 16.6           |
| 26.0                          | 1.86                               | 186                            | 4.50                      | 0.138                         | 1.18                  | 7.0                          | 0.11                    | 63.0                         | 56.9                          | 6                               | 2.96                           | 3.074                                          | 0.080              | 16.6           |
| 35.00                         | 1.79                               | 179                            | 4.50                      | 0.134                         | 1.15                  | 7.0                          | 0.12                    | 52.7                         | 49.8                          | 7                               | 1.39                           | 3.020                                          | 0.079              | 16.2           |
| 35.0                          | 1.81                               | 181                            | 4.50                      | 0.134                         | 1.15                  | 7.0                          | 0.11                    | 63.0                         | 58.4                          | 6                               | 2.20                           | 3.015                                          | 0.078              | 16.3           |

Figure 21: 130x100x32 mm 16 FPI Single Pass Core Data (Motor Side)

|      |      |     |      |       |      |      |      |      |      |    |      |        |        |      |
|------|------|-----|------|-------|------|------|------|------|------|----|------|--------|--------|------|
| 26.0 | 0.92 | 92  | 3.00 | 0.092 | 1.18 | 13.0 | 0.22 | 34.0 | 33.4 | 19 | 0.55 | 2.630  | 0.068  | 7.6  |
| 26.0 | 0.93 | 93  | 3.00 | 0.092 | 1.18 | 13.0 | 0.21 | 46.0 | 44.5 | 16 | 1.37 | 2.628  | 0.068  | 7.6  |
| 35.0 | 0.89 | 89  | 3.00 | 0.090 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 19 | 0.07 | -2.571 | -0.067 | 7.4  |
| 35.0 | 0.91 | 91  | 3.00 | 0.090 | 1.15 | 13.0 | 0.21 | 46.0 | 45.2 | 16 | 0.73 | 2.569  | 0.067  | 7.4  |
| 26.0 | 1.80 | 180 | 4.50 | 0.138 | 1.18 | 13.0 | 0.22 | 34.0 | 33.2 | 19 | 0.74 | 3.540  | 0.092  | 10.2 |
| 26.0 | 1.83 | 183 | 4.50 | 0.138 | 1.18 | 13.0 | 0.21 | 46.0 | 44.0 | 16 | 1.84 | 3.537  | 0.092  | 10.2 |
| 35.0 | 1.75 | 175 | 4.50 | 0.134 | 1.15 | 13.0 | 0.22 | 34.0 | 34.1 | 19 | 0.09 | -3.467 | -0.090 | 10.0 |
| 35.0 | 1.78 | 178 | 4.50 | 0.134 | 1.15 | 13.0 | 0.21 | 46.0 | 44.9 | 16 | 0.99 | 3.464  | 0.090  | 10.0 |

Figure 22: 130x100x26 mm 16 FPI Single Pass Core Data (Inverter Side)

The possibility of a dual-pass configuration was also explored. Dual-pass configuration shown in figure 23 showed better cooling overall, but the water side pressure drops associated with this configuration were quite

high compared to that of the current radiator therefore it was decided to keep this in mind for calculations and later choice. A different fin density was also taken into consideration, therefore a 25 FPI radiator was scoped as well with both single and dual pass configurations. Relevant data for these options can be shown in the comparison on table 5.

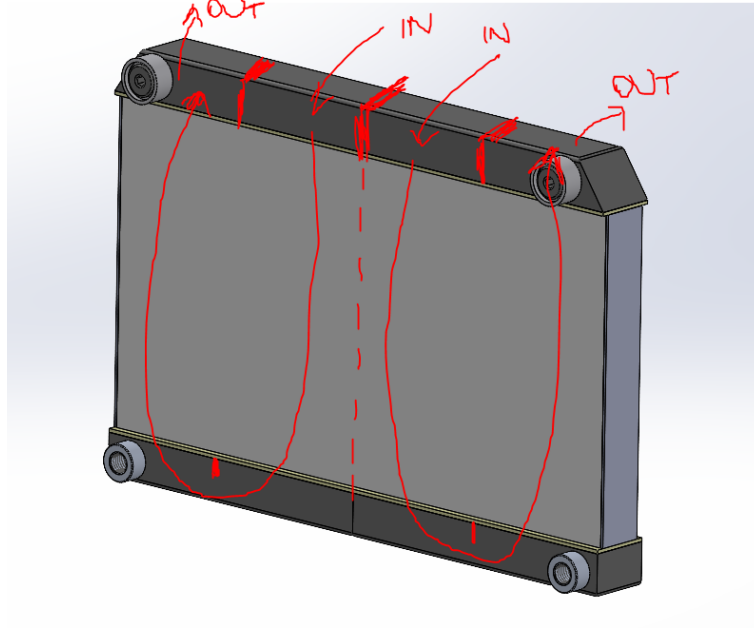


Figure 23: Dual-Pass Radiator Setup

### 3.2 Radiator Sizing and Configuration Decision

This concluded what could be edited on the radiator core giving four options to choose from all being 260x200x26 mm, these are a single pass 16 FPI core, a dual pass 16 FPI core, a single pass 25 FPI core or a dual pass 25 FPI core. Temperatures, pressure drops and efficiencies associated with these options is shown in table 5 as well as for the current radiator shown in table 5 for comparison.

Table 5: PWR Radiator Comparison

| <b>Radiator</b>                                   | <b>Total Air Side Pressure Drop @ 4.5 m/s Flow (Pa)</b> | <b>Motor Water Side Pressure Drop @ 7 LPM Flow (bar)</b> | <b>Inverter Water Side Pressure Drop @ 13 LPM Flow (bar)</b> | <b>Motor Inlet Temperature @ Air temp of 35°C, Air Flow 4.5 m/s (°C)</b> | <b>Motor Outlet Temperature @ Air temp of 35°C, Air Flow 4.5 m/s (°C)</b> | <b>Inverter Inlet Temperature @ Air temp of 35°C, Air Flow 4.5 m/s (°C)</b> | <b>Inverter Outlet Temperature @ Air temp of 35°C, Air Flow 4.5 m/s (°C)</b> | <b>Total Efficiency (%)</b> |
|---------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------|
| <b>Current Single Pass 16 FPI (328x175x26 mm)</b> | 146                                                     | 0.006                                                    | 0.014                                                        | 52.7                                                                     | 49.9                                                                      | 46                                                                          | 44.9                                                                         | 25.6                        |
| <b>Single Pass 16 FPI (260x200x26 mm)</b>         | 146                                                     | 0.009                                                    | 0.022                                                        | 52.7                                                                     | 50                                                                        | 46                                                                          | 45                                                                           | 24.3                        |
| <b>Dual Pass 16 FPI (260x200x26 mm)</b>           | 146                                                     | 0.043                                                    | 0.111                                                        | 52.7                                                                     | 49.8                                                                      | 46                                                                          | 44.9                                                                         | 26.3                        |
| <b>Single Pass 25 FPI (260x200x26 mm)</b>         | 240                                                     | 0.018                                                    | 0.044                                                        | 52.7                                                                     | 49.3                                                                      | 46                                                                          | 44.8                                                                         | 30.2                        |
| <b>Dual Pass 25 FPI (260x200x26 mm)</b>           | 240                                                     | 0.093                                                    | 0.254                                                        | 52.7                                                                     | 49.2                                                                      | 46                                                                          | 44.7                                                                         | 31.5                        |

Upon reviewing the values, we can cancel out using the single pass 16 FPI option as it has less cooling capacity than all other radiators shown. More calculations had to be done for the rest of the options to decide whether the air flow and water flow are viable using them in the cooling loops using operating points on both the EBP40 flow curve and the 7.5-inch SPAL fan flow curve.

Firstly, resistances were to be calculated, the resistance of the tubes and attachments used was estimated using the resistance of the current radiator with the tubes and attachments then subtracting a percentage of the



resistance for the current radiator measured when it was a dual pass radiator, 70 % was used as the distance water was to move was cut when using two separate loops in the radiator. This gave a resistance of 0.001823 which was added onto the radiator and the motor or motor controller to calculate the total resistance in each loop for each radiator option. The total resistances are shown below in table 6.

Table 6: PWR Radiator Resistance Comparison

| Radiator Option    | Motor Side Water Flow Resistance | Motor Controller Side Water Flow Resistance | Air Flow Resistance |
|--------------------|----------------------------------|---------------------------------------------|---------------------|
| Dual Pass 16 FPI   | 0.02110                          | 0.004560                                    | 7.160               |
| Single Pass 25 FPI | 0.02059                          | 0.004163                                    | 11.852              |
| Dual Pass 25 FPI   | 0.02212                          | 0.005406                                    | 11.852              |

Using these resistances we can plot the flow-pressure drop curve to the EBP40 flow curve and the SPAL fan flow curve accordingly for each of these options. These can be shown below in Figures 10, 11, 12 and 13.

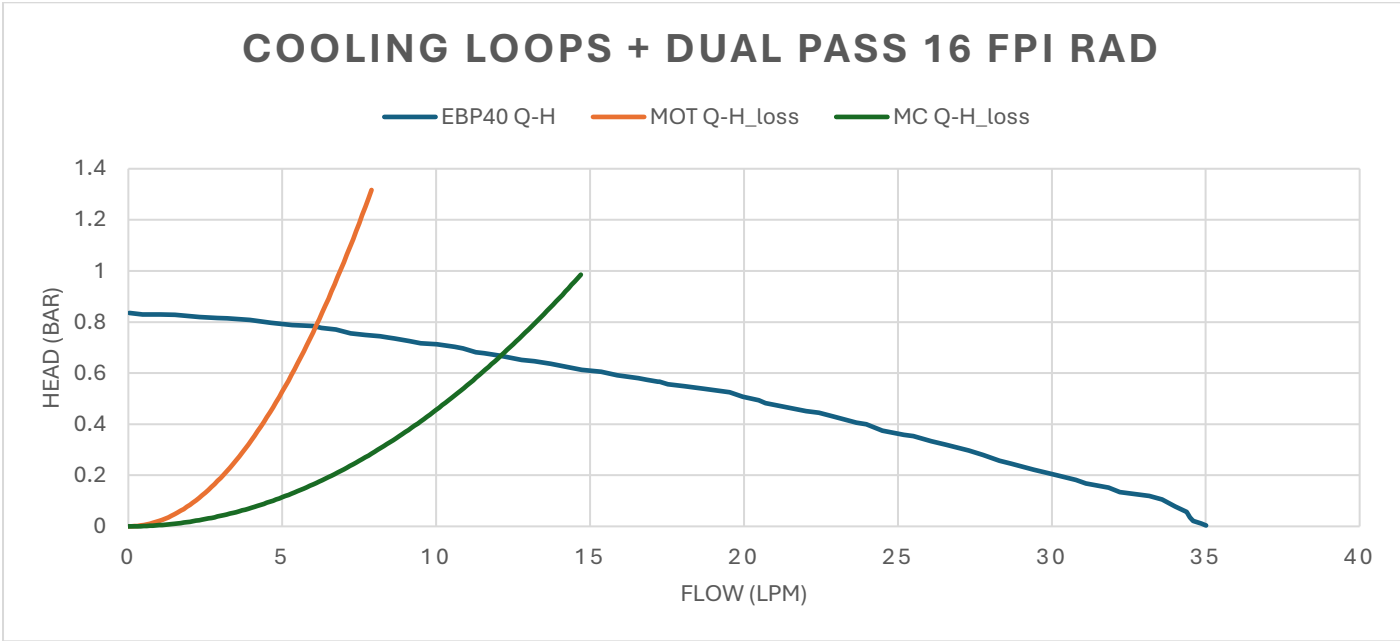


Figure 24: EBP40 Flow Curve vs System Curves with Dual Pass 16 FPI Radiator

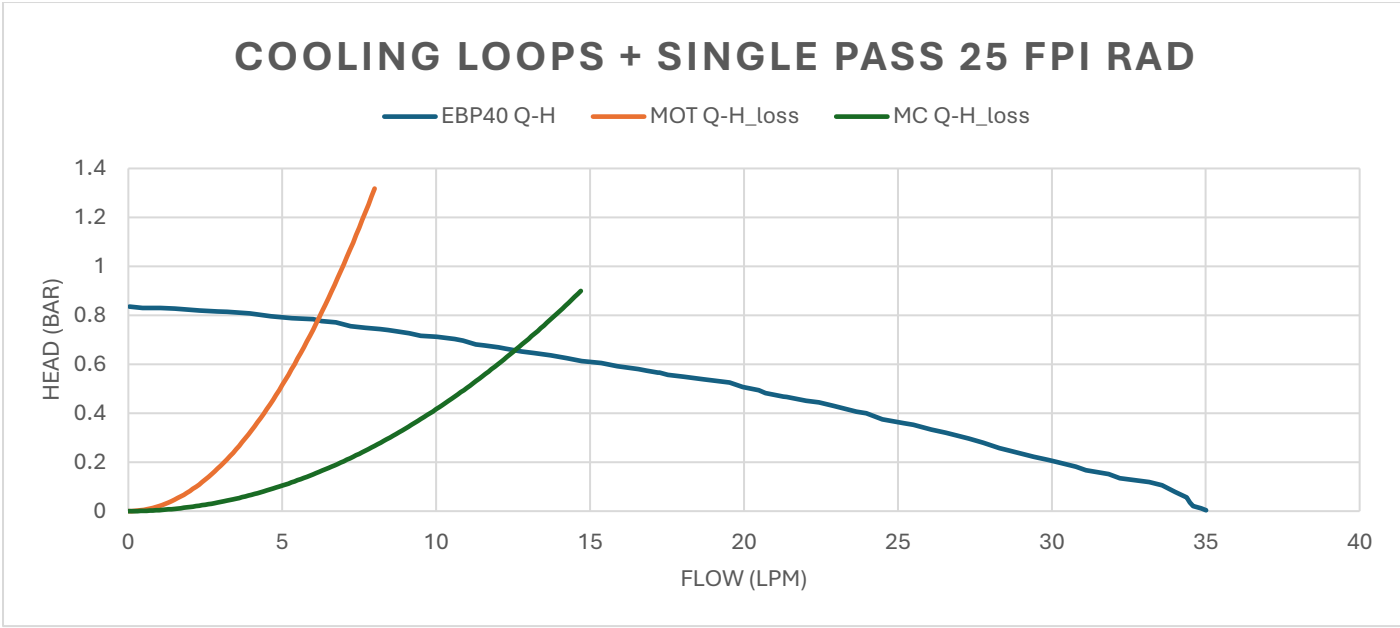


Figure 25: EBP40 Flow Curve vs System Curves with Single Pass 25 FPI Radiator

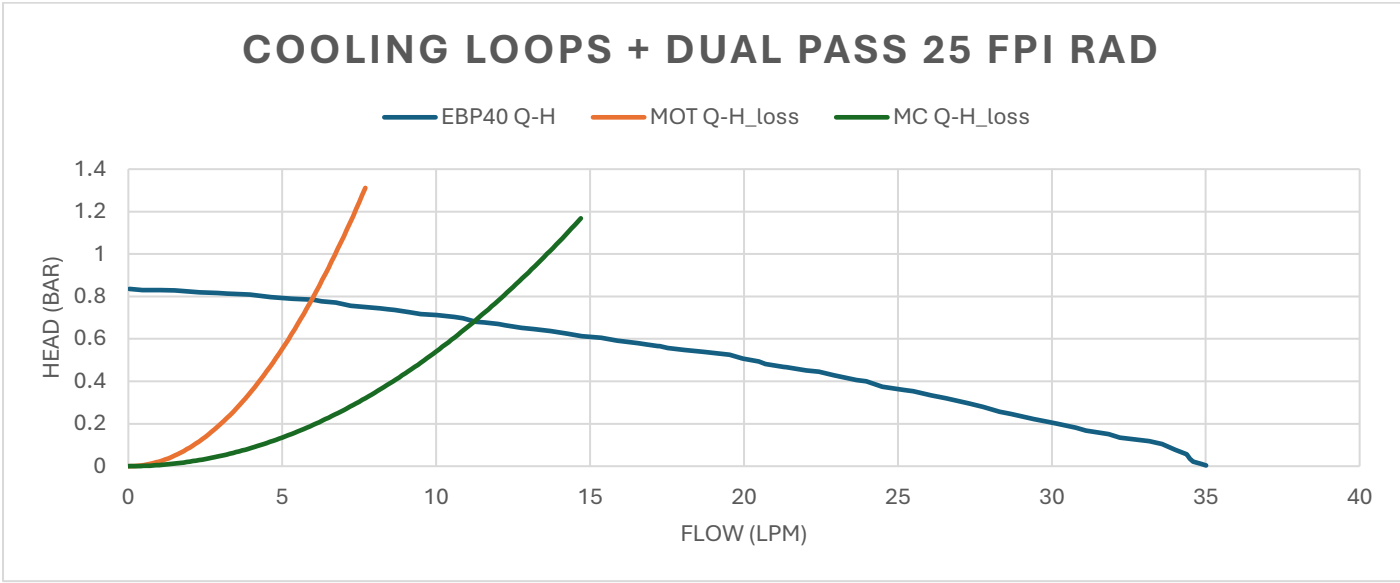


Figure 26: EBP40 Flow Curve vs System Curves with Dual Pass 25 FPI Radiator

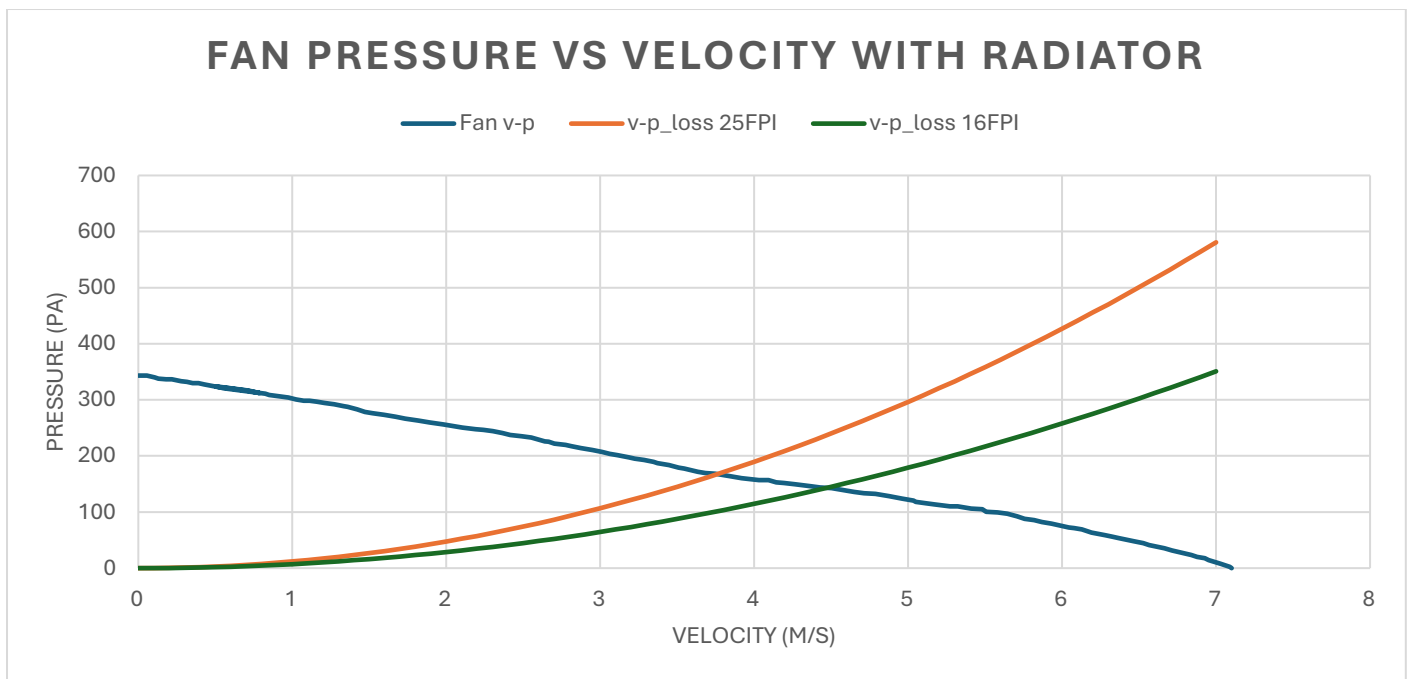


Figure 27: SPAL 7.5-inch Fan Flow Curve vs 16 FPI and 25 FPI System Curve

These flow curves show that the operating point for the 16 FPI dual pass option has 6.1 LPM of water flow for the motor side, 12.1 LPM of water flow for the motor controller side and 4.47 m/s of air flow through the radiator. The 25 FPI single pass option has 6.2 LPM of water flow for the motor side, 12.5 LPM of water flow for the motor controller side and 3.75 m/s of air flow through the radiator. The 25 FPI dual pass option has 5.9 LPM of water flow for the motor side, 11.3 LPM of water flow for the motor controller side and 3.75 m/s of air flow through the radiator.

These results show that for optimal safety to the systems we shouldn't use the 25 FPI dual pass radiator due to the operating points being less than the minimum requirements for both the motor and motor controller. The 16 FPI dual pass radiator is viable but also very close to both the minimums, therefore it was more beneficial and safer to use the 25 FPI single pass radiator.

To make sure that the 25 FPI single pass radiator was more effective at cooling at the lower air flow, interpolation was used to see a rough estimate of the outlet temperatures with the values PWR have given. This showed that the outlet temperature of the motor side was 49.7 °C from the inlet temperature of 52.7 °C

with an air temperature of 35 °C and 3.75 m/s air flow. The motor controller side saw similar results to that of the current radiator at these parameters. Although, this is shown to be not much of an improvement, overall, there will be more air flow through the radiator due to the reduction in size of the core of around 10 % with a well-designed shroud and even without a shroud. The cooling capabilities also increase more exponentially than that of the current radiator if a higher air flow is to go through the radiator thus giving more potential if the radiator is placed in a spot where air velocity from the electric race car moving is implemented.

This new radiator was purchased for roughly \$800 with welded barbs which have a lower pressure drop than threaded inserts and two filler points which allow for more accurate bleeding for each system individually.

Unfortunately, due to this process of emailing PWR, deciding configurations for a radiator, manufacture and shipping took quite some time it was best decided to run the NU23 radiator to allow for more driver training and tests on other ever changing and important components. This radiator had shown to work for previous years with no errors showing its reliability and since this radiator with the current system was validated and shown to run at safe operating temperatures and flows there was no issue in keeping this new radiator for next year's car, NU25.

## 4 Radiator Mounts and Brackets

The radiator was proposed to be rear mounted due to no aerodynamic parts spoilers or side skirts being added to the body kit, therefore if a side mounted radiator was added it would increase drag thus decreasing the performance of the race car. A rear mounted radiator would reduce cooling capacity, but this has shown to work and be efficient enough as last year also ran this configuration.

### 4.1 Radiator Mounts

Constant vibrations from the race car being driven may cause the radiator to become damaged thus reducing the cooling capacity or could also make the radiator completely unusable. Due to this issue anti-vibration mounts using rubber are a must, for our use case it seemed suitable to use cylindrical rubber mounts of 20 mm diameter and 20 mm length with a female and male threads of size M6. These rubber mounts are displayed in figure 28 below.



Figure 28: Anti-Vibration Mounts

Last year's radiator which was decided to be used for NU24s cooling system already had tabs welded onto it from last year's mounting solution, therefore the usage of these tabs allowed for less manufacturing needed, also allowed for the brackets to be designed around these tab positions.

## 4.2 Radiator Brackets

Originally, bracket designs for a smaller radiator of 200 mm of length was proposed and designed, this design also was to not include a jacking point connection as it was thought that the lower rear impact structure would be compliant with the FSAE rules as a jacking point. This design followed a similar layout as last year's radiator brackets which had a total mass of 0.85 kg, except the increase in height position was sought after as this would in turn increase the height of the fill point ensuring better bleeding for the entire cooling system. This design would have connections to not only the radiator but the pumps and overflow allowing for a simple and consolidated cooling system. This design can be shown in figure 29 below.

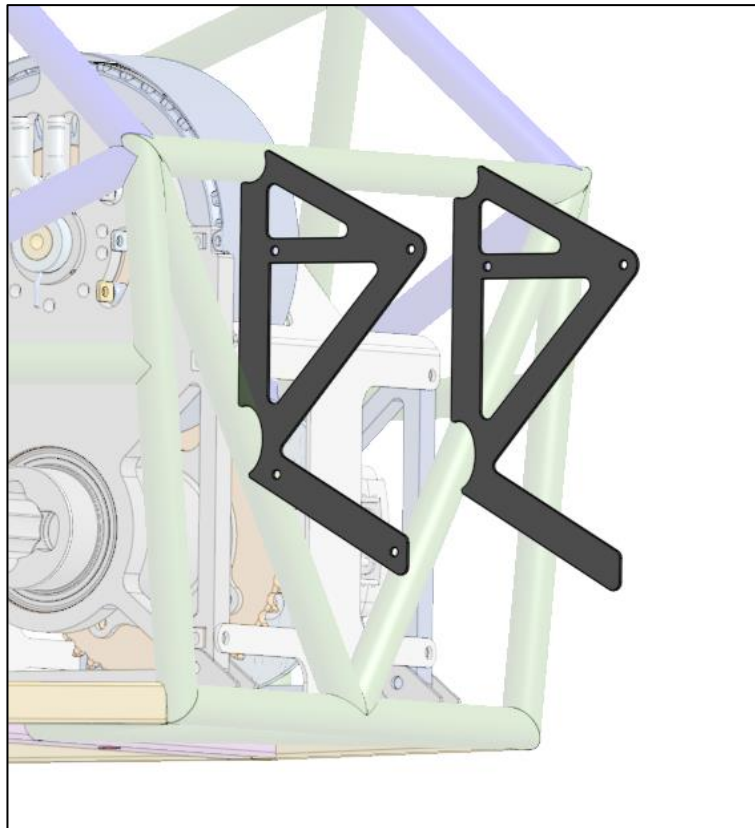


Figure 29: Radiator Brackets for 200 mm Radiator

This design with a total mass of 0.38 kg was assumed to be strong enough to hold the radiator and other cooling components, to confirm this assumption FEA was used on these brackets using the weight forces from

the mass of these components against gravity with a safety factor of up to 2. The results shown in figure 30 show that the deformation is basically negligible at a maximum of 0.002 mm, and in figure 31, the stress is also nothing to be worried about at around a maximum of 4 MPa, therefore this design was over designed meaning a reduction of area and material was proposed.

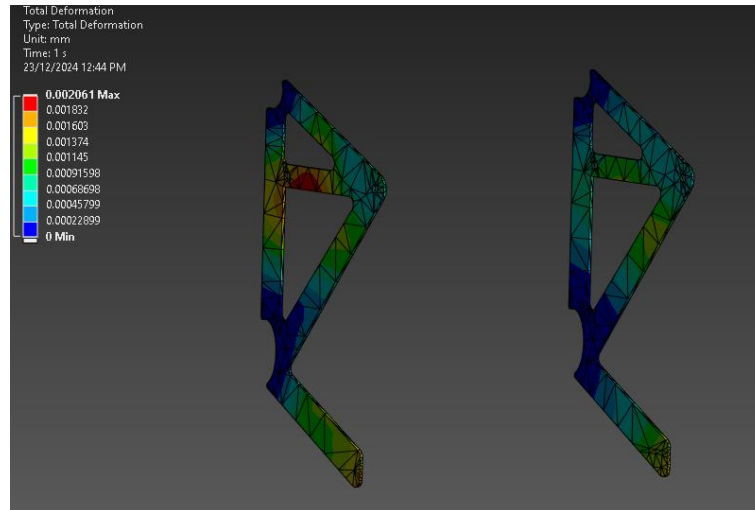


Figure 30: Deformation for Radiator Brackets for 200 mm Radiator

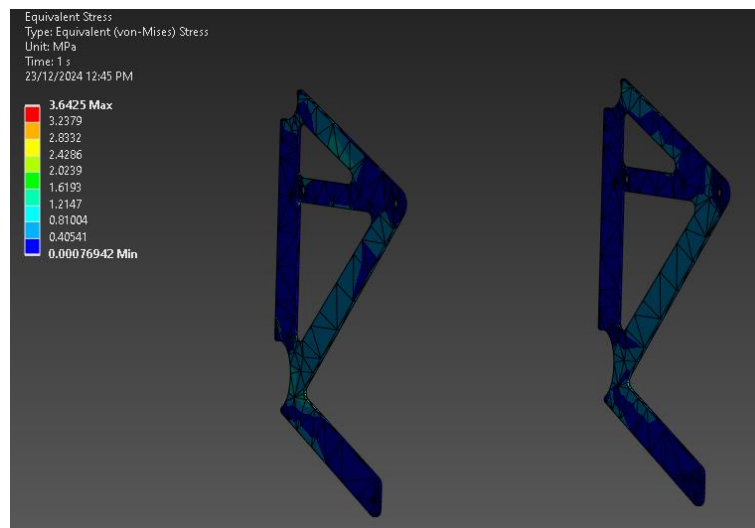


Figure 31: Stress for Radiator Brackets for 200 mm Radiator

When it was realized that the jacking point could not be a part of the rear impact structure it was decided to add the jacking point into the radiator bracket design, therefore a bracket design was created incorporating a

jacking bar and the fact that the jacking point had to be at least 300 mm in length leading to a radiator design of 260 mm in length. This new radiator bracket design was then to be welded to the bottom of the rear impact structure instead of at the top which the initial design was.

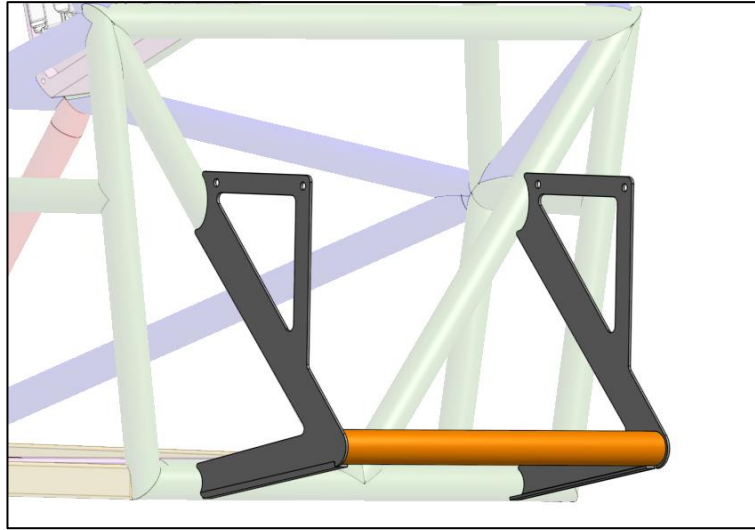


Figure 32: Radiator Brackets with Jacking Point for 260 mm Radiator

This design with a total mass of 0.75 kg also had FEA done on it except the major difference was that the majority of force applied would be at the jacking point to lift the entire back side of the car, since the car was proposed to weigh around 230 kg, therefore using a decent safety factor, a force of 3000 N was put onto the jacking bar where the results were shown to have very slight deformation in the middle of the jacking bar with a maximum of around 0.9 mm and a maximum stress of around 590 MPa which has been ruled out as this stress is created due to the jacking bar slightly protruding into the brackets bends shown in figure 35, therefore the more realistic maximum stress was roughly 200 MPa, which is much lower than the yield stress of the steel used being around 460 MPa.



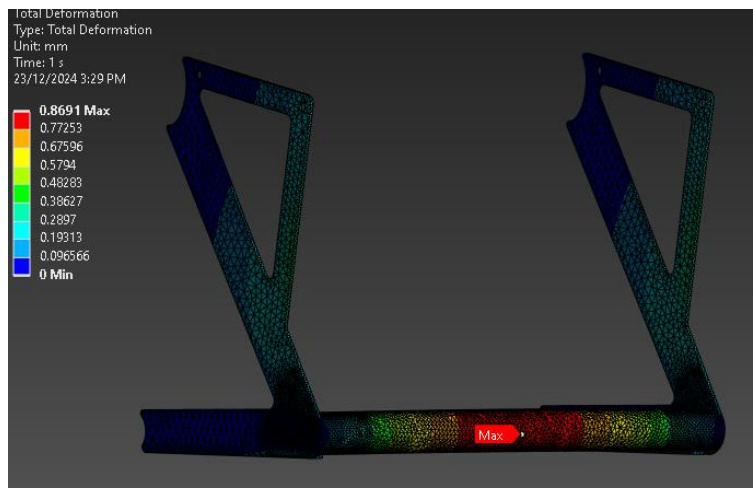


Figure 33: Deformation of Radiator Brackets and Jacking Point for 260 mm Radiator

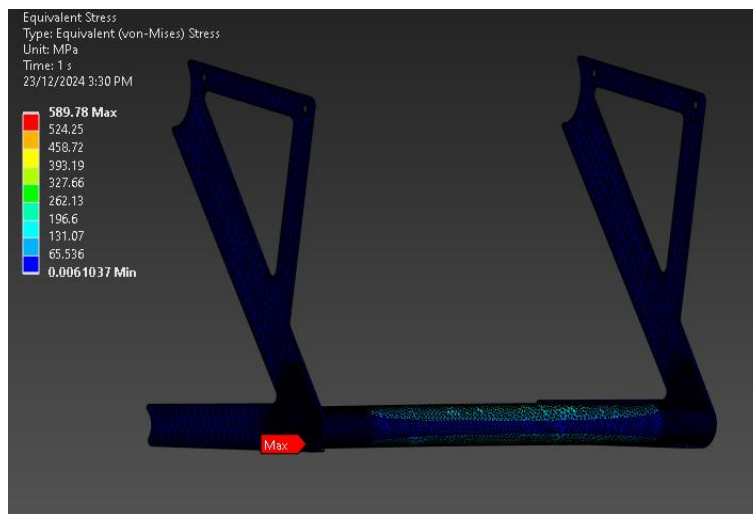


Figure 34: Stress of Radiator Brackets and Jacking Point for 260 mm Radiator

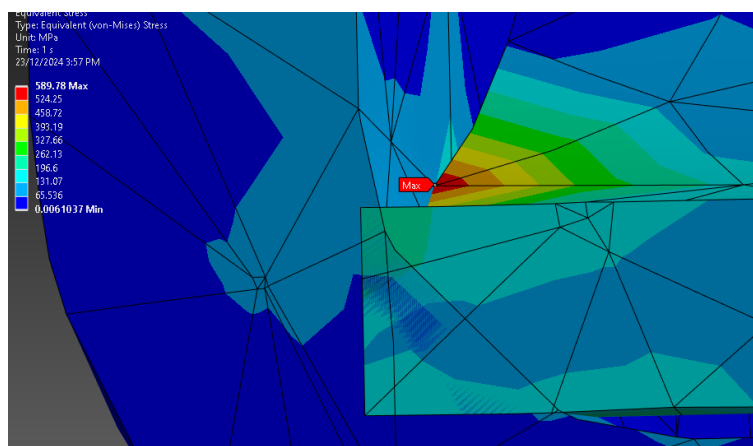


Figure 35: Close up of Stress for Radiator Brackets and jacking Point for 260 mm Radiator

Due to the new radiator of 260 mm in length taking too long to sort out, manufacture and be delivered on top of the swiftness of the cars progress, last year's radiator of 328 mm in length was to be used. Therefore, radiator brackets to incorporate the size of this radiator were to be designed quickly, these followed a very similar design to that of the brackets designed for the 260 mm radiator. This design is shown in figure 36 below.

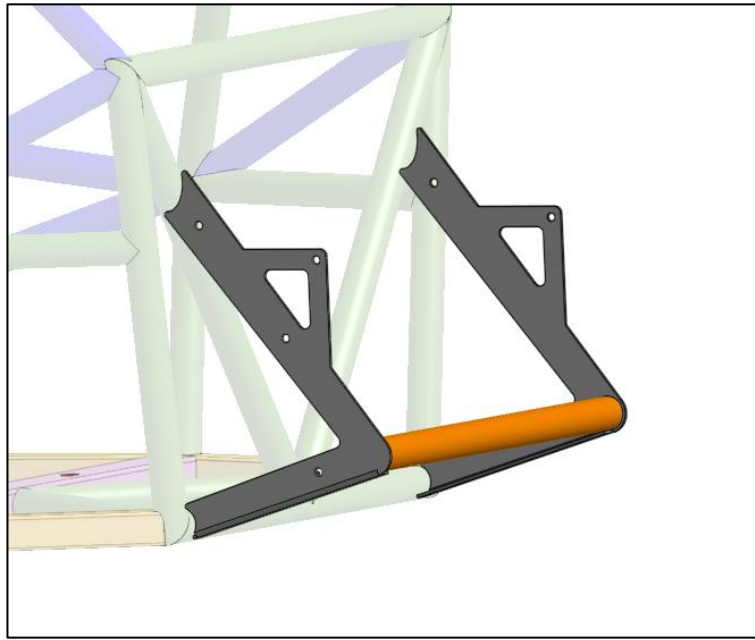


Figure 36: Radiator Brackets with Jacking Point for 328 mm Radiator

This design was also put into FEA to determine its ability to lift 3000 N of force, where we saw that the deformation and stress had an increase which was to be expected with a design with the connections being further away from each other, the maximum deformation was shown to be roughly 1.5 mm which is decently high but considering the jacking point will almost never need see forces this high as the car weighs much less and it would only need to lift most of what's in the rear of the race car, due to these reasons this was determined to be adequate. The maximum stress also showed the same problems as on the 260 mm radiator brackets with intersections showing a wrong maximum, where on close inspection the maximum was determined to be roughly 300 MPa.

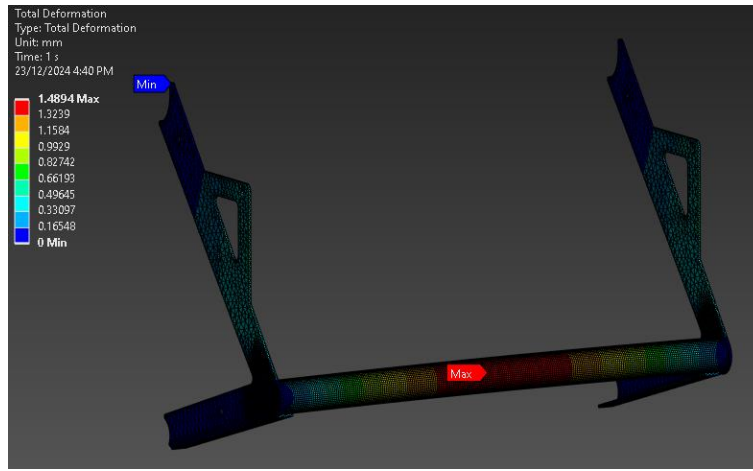


Figure 37: Deformation of Radiator Brackets and Jacking Point for 328 mm Radiator

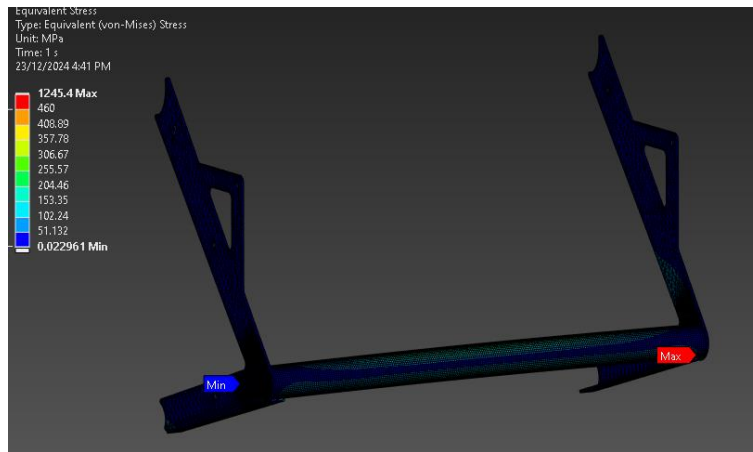


Figure 38: Stress of Radiator Brackets and Jacking Point for 260 mm Radiator

This design saw a reduction in mass from last year of around 10 % where this radiator bracket design weighs 0.78 kg, the design was also much more compacted to the rear of the car compared to that of last year which can be shown in figure 39.

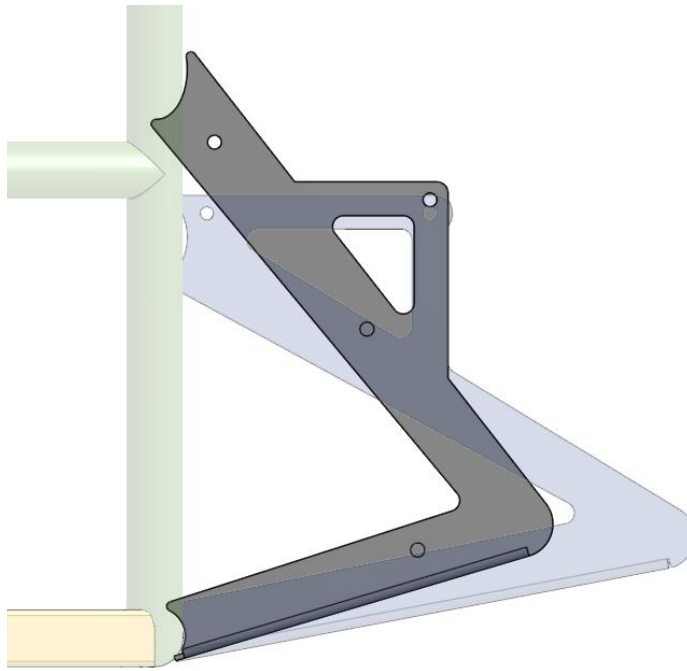


Figure 39: NU24 Radiator Brackets vs NU23 Radiator Brackets (Shaded Out)

This design was sent off to one of our sponsors Hancock Speedway to be laser cut and then bent on the bottom edge of the brackets which can be shown in figure 40. These brackets were then sent with the chassis to the Mechanical Engineering Workshop on campus to be welded onto the back of the rear impact structure where the finished product after powder coating is shown in figure 41.



Figure 40: Radiator Brackets

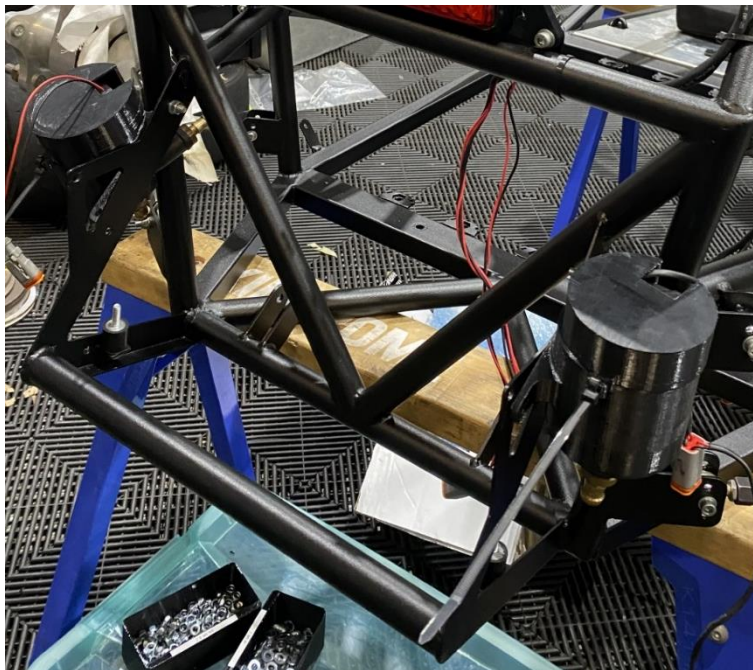


Figure 41: Welded on Radiator Brackets with Jacking Point

## **5 Pump Mounts and Catch Can/Overflow**

Both the pumps and overflow were to be mounted to the radiator brackets to keep everything compact and together. The mounting was designed in conjunction with the design of the bracket to make sure everything fit together and mounted correctly.

### **5.1 Pump Mounting**

The two EBP40 pumps used are to be mounted to the top part of the bracket where one bolt hole for them was to be the same the radiator is connected to, allowing for less fasteners for the entire system. A 100 mm distance between bolt holes was determined to be the minimum with an enclosure or mount for the pump to allow for ease of fastening. This would be added to the bracket in an incline of around 15 degrees as shown in previous figures of the brackets. A pump enclosure was proposed to mount and secure the pumps while also allowing for protection to environmental factors.

The original design for the pump enclosure included a pump holder and a lid, the pump holder was chosen to hold the pump vertically as shown in figure 42. This design had a chunk out of it for the pump outlet to slot in and still has room to move around slightly to allow for changes in direction if needed and ease of maintenance allowing for the pump to be easily pulled out the top of the holder. The lid was then designed around the holder with a closer fit to the pump to keep the lid from easily flipping open. The lid was connected to the holder with a zip tie.

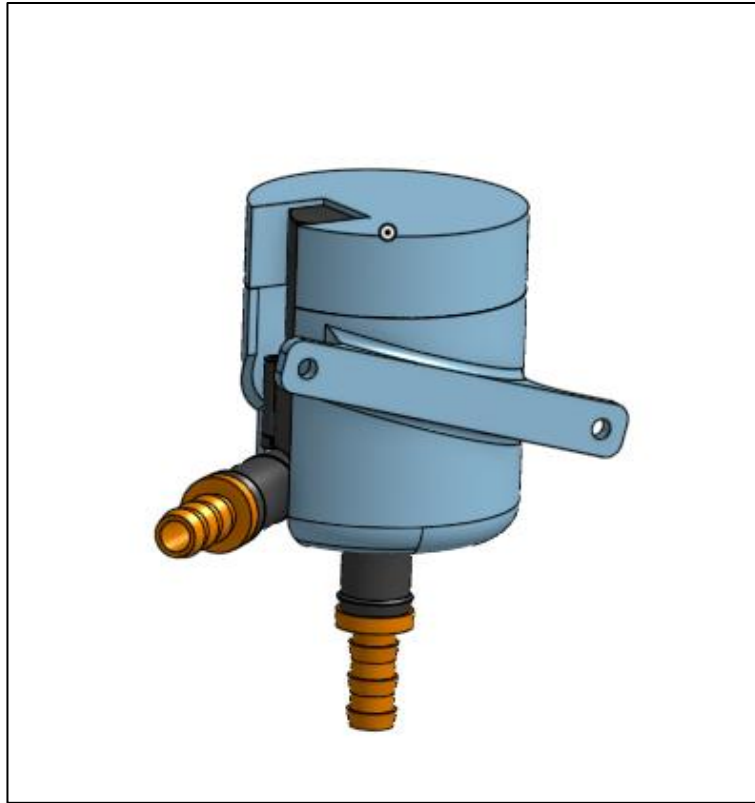


Figure 42: Pump Enclosure CAD Model

The whole pump enclosure was to be made using the 3D printers in TA using PETG for its high glass transition temperature of 85 degrees Celsius which is much above that which is experienced close to the radiator. The use of these 3D printers allowed for design changes and quick manufacturing and rapid prototyping with also allowing for a very lightweight enclosure of roughly 165 grams each. The finished product is shown below in figure 43.





Figure 43: Pump Enclosure with Pump on Car

While testing, it was noticed that since the bracket was quite wide compared to the rear of NU24 the pump did not have a very easy way of sliding out of the pump enclosure, there was also problems with bleeding which was later discovered that the pump was positioned the opposite orientation to what is safe for the pump while bleeding the cooling system, since there was not enough time and space to create new mounting at the correct positions for the pump it was decided to use the rubber pump mounting brackets which were supplied by the manufacturer with the pumps shown in figure 44.





Figure 44: EBP40 Mounting Bracket (Davies Craig, 2024)

This pump holder became a better fit for the mounting solution of the pumps due to the rubber providing anti-vibration to the pump keeping the motor inside safe, the mount also allowed for a slight angle to have the orientation of the pump closer to correct shown in figure 45. This pump mounting bracket was also roughly half the mass of the 3D printed enclosure. Environmental factors were still potentially a problem but with the pump's enclosure rating of IP67 meaning protection from dust and water it was deemed to be unnecessary to have the pump protected in an enclosure.

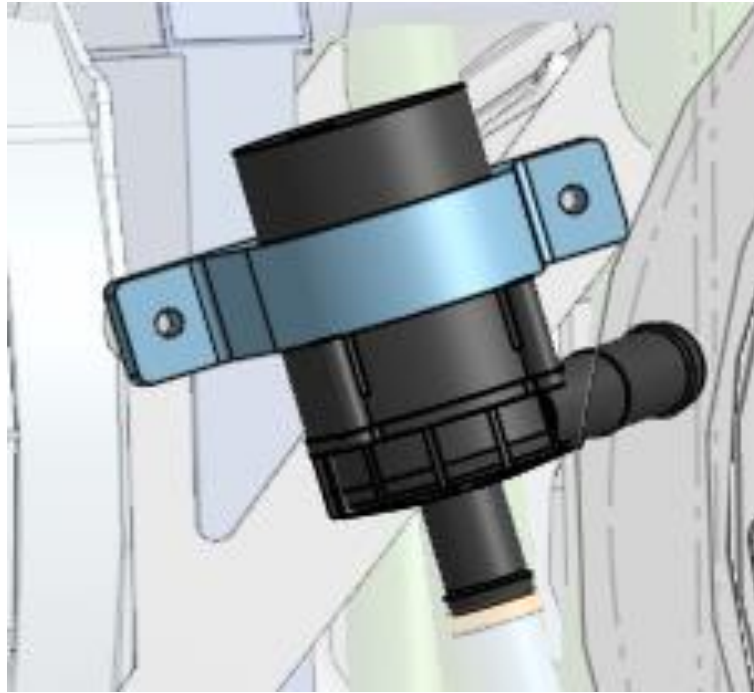


Figure 45: Pump in Rubber Mounting Bracket CAD Model

## 5.2 Catch Can/Overflow Mounting

The catch can which was chosen to be used was the one used last year, a SAAS Oil Catch Tank made from black billet aluminium weighing roughly 1 kg dry shown in figure 46. This overflow does work in line with the FSAE-A rules as well with a 500 ml capacity.



Figure 46: Catch Can

To mount this catch can to the radiator brackets the welded-on mount used last year had to be removed from the catch can and a new aluminium, welded-on mount attached, to connect and sit better on the new radiator bracket design. The design of the new mount followed the shape of the brackets angle with a two-bolt connection to the left bracket. The mount would be welded to an edge determined by a correct catch can position which was decided by trial and error on CAD models checking distances from the lowest point on the car and the pump above where an optimal position was found shown in figure 47.

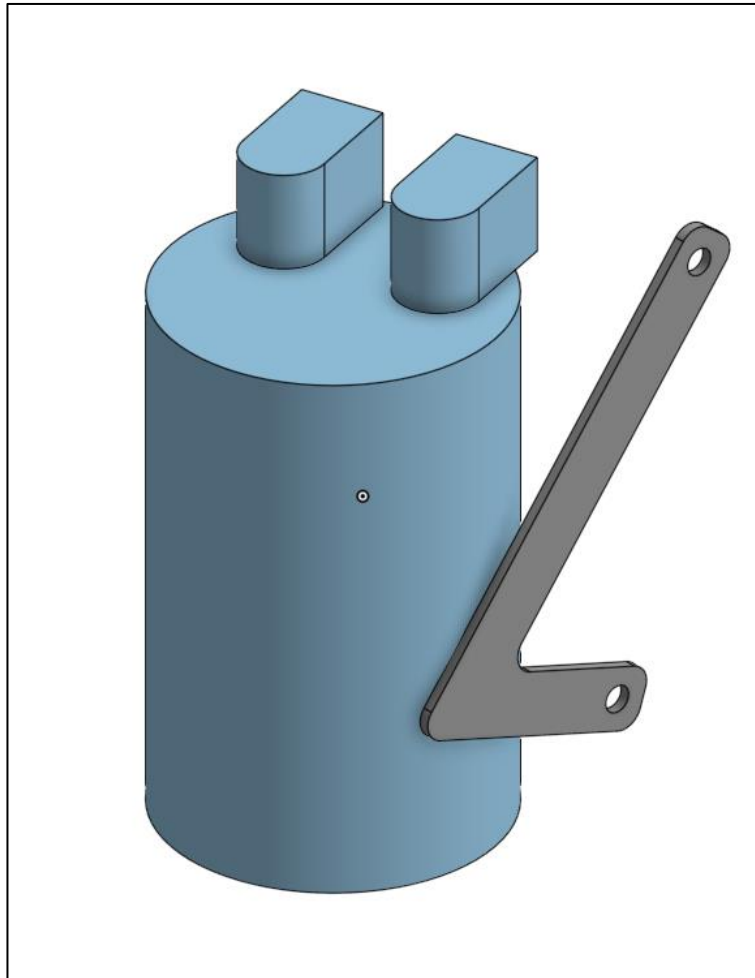


Figure 47: Catch Can with Welded Mount CAD Model

To confirm the fitment of the catch can mounted on the bracket, a 3D printed model shown in figure 48 was created before the mount was welded to the catch can. This 3D printed model was fastened to the bracket and showed to be correct and fit perfectly therefore, welding drawings were sent off to the Mechanical Engineering Workshop on campus with both the mount and the catch can for welding.



Figure 48: Catch Can with Mount 3D Print

The catch can was then fitted to the bracket and spray painted black to look more aesthetic against the powder coated bracket to be used for the FSAE-A competition. The final assembled catch can setup connected to the radiator bracket is shown in figure 49 below.



Figure 49: Catch Can Connected to Radiator Bracket

## **6 Shroud Study**

Due to the limited testing and inconclusive results of shroud usage from last year on NU23 (Symons, 2023), the addition of a much more powerful motor and possibility of regenerative braking, it was decided to test a shroud again this year to see if results would be of better worth. A shroud is thought to make a large difference to cooling efficiency of cooling systems overall but it may not be of much importance to our situation and requirements from the cooling system. This study would consist of some theoretical data or CFD with some real data from track days to reinforce findings and thus decide whether a fan shroud would be required for our cooling system.

### **6.1 Shroud Design**

The shroud design was made to accommodate the mounting tabs and additional welded on parts from the radiator used on NU23, following a similar design to what was tested last year. This shroud was chosen with a fan distance to radiator of approximately 35 mm which allows the fan to still be close enough to the radiator to allow optimal airflow. The design was chosen to be fabricated with PETG and thus 3D printed, which allows for rapid manufacture and allows constant edits to be made to the design. To be printed on the 3D printers in TA's print farm, the shroud would have to be divided into four sections which can be shown in the CAD model design shown below in figure 50.

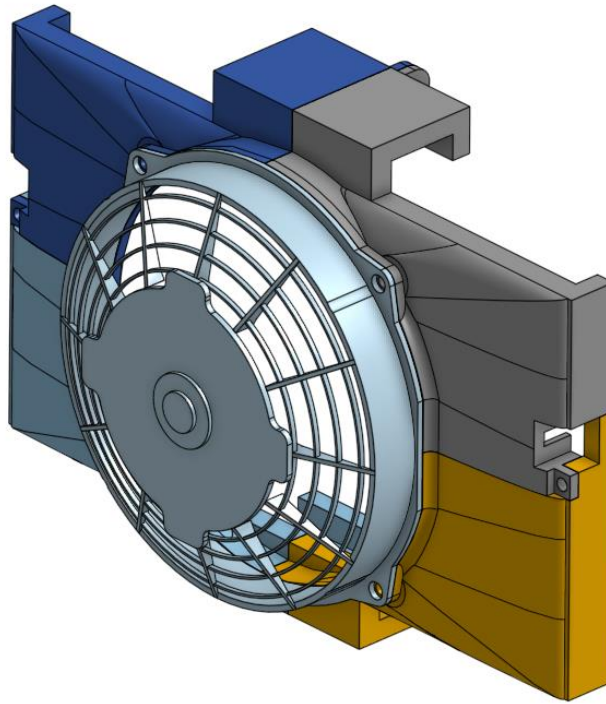


Figure 50: Fan Shroud CAD Model

This design does have one minor flaw, which has to do with the mounting tabs and where they are situated. These mounting tabs made it difficult to have a full shroud without either cutting off a section or allowing a gap in the side which would reduce airflow through the radiator. Therefore, the design which was created accounts for this and has a small section which allows the mounting tab and anti-vibration mount to sit in between sections, this allowed a small hole in the side for air to flow through but with the usage of foam tape this would be very minimal. The 3D printed shroud on the radiator can be seen in the figure below.



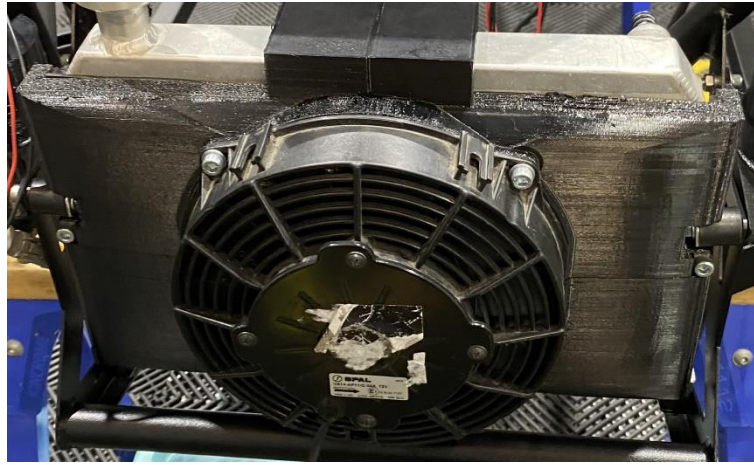


Figure 51: Fan Shroud on Radiator

## 6.2 Shroud CFD

The purpose for this shroud CFD study is to achieve a theoretical result for whether or not a shroud will make a significant impact on the cooling capacity of the radiator being used. This may also give a good indication on the usage of a shroud for future designs and allow for decisions based on cooling efficiency, weight reduction and component optimization.

### 6.2.1 Shroud CFD Design

To begin the shroud CFD analysis, the radiator fan had to be modelled, as the fan enclosure was the only CAD model provided by the company or modelled previously. This fan model will allow an accurate simulation of airflow. A fan blade was modelled using our physical radiator fan and other fan models as references, then with a circular pattern the whole fan was created and ready for CFD as shown below in figure 52.

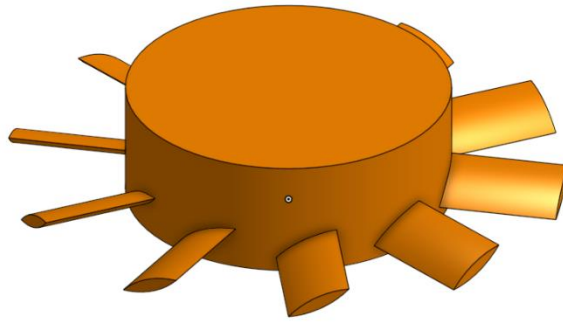


Figure 52: 7.5 Inch Fan CAD Model

A simplified model of the airflow volume for both the shrouded and non-shrouded configurations was created with the gaps between the fins being represented by a smaller number of tubes which was put into an assembly with the fan model inserted inside, as shown in figures 53 and 54 below.

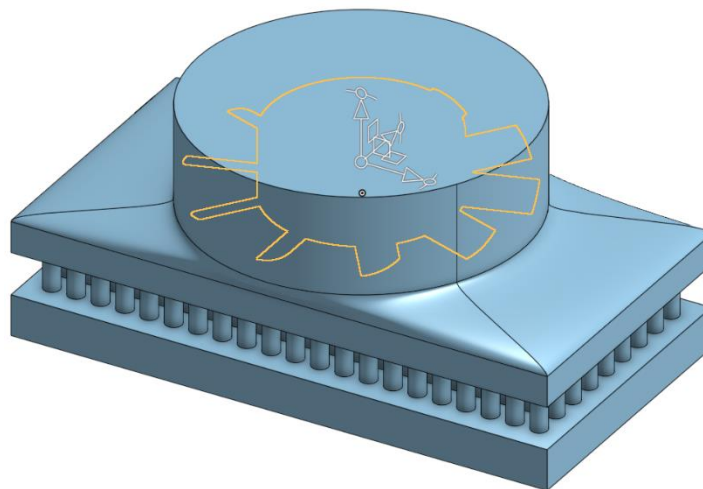


Figure 53: Simplified Fan Shroud Airflow Model

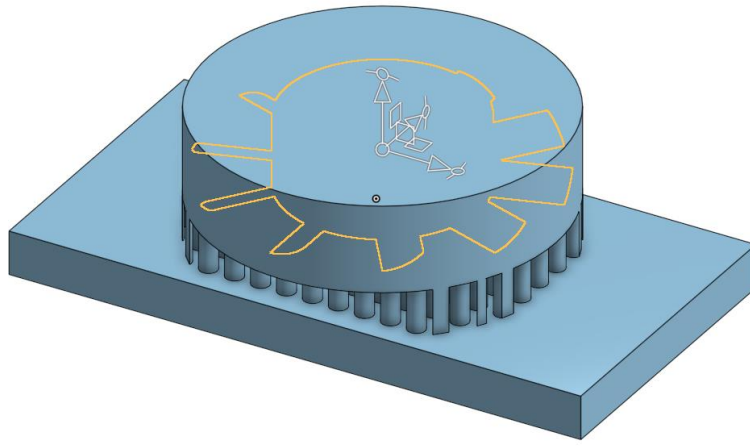


Figure 54: Simplified Fan (No Shroud) Airflow Model

These models were imported into Ansys CFX, where using SpaceClaim a Boolean operation was used to remove the area from the fluid volume where the fan occupied, then an enclosure was put around the tubes to represent a radiator core. This enclosure in green can be seen in figure 55 surrounding the tubes. A cylinder was also designed around the fan and another rectangular prism around the tubes to assist in refining the mesh later.

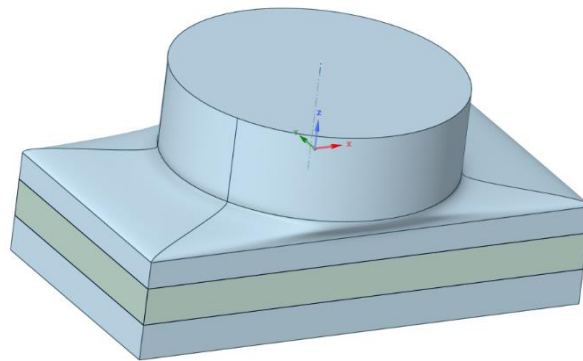


Figure 55: Fan Shroud CFD Model from SpaceClaim

### 6.2.2 Shroud CFD Meshing

The meshing for these models needed to be quite fine to have accurate representations of airflow and heat transfer therefore the use of body sizing and inflation was used to achieve this.

For our model with a shroud a number of elements of 15 million was achieved using mesh refinement tools. Body sizing was applied, using a body of influence with the cylinder designed in SpaceClaim around the fan to achieve an element size of 1 mm around the fan. Inflation layers were then used for the walls surrounding the shroud and fan with a first layer height of 1 mm and 7 layers and the tubes had a first layer height of 0.5 mm and 5 layers, this allowed accurate wall boundary conditions for airflow. Another body of influence body sizing was used but for the tubes making them have an element size of 1 mm. The mesh can be seen in figure 56 with a closeup of the inflation layers inside the tubes shown in figure 57.

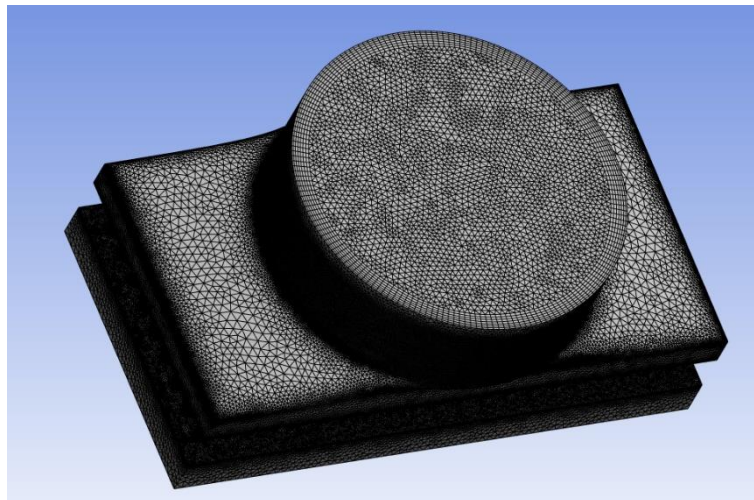


Figure 56: Fan Shroud CFD Model Mesh

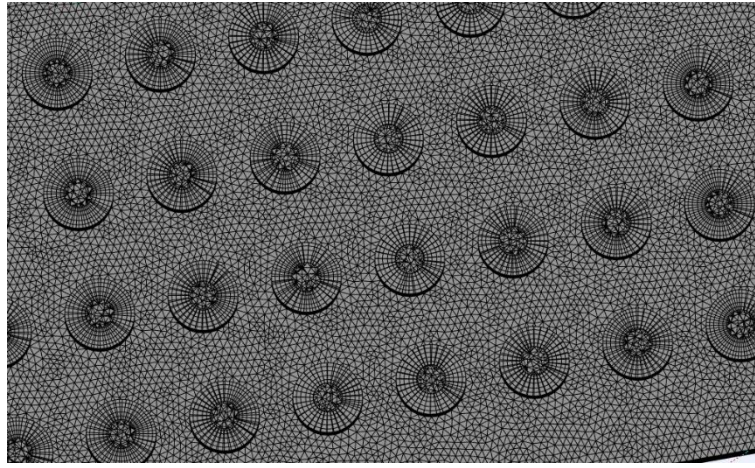


Figure 57: Fan Shroud CFD Mesh Closeup of Tubes

To ensure that this mesh was in fact viable, it was necessary to check the element quality, aspect ratio and skewness. These can be shown in figure 58 below, these graphs show that most elements for the element quality and aspect ratio were closer to 1 and for skewness closer to 0, these numbers indicate a good mesh.



Figure 58: Element Quality (Top), Aspect Ratio (Middle) and Skewness (Bottom) of Shroud Mesh

The meshing for the model without a shroud followed the exact same layout as the model with the shroud leading to roughly 13 million elements, with some minor changes. The inflation layers were the same for both the tubes and fan outer, the first layer thickness was chosen to be 0.1 mm with 10 layers. The default element size for the entire model was also edited to be 5 mm. This mesh can be seen in figure 59.

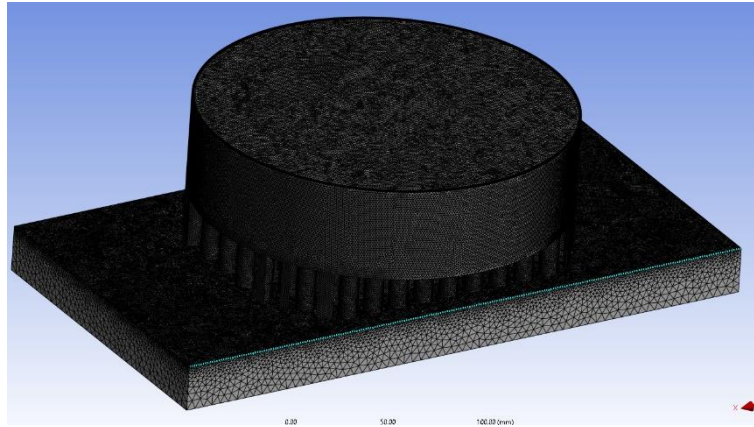


Figure 59: No Fan Shroud CFD Model Mesh

The element quality, aspect ratio and skewness were also investigated with this model. For most of the elements, the element quality and aspect ratio they were both close to 1 and the skewness was closer to 0 shown in figure 60 indicating a good mesh.



Figure 60: Element Quality (Top), Aspect Ratio (Middle) and Skewness (Bottom) of Non-Shroud Mesh

### 6.2.3 Shroud CFD Setup

To setup these models to give results CFX-Pre was to be filled out to simulate motion of fluid, to accomplish this, multiple domains were created one for each component, fan, fluid (airflow) and the solid block around the tubes representing the radiator core with a domain interface for connections between tubes and the solid block. Each of these domains had different settings and conditions.

The fan domain was made to be an immersed solid with rotating domain motion where a value of 4000 rpm was given to the fan spinning on the z-axis, this value may not have been the rpm which the 7.5-inch fan could



achieve but for this study if the value was constant, it would still give valuable results.

The fluid domain where the fluid used was air, kept the turbulence model as shear stress transport to accommodate to the odd geometry and wall boundary conditions, thermal energy was also chosen for heat transfer to allow for heat to be transferred through the solid and fluid. Two openings were made for this domain at both ends of the fluid flow.

The solid domain which was interfaced with the fluid domain was given aluminium as the material, for this study a nonoverlapping temperature was given to the solid walls of 50 degrees Celsius, the interface between these had a heat transfer configuration of conservative interface flux. Once these domains and interfaces were configured it was time to solve this simulation.

#### 6.2.4 Shroud CFD Results

Before checking the temperature results for both the shroud and non-shroud model it was necessary to check the velocity and make sure that the airflow is working how it should. Looking at figure 61 below we can see that for both models the air is being pulled from the other side of the radiator core and the velocity is higher for the non-shroud model, this is expected and therefore the model is viable.

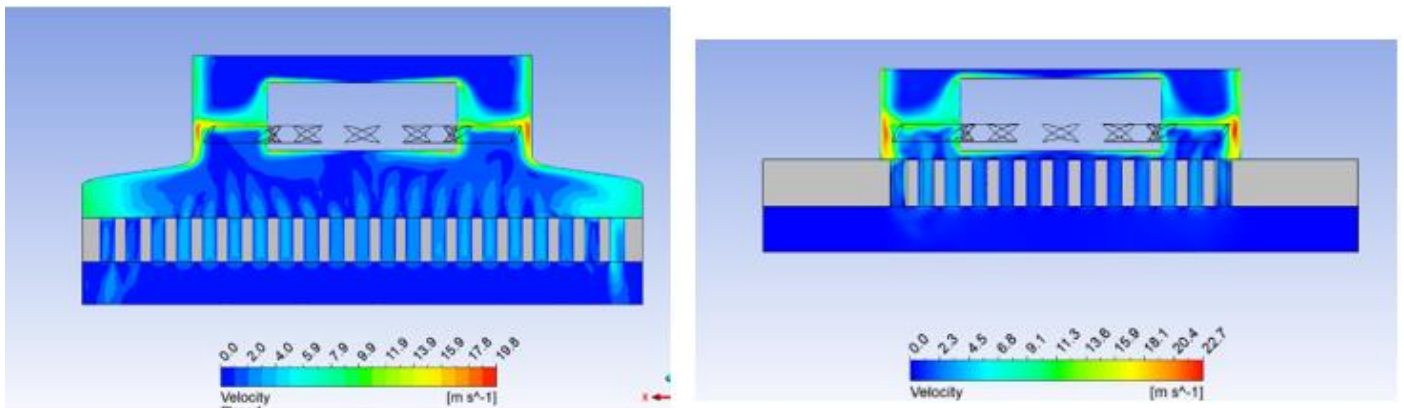


Figure 61: Velocity Plane Contours for Shroud (Left) and Non-Shroud (Right) Models

Knowing that the airflow is reacting correctly in this simulation, a temperature contour was created to examine the temperature changes on the radiator core and as expected the shroud creates a much more distributed

temperature change whereas the non-shroud model only has the distribution only below the fan. Looking at figure 62 we see that the overall temperature change is better for the shroud than the non-shroud.

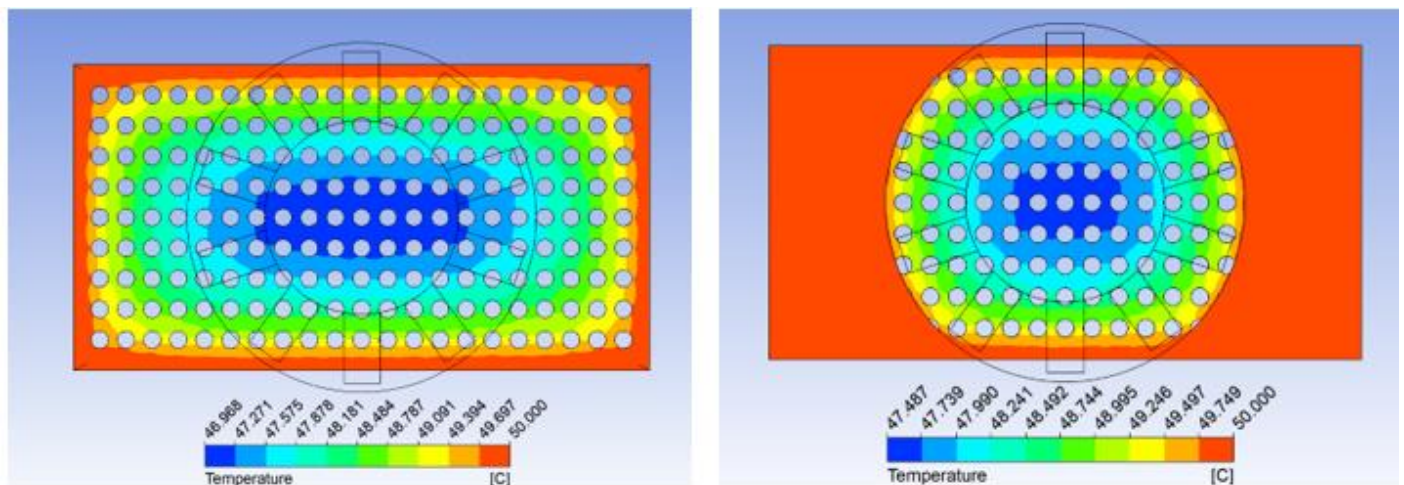


Figure 62: Temperature Contour for Radiator Core on Shroud (Left) and Non-Shroud (Right) Models

To show that the heat was being transferred to the air, another plane contour was used but for temperature this time where it shows the cross section of the radiator core where you can see the heat being transferred and pulled to the fan. This can be shown in figure 63.

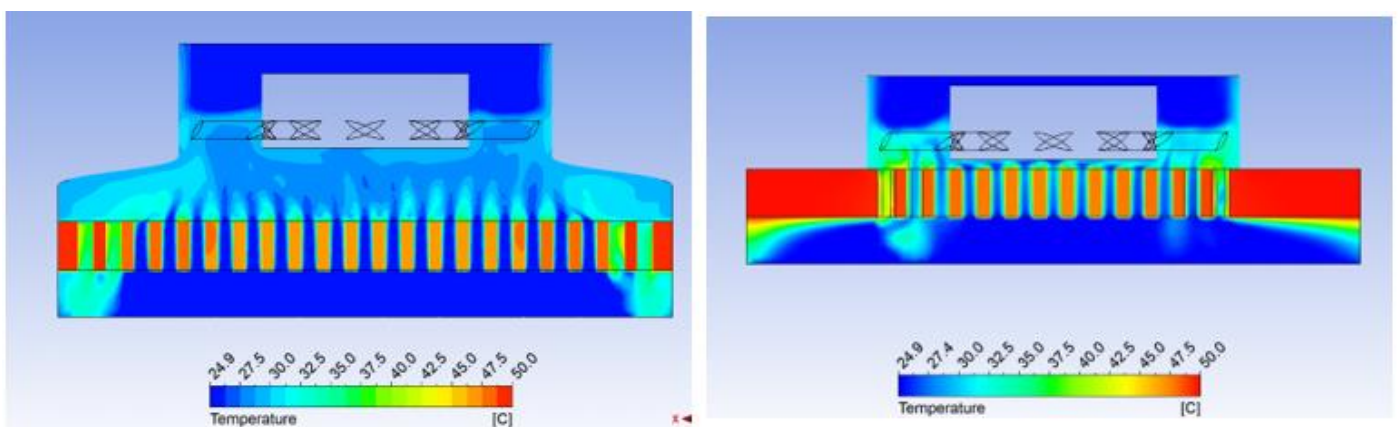


Figure 63: Temperature Plane Contour for Radiator Core on Shroud (Left) and Non-Shroud (Right) Models

As shown throughout these results we see that a shroud does increase cooling capacity of a radiator which it should, although it doesn't seem to have all that much of an effect. More tests will have to be conducted

including a real life setting to conclude whether to use a fan shroud.

### 6.3 Shroud Track Day Testing

To validate results from CFD, track days where the shroud was to be tested against no shroud was proposed, unfortunately due to problems with the motor controller setting us back 2-3 months and rapid testing and changes with other components it was almost impossible to get accurate data for comparison.

We fortunately did have at least one day where it was tested accurately, this track day composed of two sets of endurance runs where the first set was with a fan shroud equipped and the second set without. These runs were done with the same drivers where a driver swap would occur midway, the drivers were told to drive very similar to that of their first run. The track which was used for these endurance runs is shown below in figure 64.



Figure 64: Track for Endurance Runs

These endurance runs ran quite well apart from the last one where the second half of the run the car hard faulted leaving only the first half of each endurance runs to be compared. The data was then put into excel to compare them, which is shown in figure 65. Keeping in mind that the run without the shroud started 5 degrees Celsius above the run with the shroud we can see that there is not much of a noticeable difference in coolant temperature change for either, the temperature change without the shroud does increase slightly more before

reaching steady state than with the shroud but is very minimal.

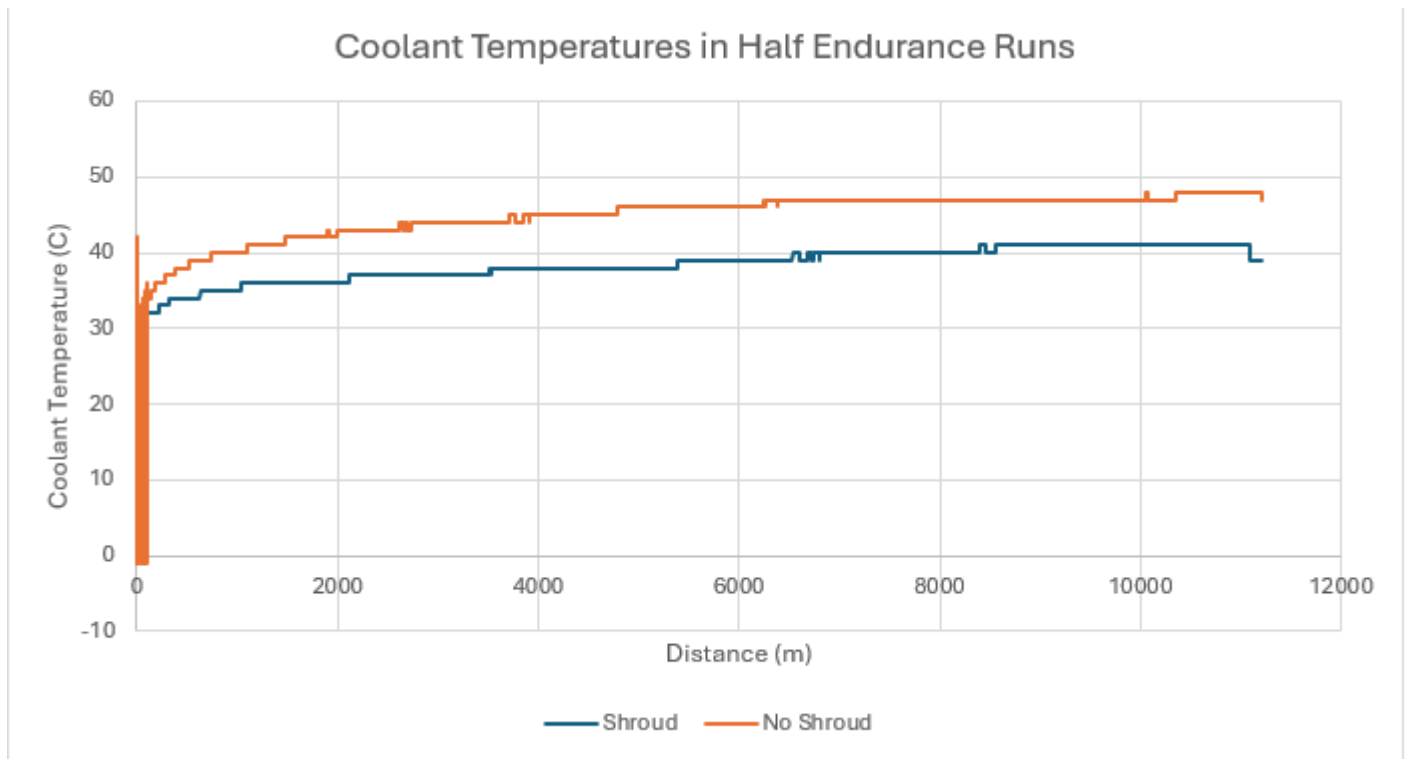


Figure 65: Coolant Temperature Comparison in Half Endurance Run

When the run without the shroud was conducted the ambient temperature was roughly 36 degrees Celsius and coolant temperature never hit the maximum limit for the motor nor the motor controller therefore, it was decided to not use the shroud to reduce part count and weight.



Figure 66: Cooling System without Fan Shroud

## 7 Radiator Fan Decision

A radiator fan is a very important part of a cooling system therefore the need for the best fan is necessary to maximize the cooling capacity. Therefore, a comparison of different fans to the current 7.5-inch fan which was used last year was conducted. To compare the abilities of different fans, a flow curve against system curve was created over a set few fan sizes and configurations including an 8-inch fan, different 10-inch fans, different 12-inch fans and two 4-inch fans in parallel. These fans were compared using velocity through both a 16 FPI and a 25 FPI radiator, for this initial comparison these fans would also have no shroud and sit directly on top of the radiator. Also, the venturi effect will be explored for the use of a shroud and a bigger fan.

### 7.1 Radiator Fan Comparison

A different fan manufacturer was explored as well as different sizes to widen the selection of fans and data. First, an 8-inch fan from Maradyne was investigated as a similar size to the current radiator fan. This fan achieved a velocity of roughly 3.9 m/s in the 16 FPI radiator and 3.4 m/s in a 25 FPI radiator as shown in figure 67.

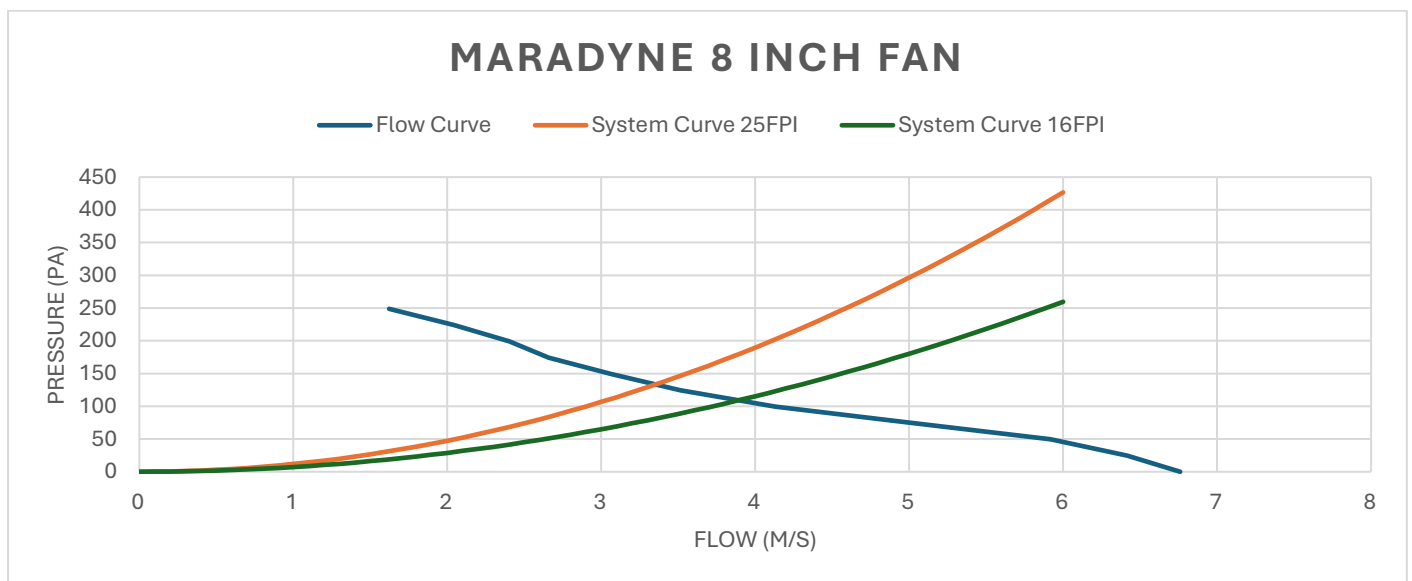


Figure 67: MARADYNE 8-Inch Fan Flow Curve

Next, 10-inch fans from both Spal and Maradyne were explored where the Spal fan achieved a 3.6 m/s velocity for a 16 FPI radiator while the Maradyne fan achieved a 4.2 m/s velocity for the same radiator. For a 25 FPI radiator the Spal fan had a velocity of 3 m/s and the Maradyne a velocity of 3.5 m/s. Showing that for this size of fan Maradyne definitely is better, but weight and power usage should be considered.

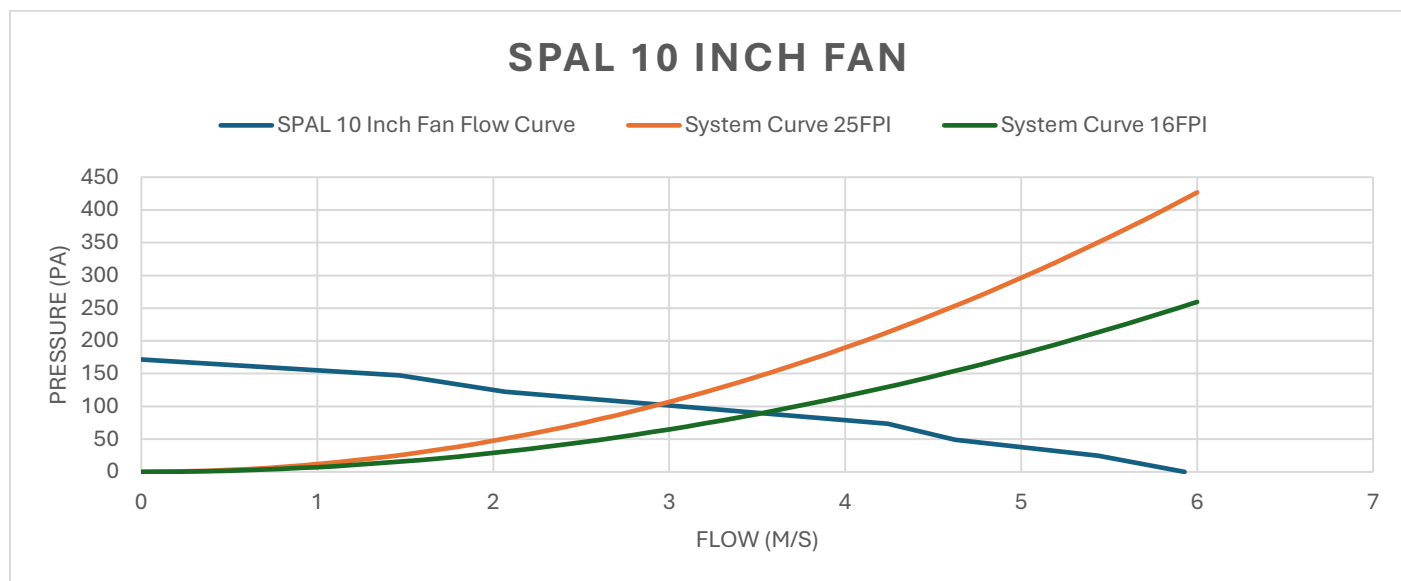


Figure 68: SPAL 10-Inch Fan Flow Curve

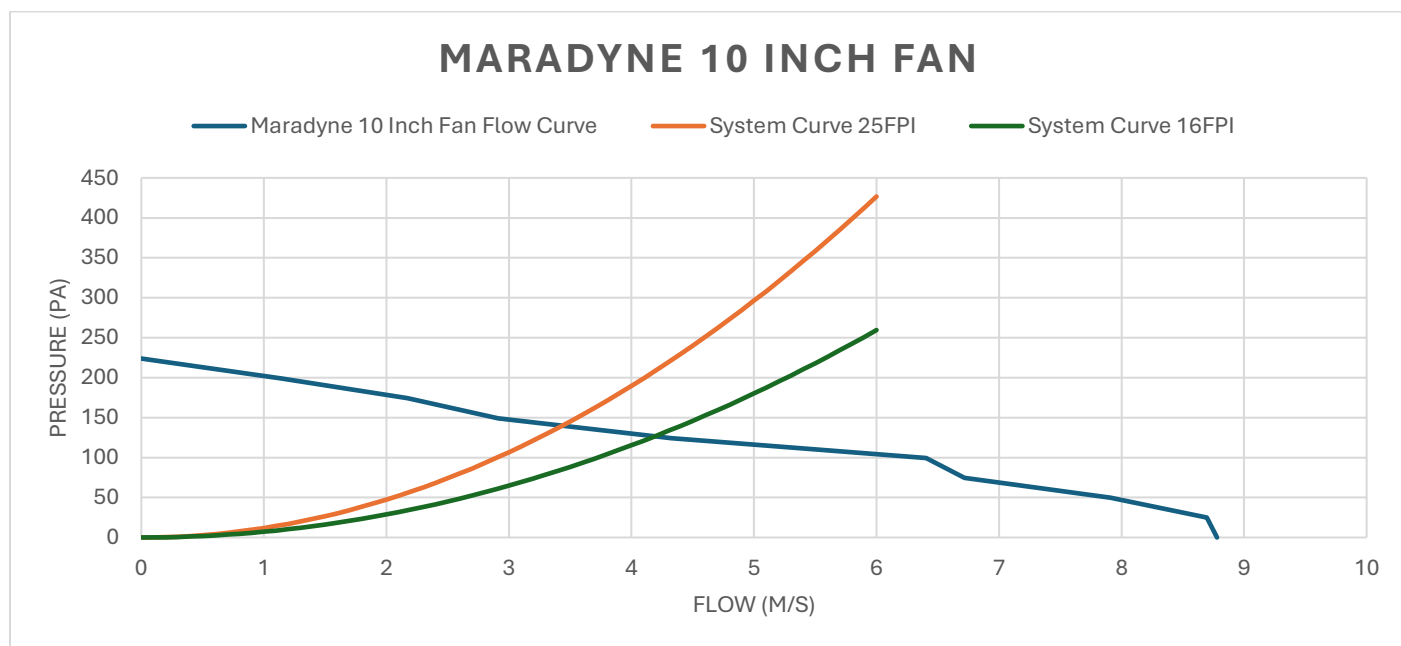


Figure 69: MARADYNE 10-Inch Fan Flow Curve



12-inch fans from Spal and Maradyne were also investigated as shown in figure 70 and 71 respectively, the results for velocity for a 16 FPI radiator was 3.3 m/s for Spal and 3.5 m/s for Maradyne, a 25 FPI radiator shows a velocity of 2.8 m/s for Spal and 2.9 m/s for Maradyne.

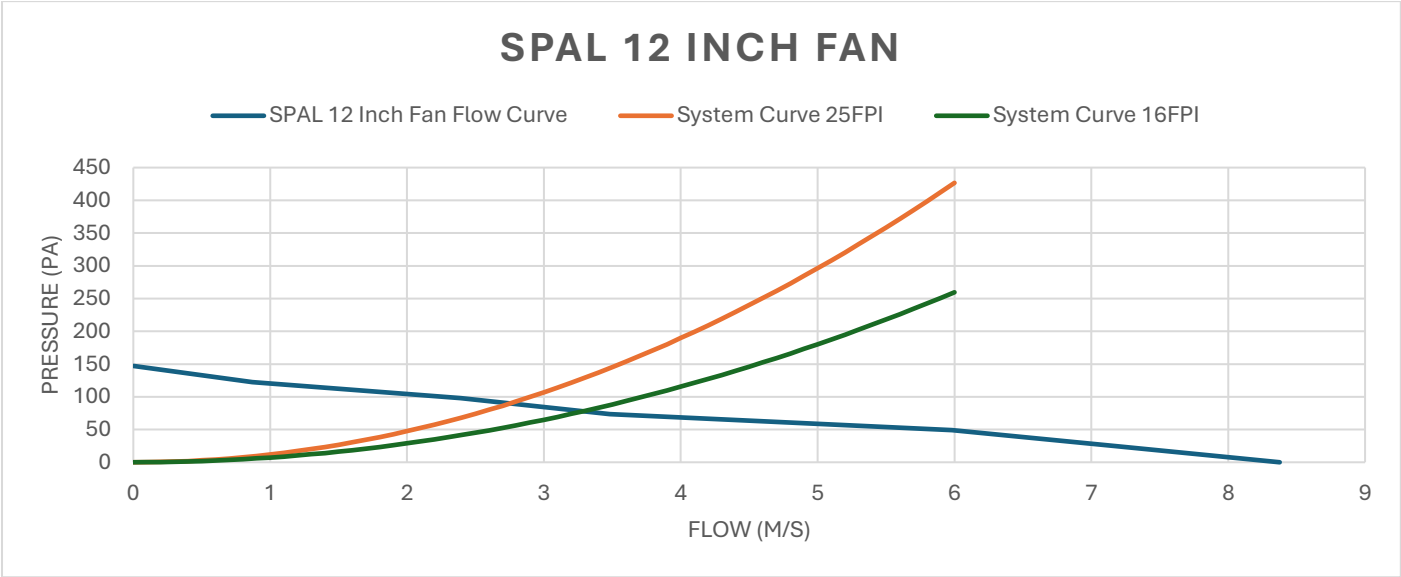


Figure 70: SPAL 12-Inch Fan Flow Curve

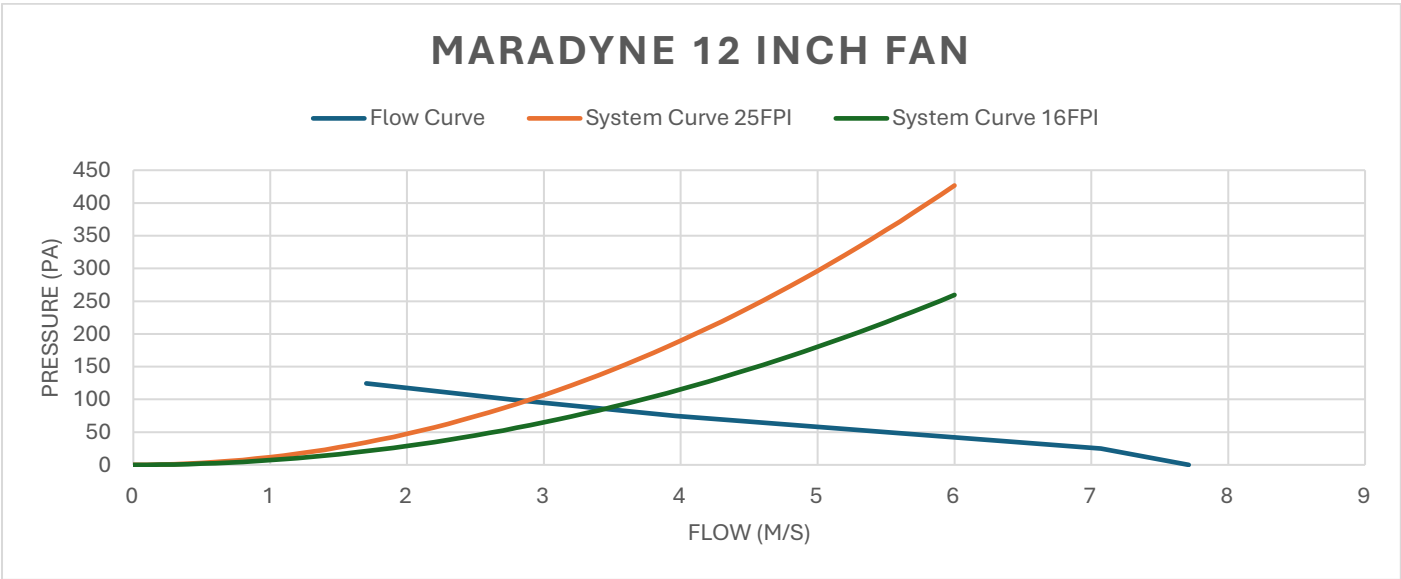


Figure 71: MARADYNE 12-Inch Fan Flow Curve

Additionally, a different configuration was also explored with fans in parallel or right next to each other similar



to that shown in figure 72. Fans in parallel increase the volume flow rate by double therefore when constructing the flow-pressure curve this was taken into consideration.

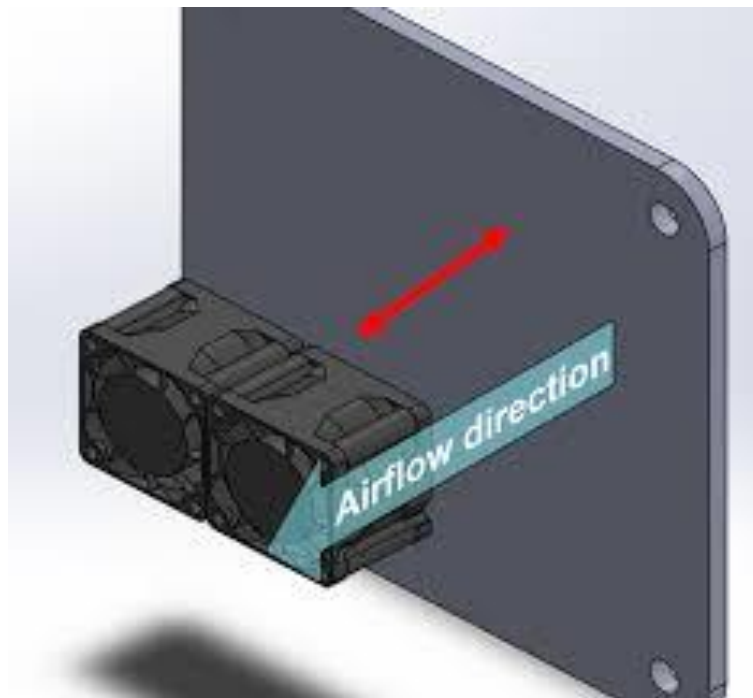


Figure 72: Cooling Fans in Parallel

This configuration was investigated using two Spal 4-inch fans where in figure 73 we can see that the fans achieved a velocity of 3.8 m/s through a 16 FPI radiator and 3.4 m/s through a 25 FPI radiator.

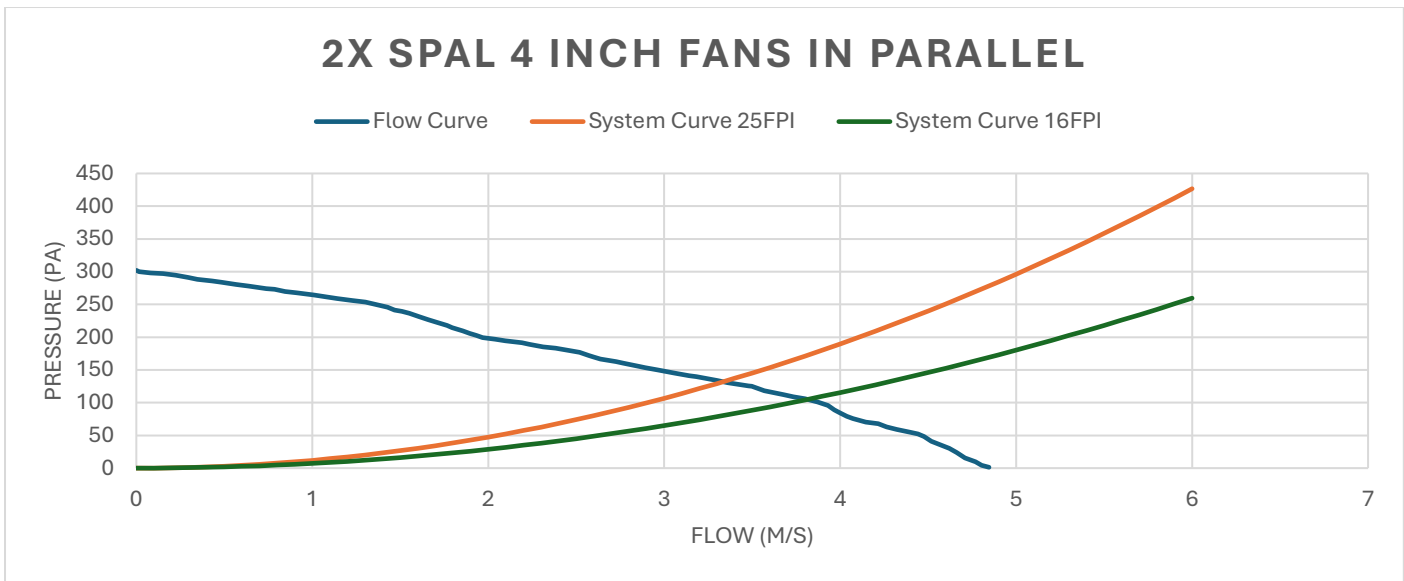


Figure 73: 2x SPAL 4-Inch Fans in Parallel Flow Curve

Table 7: Radiator Fan Comparison

| Fan                            | Operating Velocity for 16FPI (m/s) | Operating Flow for 16FPI (m <sup>3</sup> /s) | Operating Velocity for 25FPI (m/s) | Operating Flow for 25FPI (m <sup>3</sup> /s) | Mass (kg) | Current (Amps) | Price (\$) |
|--------------------------------|------------------------------------|----------------------------------------------|------------------------------------|----------------------------------------------|-----------|----------------|------------|
| SPAL 7.5 Inch Fan              | 4.5                                | 0.126                                        | 3.8                                | 0.107                                        | 0.9       | 7.3            | N/A        |
| MARADYNE 8 Inch Fan            | 3.9                                | 0.124                                        | 3.4                                | 0.108                                        | -         | 8              | 212.5      |
| SPAL 10 Inch Fan               | 3.6                                | 0.1794                                       | 3                                  | 0.149                                        | 1         | 6.2            | 193.6      |
| MARADYNE 10 Inch Fan           | 4.2                                | 0.209                                        | 3.5                                | 0.174                                        | -         | 8.3            | 199        |
| SPAL 12 Inch Fan               | 3.3                                | 0.237                                        | 2.8                                | 0.201                                        | 2.3       | 6.8            | 196        |
| MARADYNE 12 Inch Fan           | 3.4                                | 0.244                                        | 2.9                                | 0.208                                        | -         | 7.8            | 229        |
| 2x SPAL 4 Inch Fans (Parallel) | 3.8                                | 0.121                                        | 3.3                                | 0.105                                        | 0.76      | 6              | 338        |

Looking at table 7 for comparison we can instantly eliminate the 8-inch fan as its performance is less than the current fan with a similar size. The Spal 10-inch and 12-inch fans both produce less velocity and thus flow than the Maradyne fans, therefore these Spal fans are chosen not to be of use. The parallel 4-inch fans achieved

a slightly worse result than the 8-inch fan but had a bit of weight reduction and less current needed, therefore this configuration could work well as long as the cooling capacity meets requirements at the lower velocity but larger area.

This leaves the current 7.5-inch Spal radiator fan, both the 10 and 12-inch Maradyne fans and the two 4-inch parallel Spal fans. For this comparison the Maradyne fans are assumed to be the same mass as the Spal fans as they do not supply this value in their data sheets. The parallel fans cost quite a bit more than the other fans because of the need to buy multiple of the same fans. The 12-inch fan would allow for much better cooling capacity due to the increased airflow over area but weighs more than double of each of the other fans. The 10-inch fan would also lead to great cooling capacity for the cooling system and has quite a low amount of mass.

After considering performance, cost and weight, it was decided that for a fan without a shroud either the 7.5-inch or 10-inch fan would be sufficient. Although, due to budget problems with the motor controller, it was decided to stay with the current 7.5-inch fan as it has shown itself to be reliable from last year's results.

## **7.2 Radiator Fan Shroud Venturi Effect**

With the choices of the fans, a study was also conducted using the venturi effect to optimize air velocity using a shroud. To optimize the venturi effect, a fan of area larger than the face area of the radiator core is to be chosen. The venturi effect described the increase in velocity and decrease in pressure when cross sectional area is decreased. The continuity equation allows us to determine the amount of an increase or decrease in velocity due to the change in area, this can be seen below (Contributors, 2019).

$$Q_1 = Q_2$$

$$A_1 V_1 = A_2 V_2$$

$$\frac{A_2}{A_1} = \frac{V_1}{V_2}$$

Where  $A_1$  = Area of radiator core face,  $A_2$  = Area of fan,  $V_1$  = Velocity at radiator core face and  $V_2$  = Velocity

at fan.

Therefore, a fan with an area larger than  $57400 \text{ mm}^2$  for the current radiator and  $52000 \text{ mm}^2$  for the new radiator is required. Any fan above 10 inches would be sufficient for supplying extra velocity, for this study a 12-inch fan was chosen where the area of the fan would be  $72965.88 \text{ mm}^2$ . For the current radiator the increase in velocity is shown below.

$$\frac{72965.88}{57400} = \frac{V_1}{V_2}$$

$$\frac{V_1}{V_2} = 1.27$$

$$V_1 = 1.27V_2$$

For the new radiator,

$$\frac{72965.88}{52000} = \frac{V_1}{V_2}$$

$$\frac{V_1}{V_2} = 1.40$$

$$V_1 = 1.40V_2$$

This shows that for the current radiator the velocity at the core face would be 1.27 times the amount at the fan while the new radiator would be 1.40 times. Both of these results would have an increase in cooling capacity for the cooling system but with a shroud and a bigger fan the weight would increase by quite an amount.

CFD was done for this study as well to compare airflow results using the sizing from the new radiator as the core face, two models were created with one being the 7.5-inch fan shown in figure 74 and the other the 12-inch fan shown in figure 75. Both of these models had the fan modelled and put inside to simulate the fan rotating in CFD.

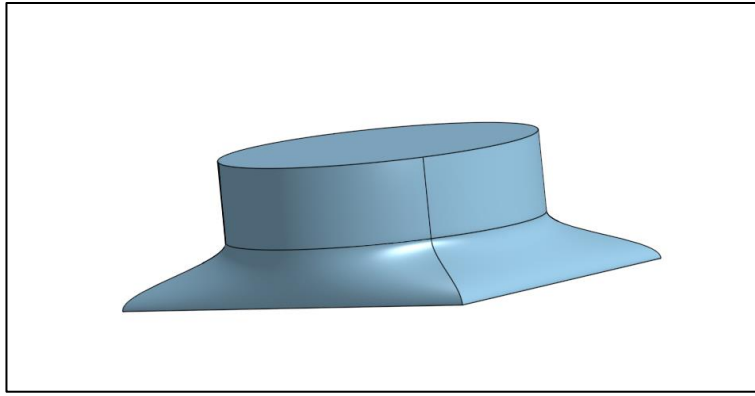


Figure 74: CFD Shroud Model with 7.5 Inch Fan

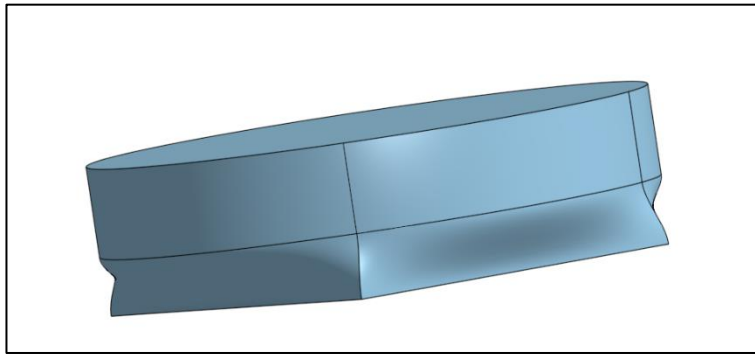


Figure 75: CFD Shroud Model with 12 Inch Fan

Using the same CFD configurations and tools for the shroud study results were created for both these models where the 7.5-inch fan rotated at 4000 rpm and the 12-inch fan rotated at 2800 rpm, these values were not supplied by the manufacturer therefore were chosen as standard rpm for a cooling fan at the sizes used.

The results which we were looking for were the comparisons of velocity for these two fans at the core face, expecting to see that the 12-inch fan has a higher air velocity. In figure 76 the 12-inch fan models' velocity is shown to be higher on average than that of the 7.5-inch fan model therefore allowing the option for a configuration of a 12-inch fan with a shroud to be used in a cooling system if the cooling system is limited with the current fan.

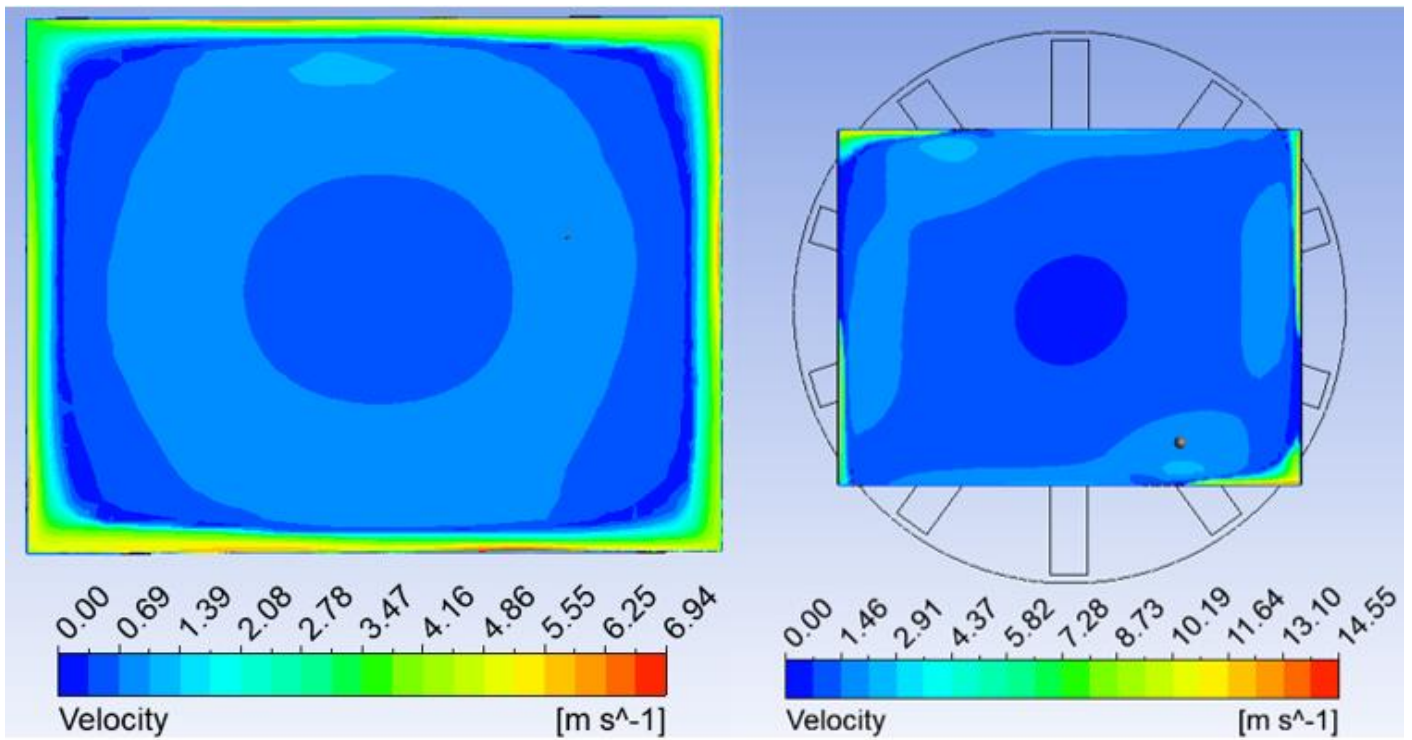


Figure 76: Core Face Velocity of 7.5 Inch Fan Shroud Model (Left) and 12 Inch Fan Shroud Model (Right)

## 8 Enclosures and Manifolds

In addition to working on the powertrain cooling system other tasks including the design and manufacture of various enclosures and manifolds to hold different components for NU24, these enclosures included an enclosure for the Central Electronic Node (CEN) which was the low voltage motherboard for the car, also a manifold for the cooling fans used for the accumulator or battery pack.

### 8.1 CEN Enclosure Design

Manufacturing the CEN enclosure design was a significant aspect of the project. The enclosure allows the safety of the CEN circuit board which is an indispensable part of NU24. This enclosure was to be designed so the board was secured inside of it allowing for minimal material usage. The enclosure was 3D printed using PETG to allow for ease of manufacturability, low cost and production of multiple prints if needed.

The CEN enclosure initially was to have two separate lids with one having constant connection with the board, being screwed into the Deutsch DT connectors as shown below.

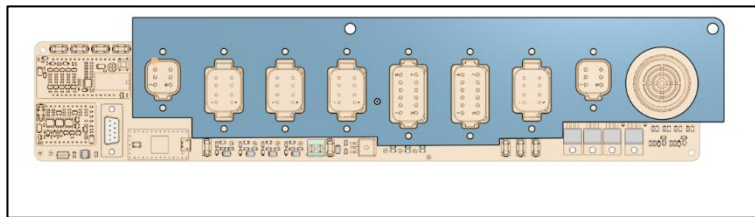


Figure 77: First CEN Enclosure Lid Design

This initial design was flawed though as there was no major connection on the bottom side meaning it could be deformed easily. Therefore, a single lid solution was proposed and designed. This lid was to sit as low as possible on the board, meaning a hollow protrusion was to be used to accommodate some components which protruded higher.

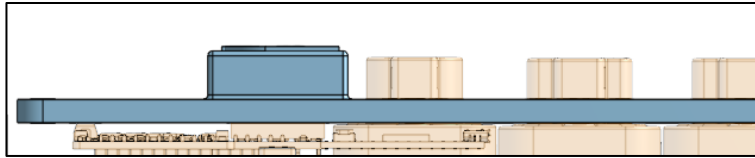


Figure 78: Protrusions on CEN Enclosure Lid

The housing was designed to surround the rest of the circuit board, with a hole on the side of it to allow a micro-USB cable to go through for ease of teensy code edits. The housing was connected to the lid via M5 bolts and threaded inserts at the bottom of the housing with the enclosure being connected to the car using the same method but from the side of the housing.

For printing, the housing and lid were split into two halves and glued together after. The enclosure is shown in the figure below.



Figure 79: 3D Printed CEN Enclosure

There were a few modifications to the lid after the final design was printed due to a few parts of the CEN sitting higher than originally thought and shown in the CAD model. Therefore, a few hollow sections on the underside of the lid were designed and printed to allow for the lid to sit flush and level.



This CEN was to be tested on NU23 therefore a temporary mounting system was to be used on the enclosure, the solution was to use zip ties to hold it in place onto the tractive systems housing. This was a simple yet effective method and showed to hold quite well as shown below in figure 80.

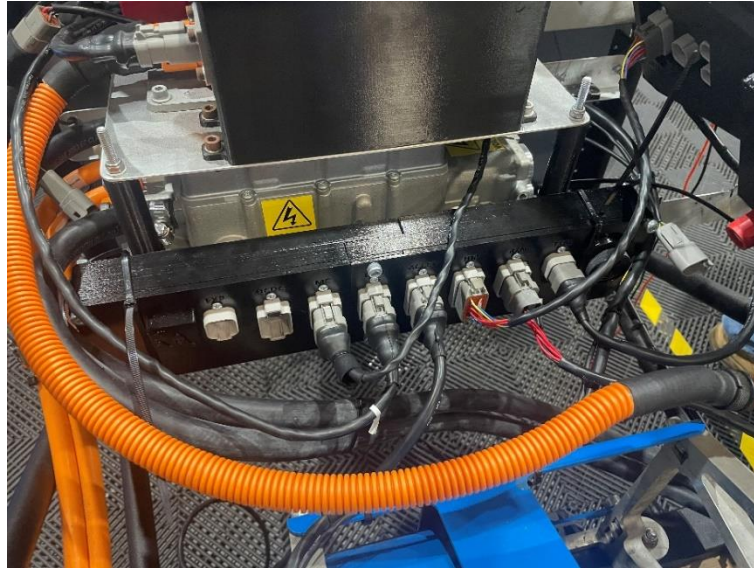


Figure 80: CEN Enclosure on NU23

Due to the constant mounting and design changes with the tractive system the job of designing the CEN enclosure was offloaded to the person handling the tractive system, this allowed the CEN enclosure to be easily incorporated into the whole design without having to go back and forth while designing. The new design shown in figure 81 below, was very similar to the temporary one created with only small edits and additions while being mounted to the firewall.

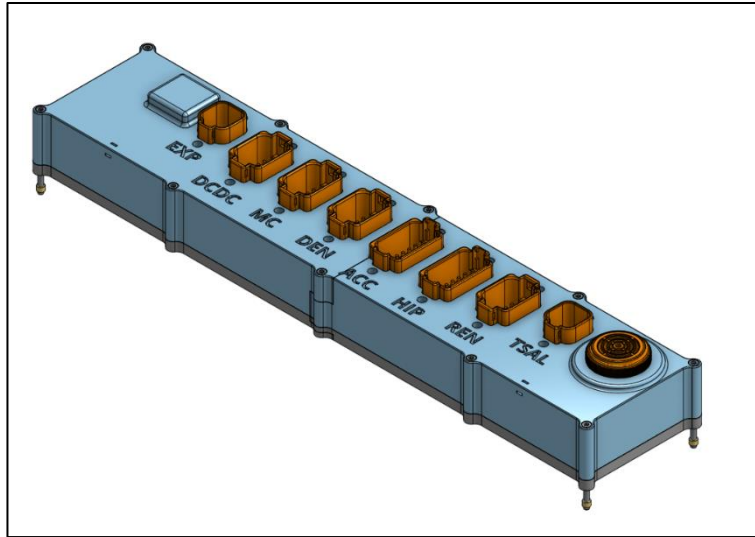


Figure 81: Final CEN Enclosure Design CAD Model

## 8.2 Accumulator Cooling Fans Manifold

Due to the changes in the accumulator container and lid from last year's design, the cooling fans manifolds had to be changed as well to be incorporated into this new design. The main accumulator enclosure design change that affected the design of the cooling fans manifolds was the connections to the lid which was enlarged due to the material change from steel to aluminium. Design changes from the accumulator enclosure are shown in figure 82 with the connections circled.

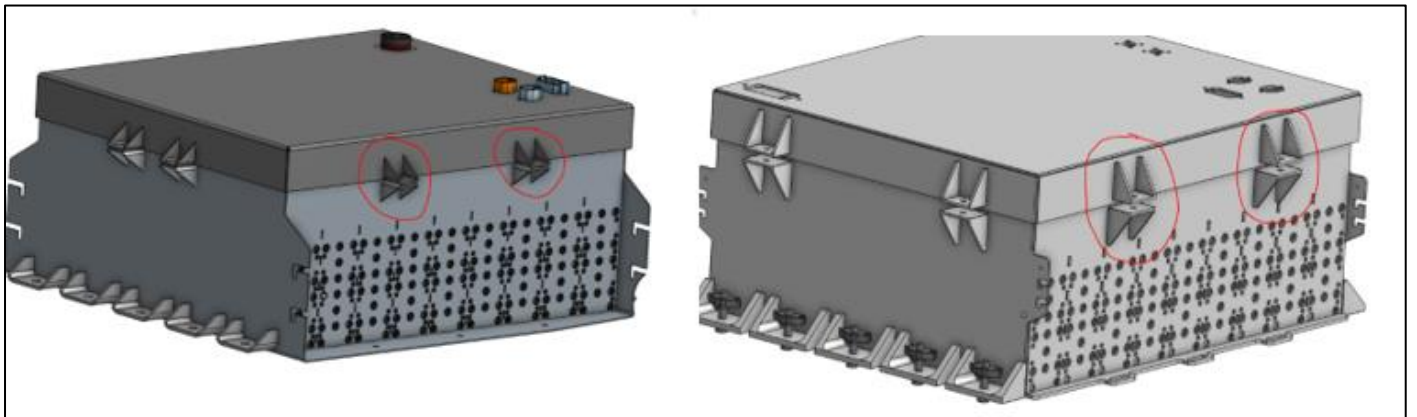


Figure 82: Accumulator Enclosures, NU23 (Left) and NU24 (Right)

Therefore, a fan manifold was required to be made around these angled connections. The usage of last year's fans, wires and Deutsch DT connectors allowed for only the manifold to be created. This manifold was to follow similar design to last year except slots were made out of the top channel which housed the wires to allow for an easy and reliable fit to the side faces of the accumulator container. The usage of PETG and 3D printing allowed for ease of manufacture, ability for multiple design changes if required, low costs and less weight requirements. Three sets of channel and housing were made due to the 3D printer dimension restrictions, where small dowels with glue were used to connect these sets to create the manifold shown below in figure 83.

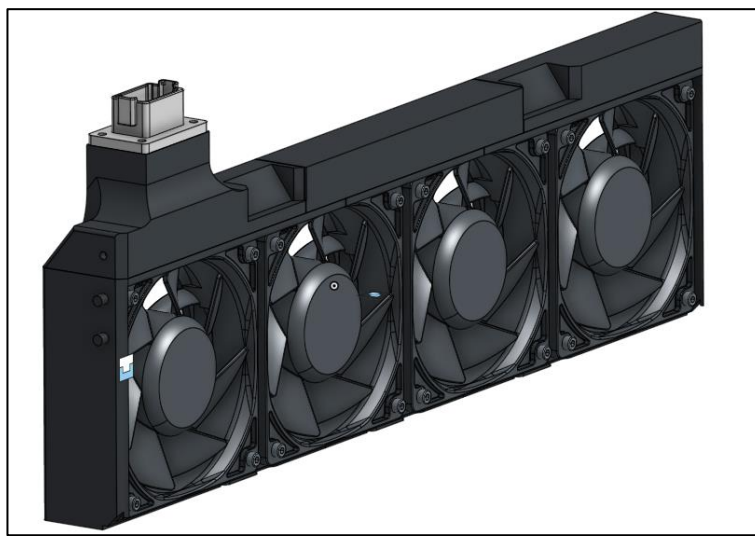


Figure 83: Accumulator Fan Manifold CAD Model

This manifold was to be mounted to opposite sides of the accumulator container using dowels on the sides and the bottoms. Additionally, wire mesh was added to the face of the fans and foam tape to protect the aluminium and batteries from environmental factors, anti-vibration was also a reason for the foam tape usage to allow safety to the fans and the container. This design was then 3D printed shown in figure 84, then fitted to the accumulator enclosure to allow for cooling to the batteries.

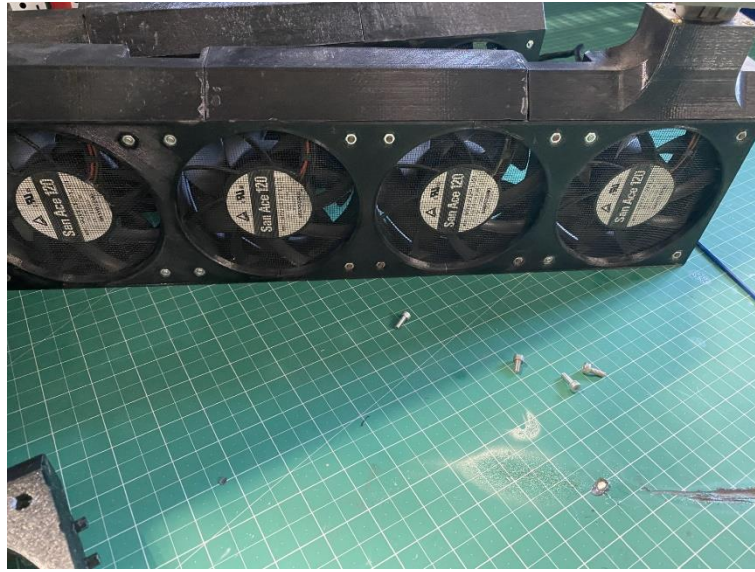


Figure 84: 3D Printed Accumulator Fan Manifold

## 9 Cost Report with Wiring and Hose CAD Models

A major part of the FSAE-A competition is the cost report which accounts for 10% of the entire points possible. The cost report provides an overview of the costing of manufacturing the entire car using the currency values provided by FSAE. Each senior team member of NU racing had to play a part in completing the cost report describing how their own system and some other systems including but not limited to; brake systems, engine and tractive system, electrical, etc, would be manufactured. This ranged from something as small as a fastener to as big as for example die casting, though the larger processes were not necessary for NU24.

The cost report included assemblies in their respective systems and how they are assembled with the parts included. Each part was also detailed how they were assembled and manufactured. An example of how an assembly looks in the cost report can be seen in figure 85 below showing the parts and their respective costs then processes with relevant process multipliers depending on material, fastener or machining usage and all fasteners used to assemble said assembly.

|                            |                              |                                                           |                                     |               |            |
|----------------------------|------------------------------|-----------------------------------------------------------|-------------------------------------|---------------|------------|
| University                 | University of Newcastle      | A/N                                                       | E03-20-DR-031000-01                 | Part Cost     | N/A        |
| Competition Code           | FSAE-A                       | System                                                    | Engine/Tractive Path and Drivetrain | QTY           | N/A        |
| Year                       | 2024                         | Assembly                                                  | Cooling                             | Extended Cost | \$ 233.81  |
| Car #                      | E03                          | Part                                                      | N/A                                 |               |            |
| <b>Parts</b>               |                              |                                                           |                                     |               |            |
| IC                         | P/N                          | Description                                               | Part Cost                           | QTY           | Sub Total  |
|                            | E03-20-DR-052001-01          | Radiator                                                  | \$ 115.02                           | 1             | \$ 115.02  |
|                            | E03-20-DR-052002-01          | Water Pumps                                               | \$ 20.49                            | 2             | \$ 40.98   |
|                            | E03-20-DR-052003-01          | Radiator Fan                                              | \$ 30.00                            | 1             | \$ 30.00   |
|                            | E03-20-DR-052004-01          | Radiator Fan Shroud                                       | \$ 17.36                            | 1             | \$ 17.36   |
|                            | E03-20-DR-052005-01          | Catch Can                                                 | \$ 16.44                            | 1             | \$ 16.44   |
| Total                      |                              |                                                           |                                     |               | \$ 219.80  |
| <b>Processes</b>           |                              |                                                           |                                     |               |            |
| <b>Process Multipliers</b> |                              |                                                           |                                     |               |            |
| IC                         | Process                      | Use                                                       | Unit                                | QTY           | Multiplier |
|                            | Assemble, 3 kg, Interference | Assemble Radiator into position                           | unit                                | 1             | None       |
|                            | Assemble, 1 kg, Loose        | Washer onto bolts                                         | unit                                | 1             | None       |
|                            | Hand - Start Only            | Hand start radiator fasteners                             | unit                                | 1             | None       |
|                            | Ratchet <= 6.35 mm           | Fasten into anti-vibration mounts                         | unit                                | 1             | None       |
|                            | Assemble, 1 kg, Interference | Assemble pump into pump holder                            | unit                                | 1             | None       |
|                            | Assemble, 1 kg, Interference | Assemble pump in pump holder onto bracket                 | unit                                | 1             | None       |
|                            | Hand - Start Only            | Hand start bolts into pump holders to bracket             | unit                                | 2             | None       |
|                            | Ratchet <= 6.35 mm           | Tighten bolts                                             | unit                                | 2             | None       |
|                            | Reaction Tool <= 6.35 mm     | Tighten bolts                                             | unit                                | 2             | None       |
|                            | Assemble, 1 kg, Interference | Assemble catch can onto bracket                           | unit                                | 1             | None       |
|                            | Hand - Start Only            | Hand start bolts, washers and nuts from catch can to brac | unit                                | 1             | None       |
|                            | Wrench <= 6.35 mm            | Tighten fasteners                                         | unit                                | 1             | None       |
|                            | Reaction Tool <= 6.35 mm     | Tighten fasteners                                         | unit                                | 1             | None       |
|                            | Assemble, 1 kg, Interference | Assemble top shroud part onto radiator                    | unit                                | 1             | None       |
|                            | Assemble, 1 kg, Interference | Assemble bottom shroud part onto radiator                 | unit                                | 1             | None       |
|                            | Hand - Start Only            | Hand start bolts and nuts for shroud parts                | unit                                | 1             | None       |
|                            | Wrench <= 6.35 mm            | Fasten shroud parts together                              | unit                                | 1             | None       |
| Total                      |                              |                                                           |                                     |               | \$ 13.07   |
| <b>Fasteners</b>           |                              |                                                           |                                     |               |            |
| IC                         | Fasteners                    | Use                                                       | Size 1                              | Unit 1        | Size 2     |
|                            | Bolt, Grade 12.9             | Radiator and Pump Mount bolts                             | 6 mm                                |               | 25 mm      |
|                            | Washer, Grade 12.9           | Radiator and Pump Mount washers                           | 6 mm                                |               | 0          |
|                            | Bolt, Grade 12.9             | Pump Bolts                                                | 6 mm                                |               | 25 mm      |
|                            | Washer, Grade 12.9           | Pump Washers                                              | 6 mm                                |               | 0          |
|                            | Nut, Grade 12.9              | Pump Nuts                                                 | 6 mm                                |               | 0          |
|                            | Bolt, Grade 12.9             | Catch Can Bolts                                           | 6 mm                                |               | 15 mm      |
|                            | Washer, Grade 12.9           | Catch Can Washers                                         | 6 mm                                |               | 0          |
|                            | Nut, Grade 12.9              | Catch Can nuts                                            | 6 mm                                |               | 0          |
| Total                      |                              |                                                           |                                     |               | \$ 0.94    |

Figure 85: Cost Report Section for Cooling System Assembly

Each of these parts also had a similar layout to the assemblies except the addition of materials and much more extensive processes. For example, the cost report for the radiator can be shown below in figure 86.



|                  |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          |           |
|------------------|------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------|---------------|----------------------|----------------------|-----------------------|------------|------------|-----------------|-----|----------|-----------|
| University       | University of Newcastle                                    | BMN                                                                      | 003-20-001-00200-01              | Part Cost     | \$ 16.02             |                      |                       |            |            |                 |     |          |           |
| Designation Code | P046-6                                                     | Structure                                                                | Engine/Tractor Path and Overview | QTY           | 1                    |                      |                       |            |            |                 |     |          |           |
| Year             | 2024                                                       | Assembly                                                                 | Cooling                          | Extended Cost | \$ 16.02             |                      |                       |            |            |                 |     |          |           |
| Unit             | 003                                                        | Part                                                                     | Radator                          |               |                      |                      |                       |            |            |                 |     |          |           |
| Materials        |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          |           |
| ID               | Material                                                   | Description                                                              | Size 1                           | Unit 1        | Size 2               | Unit 2               | Area Name             | Area (sqm) | Length (m) | Density (kg/m³) | QTY | Unit Co. | Sub Total |
|                  | Heat Exchanger, Air-to-Liquid                              | Radator                                                                  |                                  | 7501.5 cm³    |                      |                      | 0                     | 51750      | 26         |                 | 1   | \$ 5.26  | \$ 5.26   |
|                  | Aluminium, Normal (per kg)                                 | Tube (side and sensor)                                                   |                                  | 0.005264 kg   |                      |                      | 0.2mm Aluminium Sheet | 3600       | 2          | 2710            | 1   | \$ 4.20  | \$ 4.08   |
|                  | Aluminium, Normal (per kg)                                 | Large Tube (bottom)                                                      |                                  | 0.002544 kg   |                      |                      | 0.3mm Aluminium Sheet | 4000       | 3          | 2710            | 1   | \$ 4.20  | \$ 4.04   |
|                  | Aluminium, Normal (per kg)                                 | Pipe Flare                                                               |                                  | 0.0103241 kg  |                      |                      | 0.45mm Round Bar      | 7501.4370  | 25         | 2710            | 1   | \$ 4.20  | \$ 0.95   |
|                  | Aluminium, Normal (per kg)                                 | Pipe Tube                                                                |                                  | 0.00000000 kg |                      |                      | 0.25mm Round Tube     | 207.34510  | 30         | 2710            | 1   | \$ 4.20  | \$ 0.01   |
|                  | Aluminium, Normal (per kg)                                 | Upper Weld-on Thread Attachment                                          |                                  | 0.00400000 kg |                      |                      | 0.24mm Round Bar      | 422.35004  | 15         | 2710            | 2   | \$ 4.20  | \$ 0.11   |
|                  | Aluminium, Normal (per kg)                                 | Lower Weld-on Thread Attachment                                          |                                  | 0.00410000 kg |                      |                      | 0.20mm Round Bar      | 394.85027  | 15         | 2710            | 2   | \$ 4.20  | \$ 0.11   |
|                  | Aluminium, Normal (per kg)                                 | Temp Sensor Threaded Inset                                               |                                  | 0.00400000 kg |                      |                      | 0.22mm Round Bar      | 390.10271  | 15         | 2710            | 2   | \$ 4.20  | \$ 0.10   |
|                  | Aluminium, Normal (per kg)                                 | Sensor Cap                                                               |                                  | 0.00400000 kg |                      |                      | 0.22mm Round Bar      | 390.10271  | 15         | 2710            | 2   | \$ 4.20  | \$ 0.10   |
|                  | Adapted L.P. Barb to Male Pipe (Straight) Aluminium Rodded | M12 Straight Barb Radator Inlet and Outlet                               | 12 mm                            |               |                      | 12 mm                |                       |            |            |                 | 4   | \$ 6.21  | \$ 24.84  |
|                  | Adapted L.P. Barb to Male Pipe (Straight) Aluminium Rodded | Pipe Neck to Overflow Barb                                               | 8 mm                             |               |                      | 8 mm                 |                       |            |            |                 | 1   | \$ 3.36  | \$ 3.36   |
|                  | Flare Cap                                                  | Flare Cap                                                                |                                  | 0             |                      | 0                    |                       |            |            |                 | 1   | \$ 3.00  | \$ 3.00   |
|                  | Hose, Rubber (per m)                                       | 3/8" Hose Iron Motor                                                     | 10 mm                            |               |                      | 10 mm                |                       |            |            |                 | 0.4 | \$ 3.06  | \$ 1.22   |
|                  | Hose, Rubber (per m)                                       | 1/2" Hose Iron Motor                                                     | 10 mm                            |               |                      | 10 mm                |                       |            |            |                 | 1.4 | \$ 3.42  | \$ 4.75   |
|                  | Hose, Rubber (per m)                                       | 1/2" Hose to Pump                                                        | 10 mm                            |               |                      | 10 mm                |                       |            |            |                 | 0.8 | \$ 3.42  | \$ 2.05   |
|                  | Seal, O-Ring, O-Rings                                      |                                                                          | 4                                | 0             | 0                    | 0                    |                       |            |            |                 | 4   | \$ 0.05  | \$ 0.20   |
|                  | Mount, Vibration Dampening Sandwich                        | Anti-Vibration Mounts                                                    | 20 mm                            |               |                      | 20 mm                |                       |            |            |                 | 4   | \$ 5.40  | \$ 21.60  |
|                  | Flare, Coolant, Downward or Straight View                  | Flared Mount - Coolant                                                   | 1                                |               |                      | 1                    |                       |            |            |                 | 1   | \$ 0.10  | \$ 0.10   |
| Total            |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          | \$ 67.63  |
| Processes        |                                                            |                                                                          |                                  |               |                      | Process Multipliers  |                       |            |            |                 |     |          |           |
| ID               | Process                                                    | Use                                                                      | Unit                             | QTY           | Multiplier           | Multi. Val           | Multiplier 2          | Multi. V2  | Unit Cost  | Sub Total       |     |          |           |
|                  | Machining Setup, Install and remove                        | Setup Laser Cutter - 2 mm                                                | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.30    | \$ 1.30         |     |          |           |
|                  | Laser Cut                                                  | Laser Cut - Bottom Tab                                                   | cm                               | 12.43         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.01    | \$ 0.12         |     |          |           |
|                  | Laser Cut                                                  | Laser Cut - Side Tab                                                     | cm                               | 13.83         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.01    | \$ 0.14         |     |          |           |
|                  | Machining Setup, Change                                    | Setup Laser Cutter - 3 mm                                                | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Laser Cut                                                  | Laser Cut Large Bottom Tab                                               | cm                               | 21.70         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.01    | \$ 0.21         |     |          |           |
|                  | Machining Setup, Install and remove                        | Setup Laser and Place in Upper Weld Thread Attachment                    | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.30    | \$ 1.30         |     |          |           |
|                  | Machining                                                  | Box-out Ø70 Strip Hole                                                   | cm3                              | 1.258         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.10         |     |          |           |
|                  | Machining                                                  | Box-out Ø20 Holes                                                        | cm3                              | 0.6827        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.05         |     |          |           |
|                  | Machining Setup, Change                                    | Change tool                                                              | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Threading, Internal (inching)                              | Turn M12 thread on Upper Weld Thread Attachment                          | cm                               | 1.2           | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.10    | \$ 0.24         |     |          |           |
|                  | Machining Setup, Change                                    | Place Lower Weld Attachment in Lathe                                     | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Machining                                                  | Box-out Centre of Lower Weld Thread Attachment                           | cm3                              | 1.1250        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.09         |     |          |           |
|                  | Machining Setup, Change                                    | Change tool                                                              | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Threading, Internal (inching)                              | Turn M12 thread on Lower Weld Thread Attachment                          | cm                               | 1.2           | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.10    | \$ 0.24         |     |          |           |
|                  | Machining Setup, Change                                    | Place Lower Weld Attachment in Lathe                                     | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Machining                                                  | Box-out centre of Temp Sensor Threaded Inset                             | cm3                              | 1.0085        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.14         |     |          |           |
|                  | Machining Setup, Change                                    | Change tool                                                              | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Threading, Internal (inching)                              | Thread Hole for Temp Sensor Inset                                        | cm                               | 1.5           | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.10    | \$ 0.30         |     |          |           |
|                  | Machining Setup, Change                                    | Place Flare Pattern in Lathe                                             | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.05    | \$ 0.05         |     |          |           |
|                  | Machining                                                  | Box-out centre Ø70 hole                                                  | cm3                              | 5.6706        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.23         |     |          |           |
|                  | Machining                                                  | Box-out Ø32 hole                                                         | cm3                              | 0.3335        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.13         |     |          |           |
|                  | Machining Setup, Install and remove                        | Setup mill                                                               | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.30    | \$ 1.30         |     |          |           |
|                  | Machining                                                  | Machine Overflow Hole                                                    | cm3                              | 0.108         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.04    | \$ 0.03         |     |          |           |
|                  | Threading, Internal (inching)                              | Thread Overflow Hole                                                     | cm                               | 0.55          | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.10    | \$ 0.05         |     |          |           |
|                  | Tube and preparation for welding                           | Prep Flare Tube                                                          | and                              | 2             | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 1.50         |     |          |           |
|                  | Weld                                                       | Weld Flare Tube to Radator                                               | cm                               | 1.094         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 1.10         |     |          |           |
|                  | Weld                                                       | Weld Flare Tube to Flare Tube                                            | cm                               | 1.094         | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 1.10         |     |          |           |
|                  | Weld                                                       | Weld Upper Weld-on Thread Attachment to Radator                          | cm                               | 1.5454        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 2.25         |     |          |           |
|                  | Weld                                                       | Weld Lower Weld-on Thread Attachment                                     | cm                               | 6.2832        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 1.06         |     |          |           |
|                  | Weld                                                       | Weld Temp Sensor Threaded Inset to Radator                               | cm                               | 6.7544        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 2.03         |     |          |           |
|                  | Weld                                                       | Weld Side Tube to Radator                                                | cm                               | 6.7           | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 2.01         |     |          |           |
|                  | Weld                                                       | Weld Bottom Tube to Radator                                              | cm                               | 4.7071        | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 1.41         |     |          |           |
|                  | Weld                                                       | Weld Large Bottom Tab to Radator                                         | cm                               | 5.56          | Material - Aluminium | 1                    | 1                     | 1          | \$ 0.15    | \$ 2.07         |     |          |           |
|                  | Assemble, Tight, Loose                                     | Assemble O-Rings onto Upper Weld Thread Attachment and Temp Sensor Inset | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.06    | \$ 0.06         |     |          |           |
|                  | Hand - Start Only                                          | Start Upper and Lower Straight Barbs                                     | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.12    | \$ 0.48         |     |          |           |
|                  | Wrench (x 25.4 mm)                                         | Tighten Upper and Lower Straight Barbs                                   | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.50    | \$ 6.00         |     |          |           |
|                  | Assemble, Tight, Loose                                     | Anti-Vibration Mounts to Side and Bottom Tab                             | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.06    | \$ 0.24         |     |          |           |
|                  | Assemble, Tight, Loose                                     | Mounts onto Anti-Vibration Mounts                                        | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.06    | \$ 0.24         |     |          |           |
|                  | Hand - Start Only                                          | Start nuts on Anti-Vibration Mounts                                      | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.12    | \$ 0.48         |     |          |           |
|                  | Wrench (x 25.4 mm)                                         | Tighten nuts on Anti-Vibration Mounts                                    | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.15    | \$ 3.00         |     |          |           |
|                  | Assemble, Tight, Loose on Unit                             | Flare Cap onto Flare Barb                                                | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.10    | \$ 0.10         |     |          |           |
|                  | Hand - Start Only                                          | Start Overflow Barb                                                      | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.12    | \$ 0.12         |     |          |           |
|                  | Wrench (x 8.35 mm)                                         | Tighten Overflow Barb                                                    | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 1.00    | \$ 1.00         |     |          |           |
|                  | Tube out                                                   | Cut Radator to Pump Hose in Half                                         | cm                               | 1.9           | None                 | 1                    | 1                     | 1          | \$ 0.15    | \$ 0.29         |     |          |           |
|                  | Tube out                                                   | Cut Radator to Pump Hose to Length                                       | cm                               | 1.9           | None                 | 1                    | 1                     | 1          | \$ 0.15    | \$ 0.57         |     |          |           |
|                  | Tube out                                                   | Cut Hose to Radator Hose to Length                                       | cm                               | 1.7           | None                 | 1                    | 1                     | 1          | \$ 0.15    | \$ 0.26         |     |          |           |
|                  | Tube out                                                   | Cut Motor Controller to Radator Hose to Length                           | cm                               | 1.9           | None                 | 1                    | 1                     | 1          | \$ 0.15    | \$ 0.29         |     |          |           |
|                  | Assemble, Tight, Inference                                 | Hose onto Barb                                                           | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.19    | \$ 0.19         |     |          |           |
|                  | Assemble, Tight, Loose                                     | Slide on Hose Clamp (Loose)                                              | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.06    | \$ 0.24         |     |          |           |
|                  | Tighten (over) 1 Turn                                      | Tighten Hose Clamps                                                      | unit                             | 1             | 1                    | 1                    | 1                     | 1          | \$ 0.10    | \$ 2.00         |     |          |           |
| Total            |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          | \$ 43.47  |
| Fasteners        |                                                            |                                                                          |                                  |               |                      | Fastener Multipliers |                       |            |            |                 |     |          |           |
| ID               | Fastener                                                   | Use                                                                      | Size 1                           | Unit          | Size 2               | Unit 2               | QTY                   | Unit Cost  | Sub Total  |                 |     |          |           |
|                  | Hose Clamp, Worm Drive                                     | Fasten 3/8" Hose                                                         | 10 mm                            | 0             |                      | 0                    | 0                     | \$ 0.56    | \$ 0.56    |                 |     |          |           |
|                  | Hose Clamp, Worm Drive                                     | Fasten 1/2" Hose                                                         | 10 mm                            | 0             |                      | 0                    | 0                     | \$ 0.57    | \$ 1.72    |                 |     |          |           |
|                  | Washer, Grade 12.9                                         | Anti-Vibration Mounts                                                    | 6 mm                             | 0             |                      | 0                    | 0                     | \$ 0.01    | \$ 0.06    |                 |     |          |           |
|                  | Nut, Grade 12.9                                            | Anti-Vibration Mounts                                                    | 6 mm                             | 0             |                      | 0                    | 0                     | \$ 0.05    | \$ 0.20    |                 |     |          |           |
| Total            |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          | \$ 2.53   |
| Tooling          |                                                            |                                                                          |                                  |               |                      | Tooling Multipliers  |                       |            |            |                 |     |          |           |
| ID               | Tooling                                                    | Use                                                                      | Unit                             | QTY           | PWF                  | Frac. Incl.          | Unit Cost             | Sub Total  |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Flare Tube to Radator                                               | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Flare Tube to Flare Tube                                            | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Upper Weld-on Thread Attachment to Radator                          | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Lower Weld-on Thread Attachment                                     | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Temp Sensor Threaded Inset to Radator                               | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Side Tube to Radator                                                | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Bottom Tube to Radator                                              | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
|                  | Weld - Welding Fixture                                     | Weld Large Bottom Tab to Radator                                         | Number of Points                 | 1             | 3000                 | 1                    | \$ 500.00             | \$ 0.11    |            |                 |     |          |           |
| Total            |                                                            |                                                                          |                                  |               |                      |                      |                       |            |            |                 |     |          | \$ 1.33   |

Figure 86: Radiator Cost Report Section

Drawings were also required for every part and assembly of the race car including wiring, brake lines and cooling hoses. Therefore, these components had to be modelled to accurately depict the layout of NU24.

First, the wiring was modelled as it was the largest system of these three components. To model these wires multiple planes and sketches of circles or points were created in context of the electrical components on the

chassis shown in figure 87. This setup allowed for a 3D fit spline to be created to follow these sketches, then a sweep was applied with a sketch of a circle at the size of the wiring and then a 3D fit spline was used to create a wire. This was done for every major wire on the car with high voltage wires highlighted in orange colour. The wiring model by itself is shown in figure 88.

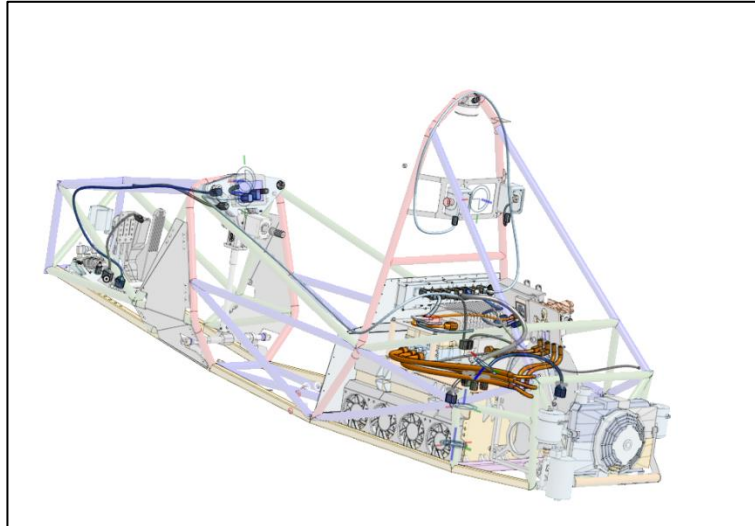


Figure 87: Wiring CAD Model with Main Assembly in Context

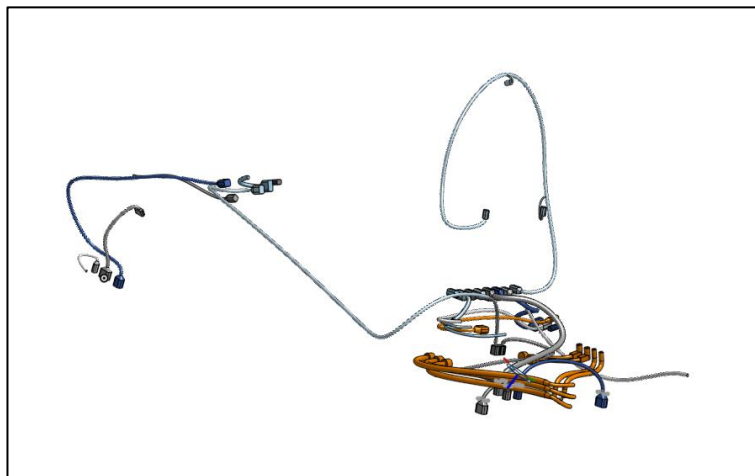


Figure 88: Wiring CAD Model by Itself

The brake lines and hoses were done in the exact same way the wiring was done except on a much smaller



scale. The brake lines CAD model can be seen in figure 89 while the cooling hoses can be seen in figure 90.

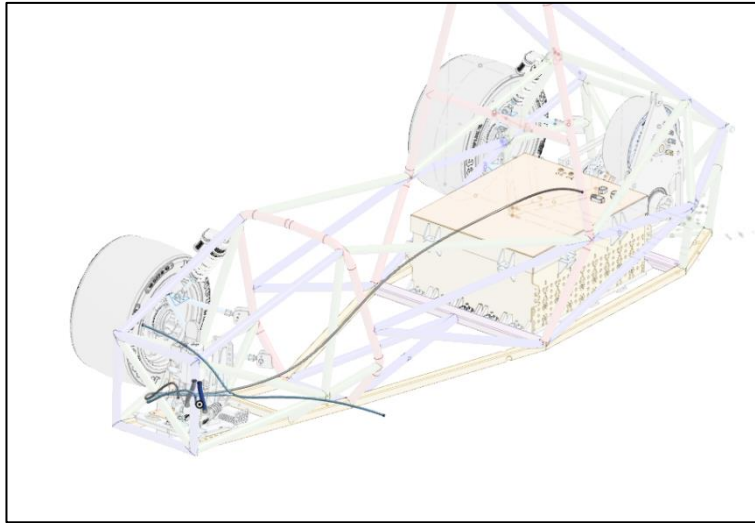


Figure 89: Brake Lines CAD Model with Main Assembly in Context

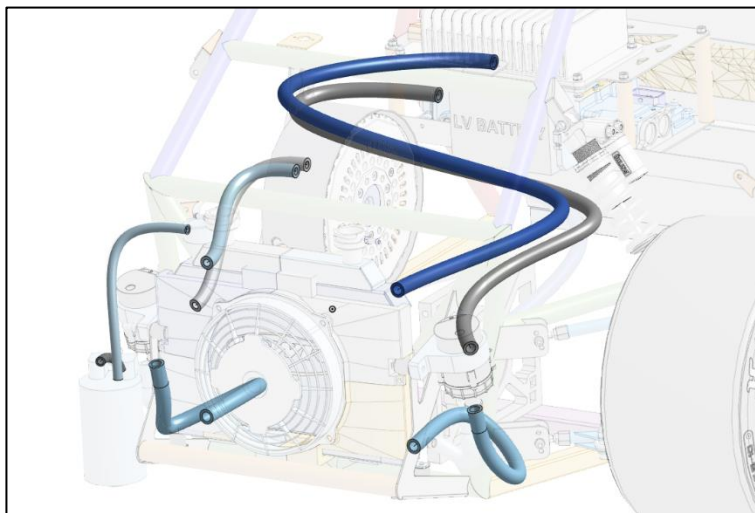


Figure 90: Cooling Hoses CAD Model with Main Assembly in Context

## 10 FSAE-A Competition 2025

At the 2024 FSAE-A competition, the cooling system for NU24's powertrain showed to perform exceptionally well, reinforcing the high quality of the cooling system designed. The cooling system's performance was determined to not be considered in the skid pad and acceleration events due to the lack of continuous driving and the ability for the car to cool down between runs. Therefore, only the autocross and endurance events would show the full ability of the cooling system designed throughout the year.

Unfortunately, during our autocross run we had an accumulator fault, meaning only one run was done for this event meaning that neither of the components which require cooling would reach anything close to a dangerous temperature leaving the data for this run unusable for cooling data.

In contrast, for the endurance event, NU24 made it through the entire twenty-two kilometers without faulting therefore leaving great data for cooling. Shown in the MoTeC data in figure 91 below, starting from around 27 degrees Celsius the highest temperatures seen from the coolant was 41 degrees Celsius which is much lower than what would be considered dangerous for the motor controller and 9 degrees below the recommended limit for the motor. These results showed that the cooling system may have been slightly over-specified, particularly for the motor controller's cooling loop allowing for potential design improvements for next year.

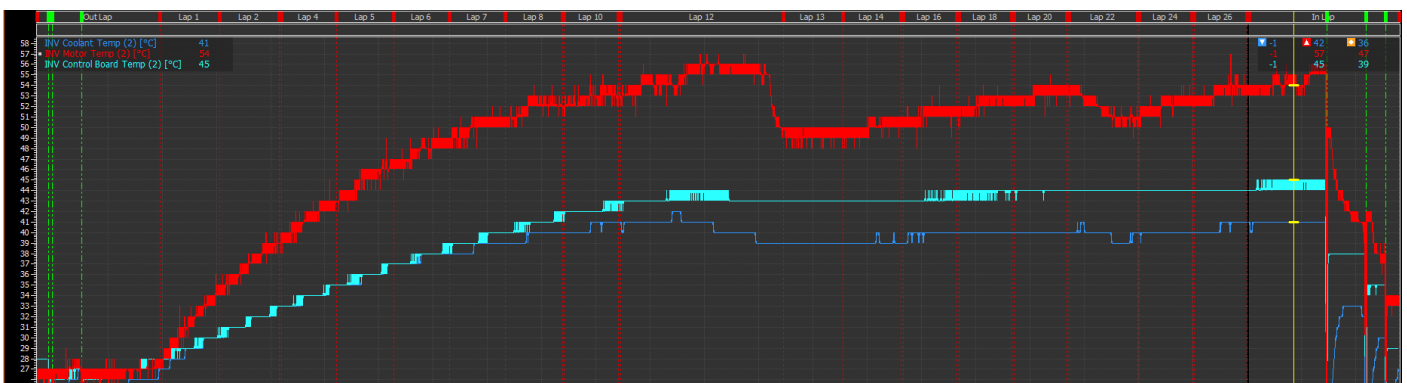


Figure 91: Competition Endurance Cooling Temperatures

## **11 NU25 Cooling System Design**

NU25 should see some upgrades in the cooling system in the form of weight reduction as the cooling system is suspected to be still quite oversized. This will be achieved with some new components including radiator, pumps, hoses, etc. Design changes to optimize safety and effectiveness of the cooling system is also a focus as bleeding has shown to be a problem in the past.

### **11.1 NU25 New Components**

To improve the cooling system, it is advised that the new radiator mentioned earlier I used, this is due to the lower mass, split cooling and welded barbs for more accurate and better cooling with less pressure.

Potential improvements could include the use of a more powerful pump much like the PCE-XL's or other pumps mentioned which have a higher head pressure with better flowrate to ensure the motor and motor controllers safety. With the pumps, if a new pumps or the old pumps being used it is advised that the use of AN-12 fittings for the pumps to be used, which will allow more convertibility to other fitting sizes and options if necessary.

The use of a shroud is something that can be researched again. Previously it has shown to be more efficient at cooling as a shroud should but may be chosen to be left off the car as did with this year due to the cooling capacity being adequate without it for NU23. With the smaller radiator in use as well, it may be even better without a shroud as the fan will occupy much more of the radiator cores face.

If cooling capacity is lacking, the use of a larger fan like a 12-inch fan could be of use as discussed earlier in this report, a 12-inch fan with a shroud could provide an increase in air velocity through the radiator core thus resulting in better cooling. However, this option would increase the mass of the cooling system.

### **11.2 NU25 Cooling System Design Concept**

Based on the knowledge for the best position of the pumps as shown in figure 92 and stated "When installing the EBP ensure the inlet of the EBP is positioned horizontally, vertically up or on an upward angle" (Electric

Booster Pump Installation Instructions, 2024). It is believed increasing the height position of the radiator and lowering the pumps is a must to allow for optimal bleeding and increased pump safety which will allow for an improved pump longevity.

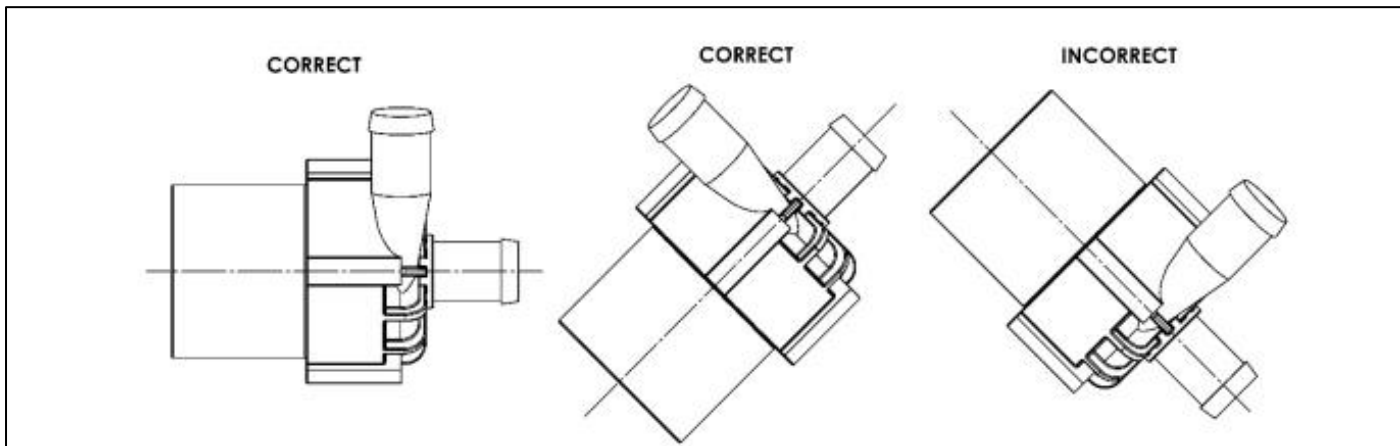


Figure 92: Correct Orientation of EBP40 (Electric Booster Pump Installation Instructions, 2024)

Therefore, splitting up the jacking point and the pumps seems to be the most optimal decision thus keeping the jacking point in line with the rules and allowing the cooling system to also work more efficiently and safer.

A new design for this setup was conceived for the use on NU25. First, the mounting was designed for the new radiator with new welded-on tabs on the sides and bottom to secure the radiator in place to a designed bracket. This design follows the previous welded-on mounts design which was seen on the last radiator with the use of anti vibration mounts, except the side mounting has changed to behind the radiator to allow easier shroud placement and design if the use of one is justified. These mounts can be seen in figure 93 below on the radiator CAD model.



Figure 93: New Radiator CAD Model with Welded-On Mounts

The radiator CAD model was then angled slightly and placed into an optimal position near the top of the rear of the chassis CAD model to make the brackets in context, these brackets were designed using as minimal amount of material as possible as they would only need to support the radiator which weighs very little even when wet, meaning almost no stress would be put on these brackets. These would just connect the anti-vibration mounts with the rear of the chassis as shown in figure 94.



Figure 94: Radiator Mounting Assembly on Rear of Chassis

Since the pumps would not occupy the radiator brackets and need to be lower than the radiators outlets it was decided to add them to the jacking point brackets. The jacking point was identified according to the rules where the jacking bar was modelled, thus allowing the creation of brackets from the bottom of the rear impact structure of the chassis to this jacking bar.

These brackets may have to be thicker than the brackets on NU24 as they would not have as much support for the amount of force to lift the car. The pumps were proposed to be mounted to the brackets using the rubber mount provided with the EBP40 pumps on purchase shown on the assembly in figure 95.

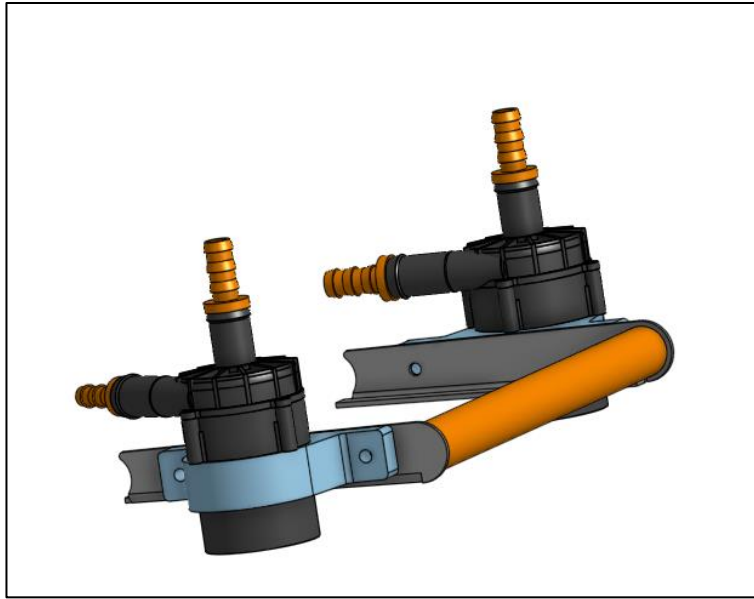


Figure 95: Jacking Point Bracket Assembly

The whole cooling system was then put into an assembly shown in figure 96 allowing the ability to visually see if everything works together and looks appealing. With these together the mass could be calculated showing that the new brackets altogether weighed roughly 220 grams lighter even with an increase in thickness for the jacking point bracket of around 25%. The creation of this assembly also allows for the creation of hoses to determine bends and kinks which could occur with the positions of the pumps and radiator.



Figure 96: Cooling Assembly on Rear of Chassis

The only problem with this setup is the bends and possible kinks it creates for the hoses, but this could be solved by creating a perfect bend using tube bending springs shown in figure 97. Since these are mainly used on exterior tubing the sizing is slightly off therefore finding the correct spring that fits well inside the hose is necessary. Using these springs, fill the inside of the hose and bend the hose into the desired shape and use a heat gun to form the hose into shape.





Figure 97: Hose Bending Spring

This concept has a few possible obstacles in practice, including the position of the radiator, as rules with the firewall shown in Appendix C state that any of the cooling systems should not be seen past the firewall under half of the drivers' head. This was explored and as shown in figure 98 and seems to comply with the rules although this was using NU24s firewall which shouldn't change all that much for NU25.

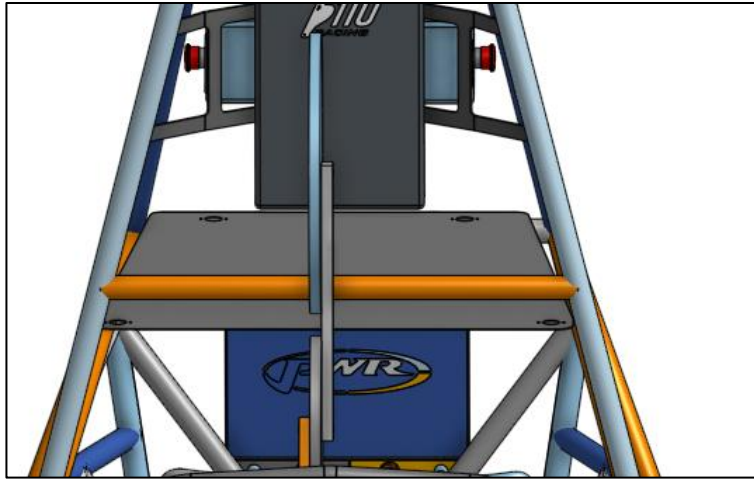


Figure 98: Firewall Rules Confirmation

If the hose placement seems to prove difficult for the position conceived, laying the pump back could give more room for a better bend. A new mounting would have to be designed for this option, or the pump could be positioned somewhere within the chassis depending on the layout, therefore, it is recommended to work closely with the engineers which are responsible for the power box and accumulator designs to allow for spaces for pumps if required.

The jacking point bracket also has the possibility of not being able to hold the weight of the car. Therefore, FEA was done on this bracket assuming 2.5 mm thick steel was used. The same parameters were used as the FEA done on NU24s radiator bracket for holding up the cars' weight, except a closer force was used of 2200 N instead of the large number of 3000 N to provide more accurate depiction of stresses and deformations. The deformations seemed to be very similar to those of NU24s brackets, which had no problems as shown with a maximum of roughly 1.4 mm shown in figure 99.

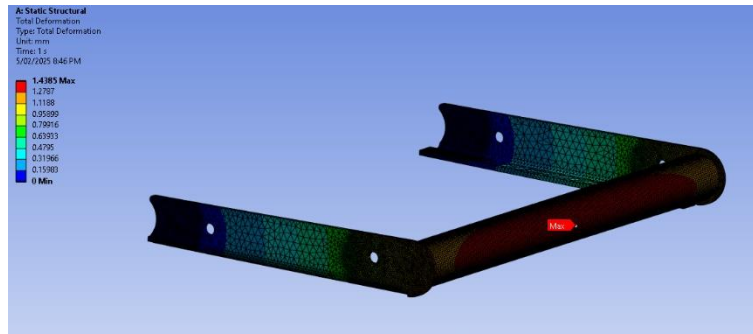


Figure 99: Deformation of Jacking Point Bracket (2200 N)

The stresses on the other hand were quite high where in figure 100 the maximum is over 1000 MPa but can be neglected due to it being intersection in the model between the jacking point and the brackets bend. Although this maximum was neglected on closer inspection on figure 101 the stress exceeded 700 MPa which was above the yield stress of the current used carbon steel of roughly 460 MPa which could cause the bracket to fail at this force.

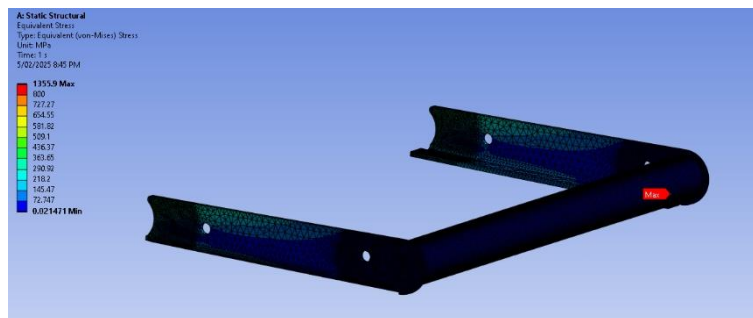


Figure 100: Stress of Jacking Point Bracket (2200 N)

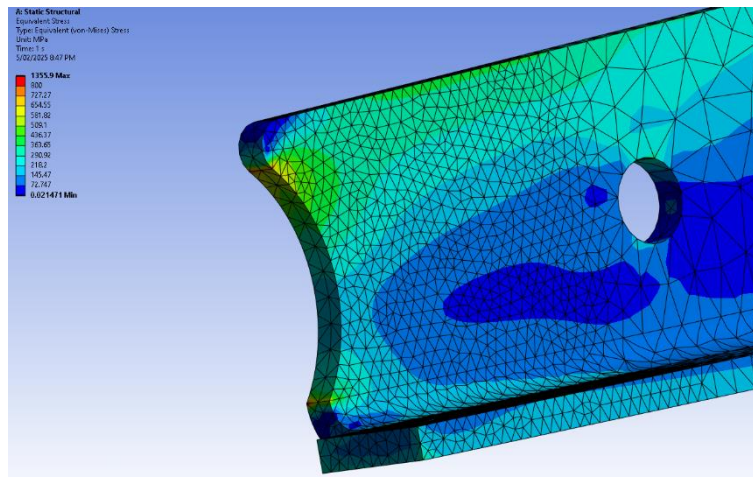


Figure 101: Close-Up of Jacking Point Bracket Stress (2200 N)

Therefore, seventy percent of the force was used due to the jacking point actually not needing to carry the entire cars weight to simulate the stress, where the results were much better showing stresses closer to the yield stress of the steel in figure 102 which is still of concern but manageable, this design could still be used as NU25 should have a reduction in weight.

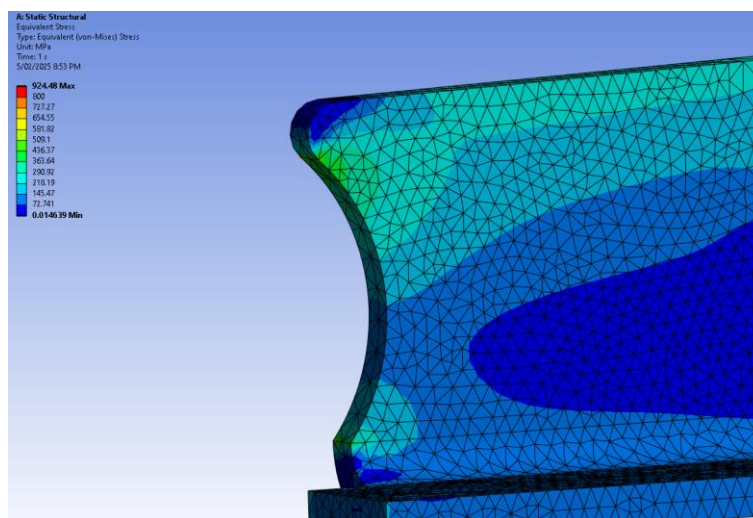


Figure 102: Close-Up of Jacking Point Bracket Stress (1500 N)

If necessary, the brackets could have an increase in thickness to reduce the amount of stress, possibly using a

higher-grade steel could also work, except an extra cost would be required and possibly increased mass. If these concepts don't seem plausible a design similar to NU24 could be used where the radiator bracket and jacking point is all connected. This design would be similar to what is shown in figure 103 below.



Figure 103: Jacking Point Connected to Radiator Brackets for New Radiator

### 11.3 Single Loop Cooling System

At the FSAE-A competition, a cooling engineer from Monash University gave some insight into the possibility of a single loop cooling system. The idea was that the motor controller does not always use all its power therefore the required flow rate from the data sheet could be dismissed. This was reinforced in cooling data collected where the motor controller never reached anything close to dangerous levels of temperature even in some of the hottest conditions.

To test the limits of this, a valve could be used to reduce the flowrate, where in a track day run the car while closely monitoring the temperatures. Another way could be to change the radiator back to a dual pass radiator and run a single loop where the fluid would go from the pump to the motor then to the motor controller and back to the radiator like the figure below.

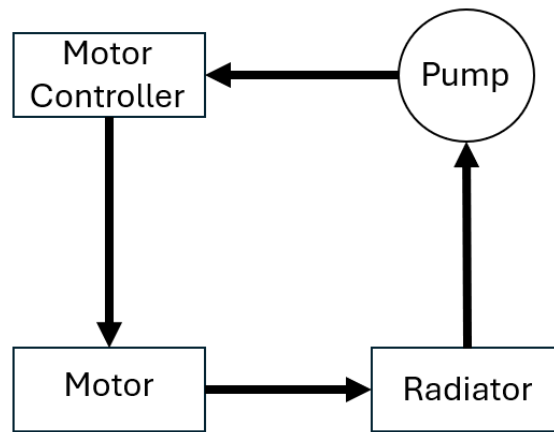


Figure 104: Single Loop Cooling System Configuration

Flow curves were created for this scenario to get an estimate of the flowrates seen using this configuration with either an EBP 40 or PCE-XL water pump. Pressure drops for a dual pass radiator were considered but the hose pressure drops were assumed the same as the dual-loop cooling system therefore real data may slightly deviate as the hose lengths and sizes may change. These flow curves can be shown in graphs below with EBP40 in figure 105 and PCE-XL in figure 106.

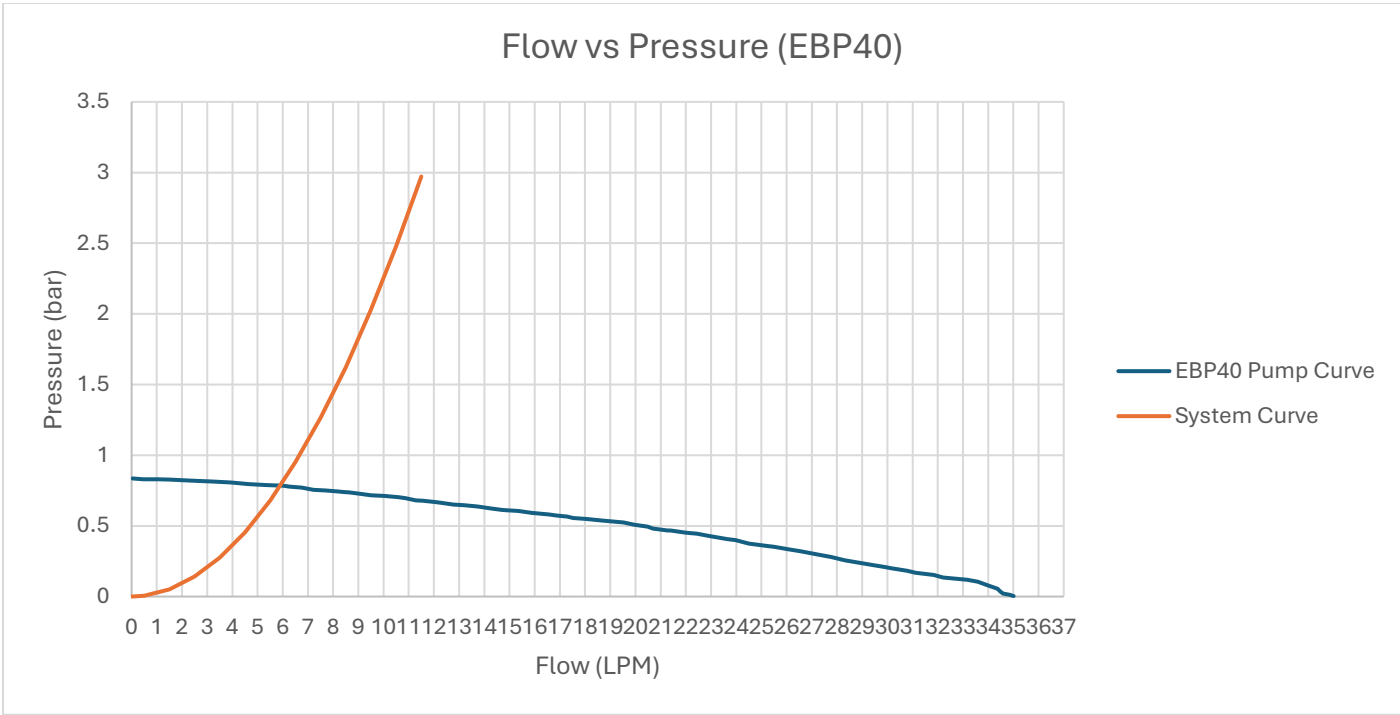


Figure 105: EBP40 Pump Curve Single Loop

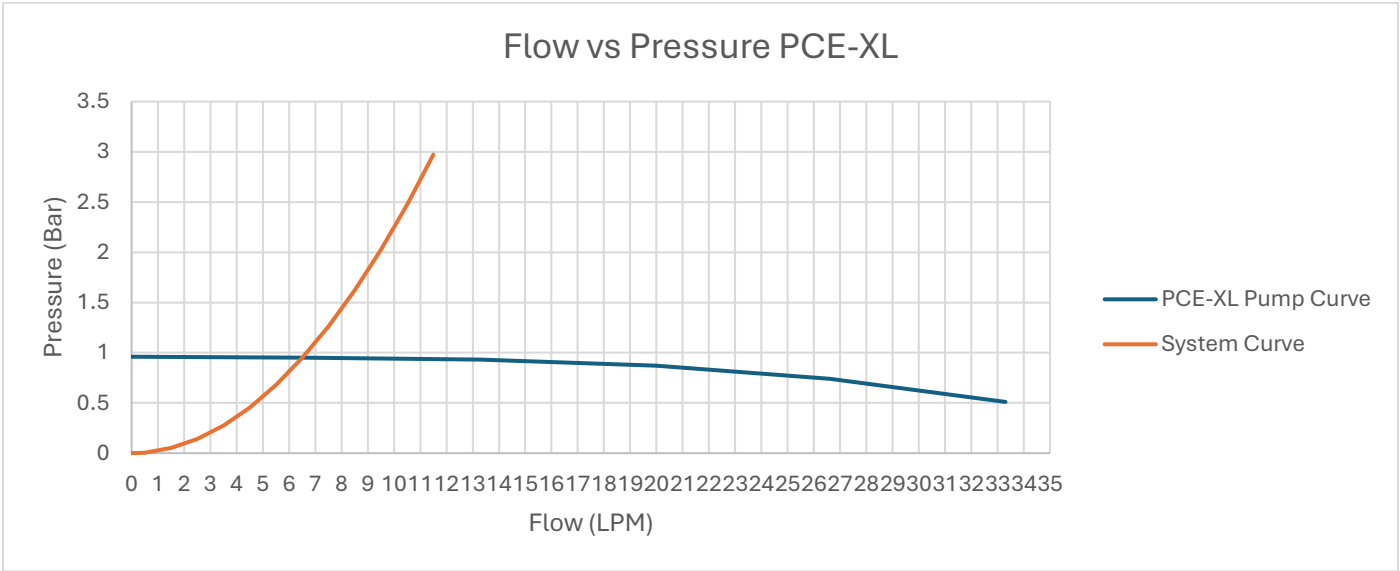


Figure 106: PCE-XL Pump Curve Single Loop

Looking at these graphs it is shown that with an EBP40 water pump in a single loop configuration the flowrate would be roughly 5.9 LPM which is slightly under what the motor requires which could become an issue but

may also be usable. Testing with close monitoring to the temperatures would be required. The PCE-XL water pump showed roughly 6.5 LPM to the system which is more adequate for safety. This is technically a quite risky test and configuration as it lies outside the safety guidelines of the motor controller therefore it is necessary to acquire confirmation by supervisors or team leaders before conducting further testing.



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## Appendix A – Flow Meter Code

```
/*  
YF-S201 Hall Effect Water Flow Meter / Sensor  
Read Water Flow Meter and output reading in litres/hour  
*/  
  
volatile int flow_frequency; // Measures flow meter pulses  
float l_hour; // Calculated litres/hour  
unsigned char flowmeter = 23; // Flow Meter Pin number  
unsigned long currentTime;  
unsigned long cloopTime;  
  
void flow () { // Interrupt function  
flow_frequency++;  
}  
  
void setup() {  
pinMode(flowmeter, INPUT);  
Serial.begin(9600);  
attachInterrupt(3, flow, CHANGE); // Setup Interrupt  
// see http://arduino.cc/en/Reference/attachInterrupt  
sei(); // Enable interrupts  
currentTime = millis();  
cloopTime = currentTime;  
}  
  
void loop () {  
currentTime = millis();  
  
// Every second, calculate and print litres/hour  
if(currentTime >= (cloopTime + 1000)) {  
cloopTime = currentTime; // Updates cloopTime  
// Pulse frequency (Hz) = 7.5Q, Q is flow rate in L/min. (Results in +/- 3% range)  
l_hour = (flow_frequency * 60 / 7.5); // (Pulse frequency x 60 min) / 7.5Q = flow rate in L/hour  
flow_frequency = 0; // Reset Counter  
Serial.print(l_hour); // Print litres/hour  
Serial.println(" L/hour");  
}  
}
```

## Appendix B – Effectiveness NTU Method Formulas

$$q_{actual} = C_{water}(T_{Hot\ Water} - T_{out})$$

Rearrange to find  $T_{out}$

$$T_{out} = T_{Hot\ Water} - \frac{q_{actual}}{C_{water}}$$

Where the heat transfer is given by

$$q_{actual} = q_{max} \cdot \varepsilon$$

Where maximum heat transfer is given by

$$q_{max} = C_{min}(T_{Hot\ Water} - T_{cold\ air})$$

And efficiency of heat transfer is given by

$$\varepsilon = 1 - e^{\left[\left(\frac{1}{C_r}\right)NTU^{0.22}\{e^{-C_rNTU^{0.78}} - 1\}\right]}$$

Our  $C_{min}$  is given by determining the lowest value between  $C_{water}$  and  $C_{air}$  while the maximum is the larger

$$C_{water} = \dot{m}_{water} \cdot C_{p_{water}}$$

$$C_{water} = \dot{m}_{air} \cdot C_{p_{air}}$$

and the ratio is given by

$$C_r = \frac{C_{min}}{C_{max}}$$

Mass flow rates are given by

$$\dot{m}_{water} = Q_{water} \cdot \rho_{water}$$

$$\dot{m}_{air} = A_{radiator} \cdot V_{air} \cdot \rho_{air}$$

Number of transfer units is given by

$$NTU = \frac{UA}{C_{min}}$$

Where UA is given by

$$UA = \frac{1}{\left( \frac{1}{\eta_o h_{air} A_{external}} + \frac{1}{h_{water} A_{internal}} \right)}$$

Where

$$A_{internal} = CoreThickness * CoreHeight * 2 * No.LiquidTubes * 2 + CoreThickness * CoreLength * 2$$

$$A_{external} = No.FinsEachStack * FinPitch * CoreThickness * (No.LiquidTubes * 2 + 1) * 2 * 2 + A_{internal}$$

$$h_{water} = \frac{Nu_{water} k_{water}}{D_{hydraulic}}$$

$$h_{water} = \frac{Nu_{air} k_{air}}{D_{hydraulic}}$$

$$\eta_o = 1 - \frac{NA_f}{A_t} (1 - \eta_f)$$

Where the fin efficiency is

$$\eta_f = \frac{(\tanh \sqrt{\frac{2 \cdot h_{air}}{k_f \cdot t}} \cdot (F_L + \frac{t}{2}))}{\sqrt{\frac{2 \cdot h_{air}}{k_f \cdot t}} \cdot (F_L + \frac{t}{2})}$$

## Appendix C – Applicable FSAE Rules

### T.1.8 Firewall(s)

A Firewall(s) must separate the driver compartment and any portion of the Driver Harness from:

- T.1.8.1 Requirement a. All components of the Fuel System, the engine oil, the liquid cooling systems, any lithium batteries
- b. (EV only) All Tractive System components other than Outboard Wheel Motors **EV.4.1.3** where mounted at the wheels or on the front control arms
- T.1.8.3 Positioning

The Firewall must extend sufficiently far upwards and/or rearwards and/or sideways where any point on the drivers body less than 100 mm above the bottom of the helmet of the tallest driver must not be in direct line of sight with any part given in **T.1.8.1 above**

### VE.2 VEHICLE EQUIPMENT

#### VE.2.1 Jacking Point

VE.2.1.1 A Jacking Point must be provided at the rear of the vehicle

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VE.2.1.2 The Jacking Point must be:

- a. Capable of supporting the vehicle weight and of engaging the organizer Quick Jacks
- b. Visible to a person standing 1 m behind the vehicle
- c. Color: Orange
- d. Oriented laterally and perpendicular to the centerline of the vehicle

- e. Made from round, 25 - 30 mm OD aluminium or steel tube
- f. Exposed around the lower 180° of its circumference over a minimum length of 280 mm
- g. Access from the rear of the tube must be unobstructed for 300 mm or more of its length
- h. The height of the tube must allow 75 mm minimum clearance from the bottom of the tube to the ground
- i. When the vehicle is raised to where the bottom of the tube is 200 mm above ground, the wheels do not touch the ground when they are in full rebound

#### **T.5.4 Coolant Fluid**

T.5.4.1 Water cooled engines must use only plain water with no additives of any kind

T.5.4.2 Liquid coolant for electric motors, Accumulators or HV electronics must be one of:

- plain water with no additives
- oil

T.5.4.3 (EV only) Liquid coolant must not directly touch the cells in the Accumulator

#### **T.5.5 System Sealing**

T.5.5.1 Any cooling or lubrication system must be sealed to prevent leakage

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T.5.5.2 The vehicle must be capable of being tilted to a 45° angle without leaking fluid of any type.

T.5.5.3 Flammable liquid and vapors or other leaks must not collect or contact the driver

T.5.5.4 Two holes of minimum diameter 25 mm each must be provided in the structure or belly pan at



the locations:

- a. The lowest point of the chassis
- b. Rearward of the driver position, forward of a fuel tank or other liquid source
- c. If the lowest point of the chassis obeys T.5.5.4.b, then only one set of holes T.5.5.4.a is necessary

T.5.5.5 Absorbent material and open collection devices (regardless of material) are prohibited in compartments containing engine, drivetrain, exhaust and fuel systems below the highest point on the exhaust system.

### **T.5.6 Catch Cans**

T.5.6.1 The vehicle must have separate containers (catch cans) to retain fluids from any vents from the powertrain systems.

T.5.6.2 Catch cans must be:

- a. Capable of containing boiling water without deformation
- b. Located rearwards of the Firewall below the driver's shoulder level
- c. Positively retained, using no tie wraps or tape

T.5.6.3 Catch cans for the engine coolant system and engine lubrication system must have a minimum capacity of 10% of the fluid being contained or 0.9 liter, whichever is higher

T.5.6.4 Catch cans for any vent on other systems containing liquid lubricant or coolant, including a differential, gearbox, or electric motor, must have a minimum capacity of 10% of the fluid being contained or 0.5 liter, whichever is higher

T.5.6.5 Any catch can on the cooling system must vent through a hose with a minimum internal diameter of 3 mm down to the bottom levels of the Chassis